

OUTAGES IN SOVEREIGN BOND MARKETS

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Abstract

We exploit outages in sovereign bond markets as natural experiments. When the euro area futures market goes down, trading volumes on the cash market decline, liquidity evaporates, and prices deviate from fundamental values. Micro-level evidence reveals two mechanisms: the loss of a hedging instrument reduces dealers' intermediation capacity, and the missing benchmark price widens information asymmetries for clients. Outages on the cash market, in contrast, merely reduce futures trading activity, implying one-way price formation and liquidity provision. Our findings highlight the trade-offs of market centralization, support cross-asset learning over symmetric arbitrage models, and demonstrate how intermediaries impose limits to arbitrage.

Keywords: Yield curve, market microstructure, natural experiment.

JEL classification: D47, G12, G14, G23

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1 Introduction

Sovereign bonds build the core of the financial system, serving as collateral, pricing benchmarks, and as safe-haven assets during crises. Recent stress episodes, however, have shown that sovereign bond markets can become fragile and may need reform.¹ Attention typically focuses on the cash market, where trading is decentralized and intermediated by dealers. Yet a key feature of sovereign bonds is the coexisting futures market, where trading is fully centralized on a single exchange.

This paper exploits technical disruptions on either of these two market segments as natural experiments. Since outages were unanticipated and exogenous to market conditions, they provide a rare glimpse into the role these segments—and the investors active on them—play in liquidity provision and price discovery.

Our first contribution is to demonstrate the vital role of bond futures for market functioning. When Eurex, the dominant futures exchange in the euro area, goes down, trading volumes on the cash market drop by over 70%. Liquidity falls by a similar amount, whether measured through executable order book volume or through the update frequency of indicative quotes. The effect on mispricing is even larger. Focusing on German bonds, the euro area’s risk-free benchmark, the remaining cash market transactions during outages occur at prices far from fundamental values. Under normal conditions, arbitrage forces keep pricing errors—the deviation of observed yields from a smooth fitted yield curve—small. When Eurex goes down, however, the average pricing error doubles and the root mean squared pricing error—[Hu, Pan, and Wang \(2013\)](#)’s noise measure—roughly quadruples, exceeding the peak of the Covid-19 market turmoil in March 2020.

Taken together, these macro-level results suggest that bond futures are vital for the

¹Recent bond market stress episodes include the Covid-19 turmoil in March 2020 (see e.g. [Haddad, Moreira, and Muir, 2021](#); [Eisenbach and Phelan, 2025](#); [Duffie, Fleming, Keane, Nelson, Shachar, and Van Tassel, 2023](#); [Duffie, 2025](#)), the UK Gilt crisis in late 2022 ([Pinter, Siriwardane, and Walker, 2024](#); [Czech and Monroe, 2025](#)), and the US tariff shock in April 2025 ([Jiang, Krishnamurthy, Lustig, Richmond, and Xu, 2025](#); [Pinter, Uslu, and Smets, 2025](#)). Ongoing bond market policy reforms include the U.S. Securities and Exchange Commission’s initiative to increase central clearing and all-to-all trading ([press release](#)), the Federal Reserve’s proposal to increase incentives for dealer banks to intermediate in U.S. Treasuries ([draft notice](#)) and the EU Commission’s work towards a consolidated tape ([press release](#)).

euro area fixed-income market to function smoothly. To establish these effects, we combine non-anonymous regulatory transaction-level data with proprietary data from major trading platforms, interdealer brokers, and indicative quote providers, and use an event-study design that compares outage windows to matched control windows on the same day one week before and after.

As our second contribution, we exploit the granularity of our transaction dataset and identify two micro-level mechanisms causing the cash market to break down: (1) outages remove a hedging instrument, reducing the intermediation capacity of dealers, and (2) outages remove a benchmark price, exacerbating information asymmetries for clients. Two stylized facts immediately point to these mechanisms. Outages reduce trading volumes most for dealer-to-client trades and least for client-to-client trades, consistent with dealers cutting back intermediation. Mispricing, in contrast, spikes most for client-to-client trades and does not increase at all for dealer-to-dealer trades. In fact, dealers trade more profitably during outages, consistent with clients relying on bond futures as benchmarks.

We corroborate the hedging mechanism by exploiting dealer-level heterogeneity in hedging strategies. Dealers with the most intense use of bond futures retract most from the cash market during Eurex outages. Dealers with a more moderate use of bond futures, in contrast, retain some intermediation capacity, but they demand higher markups when trading with clients, particularly for large trades in long-term bonds. To corroborate the benchmark mechanism, we show that less sophisticated types of clients fare worse during outages, e.g. retail investors compared to hedge funds, and that private information and relationships become more important without the public benchmark. Better connected clients cause less mispricing, pay less markup, and trade more profitably while Eurex is down. The same is true for strong dealer-client relationships. Dealers extract less profits from loyal clients during outages. Further evidence shows that outages cause severe mispricing also in interest rate swaps, despite swaps not being directly linked to Eurex-traded derivatives. This suggests that Bund futures provide the pricing benchmark not only for the bond but also the swap market.

As our third and final contribution, we document the asymmetry of outage effects, using five outages on four different cash trading platforms as natural experiments. On the cash side, these outages reduce aggregate trading activity, indicating that trading does not seamlessly migrate to alternative platforms when individual venues go down. On the futures side, trading activity also declines, but not nearly as much as in the reverse direction. Measures of market functioning react even less. In a pooled regression of all cash venue outages, only [Roll \(1984\)](#)’s bid-ask spread estimate indicates a slight decrease in bond future liquidity by 0.16 standard deviations. To address the concern that these benign effects might simply reflect the smaller scale of cash venue outages, we rescale the analysis by the affected venue’s market share in the underlying country’s cash bond market. Cash venue outages then have a stronger impact on bond future quantities: the drop in trading volumes and trade counts is significantly larger when the disrupted venue has a higher market share. Market functioning measures, however, do not exhibit comparable scaling with venue importance. Even for high-share venues, we find at most a modest increase in illiquidity; for example, our estimates imply that an outage of a venue with a 50% cash market share raises the [Amihud \(2002\)](#) illiquidity proxy for the corresponding bond future by only about 0.24 standard deviations. Event-level evidence for three high-share outages—Bloomberg in German Bunds, MTS in Italian bonds, and BrokerTec in U.S. Treasuries—confirms the benign effects on bond futures. Taken together, these findings suggest that cash venue outages primarily reduce hedging needs and thus trading volumes in futures, whereas futures outages additionally remove the centralized benchmark that disciplines prices. Price discovery and liquidity provision thus appear to run predominantly from the centralized futures market to the fragmented cash market, rather than symmetrically in both directions.

Literature Our quasi-experimental evidence is informative for three major strands in the literature. It highlights the trade-offs between a (de)centralized market structure, distinguishes competing theories about the link between liquidity and arbitrage, and lastly

shows how financial intermediaries create limits to arbitrage.

Market Centralization A key issue in market microstructure is the prevalence and desirability of trading outside of exchanges. Positive network effects push trading towards a single central exchange, but information asymmetries pull in the opposite direction. In particular, because more informed and faster traders impose adverse-selection costs on liquidity suppliers, these liquidity suppliers have an incentive to trade with uninformed traders off-exchange, potentially at a discount.² [Lee and Wang \(2025\)](#) formalize this intuition. In their model, less informed traders optimally choose the OTC market. Nonetheless, closing the OTC market raises welfare, particularly for assets traded mostly OTC. [Allen and Wittwer \(2023a\)](#) use transaction-level data to estimate a structural model of the Canadian government bond market. They find that shifting trades to a centralized platform could decrease welfare, unless competition among dealers is sufficiently strong.³

Our results provide an important qualifier. On the decentralized cash market, investors—particularly clients—free-ride on the price discovery provided by other investors on the centralized futures market. Bond futures serve as benchmarks, reducing information gaps and increasing liquidity on the cash market ([Glosten and Milgrom, 1985](#); [Duffie, Dworzak, and Zhu, 2017](#)). Welfare calculations should take these positive externalities into account. Centralizing the OTC market would have additional benefits, insofar as it reduces the dependency on the futures market and leads to self-sustained price discovery on the cash market. This is in line with [Kutai, Nathan, and Wittwer \(2025\)](#), who document that the Israeli bond market, the only major government bond market operating on an exchange, performed better during the Covid-19 crisis than most other markets operating OTC. We find similar results across euro area countries: the liquidity in Italian

²[de Roure, Moench, Pelizzon, and Schneider \(2025\)](#) provide empirical evidence in line with the price discrimination channel. They document that for German government bonds, transaction prices in the OTC market are favorable compared to the centralized MTS exchange. This ‘OTC discount’ is in line with the fact that most German bonds are traded off-exchange.

³[Dugast, Üslü, and Weill \(2022\)](#) provide a model for on vs. off-exchange trading and show that, depending on their trading capacity, some market participants benefit from a decentralized OTC market. [Biais and Green \(2019\)](#) provide historical context and document that up until World War II, most US bond trading occurred on-exchange.

bonds drops least during futures market outages, in line with the fact that most cash trading is centralized in a single limit order book. At the same time, our results show that full market centralization comes at the cost of introducing systemic risks.

Arbitrage across Time and across Markets Our natural experiments distinguish two margins of market efficiency: arbitrage *across time* (Grossman and Miller, 1988) and *across markets* (Gromb and Vayanos, 2010, 2018). The Eurex outages show that the intertemporal margin alone does not ensure market functioning. When the futures market goes dark, cash market prices dislocate, because arbitrageurs fail to fully absorb the shock by temporarily warehousing bonds. Instead, market efficiency hinges on the cross-market linkage between Bunds and Bund futures.

Cross-market arbitrage models, however, typically assume symmetry. They assume that arbitrageurs provide liquidity on one market conditional on the other being active. We test and reject this symmetry.⁴ Outages on cash market venues barely affect the futures market, suggesting that price formation and liquidity provision is somewhat of a one-way street from the futures to the cash market.⁵ This asymmetry aligns with *cross-asset learning* models (see e.g. Admati, 1985; Veldkamp, 2006; Cespa and Foucault, 2014; Asriyan, Fuchs, and Green, 2017). Those models posit that liquidity providers use some particularly informative prices — in our case bond future prices — when trading related instruments — in our case sovereign bonds and interest rate swaps. An inherently linked source of the asymmetry is the higher liquidity of the futures market. Arbitrageurs can hedge cash market exposures more easily and cheaply through futures than vice versa. Structural features of the futures market, such as the above-mentioned centralized trading, central clearing and margin-based leverage, contribute to this liquidity advantage.

⁴Harding and Ma (2010) show that outages on the US Treasury futures exchange lead to a dramatic fall in liquidity on an electronic cash market trading platform. This is similar to our finding that liquidity on MTS evaporates when Eurex is down. Harding and Ma (2010), however, cannot test for asymmetry, because there had been no suitable outages of cash trading venues.

⁵Vector error correction models confirm that bond futures ‘lead’ price discovery, reflecting new information faster than cash bonds (see e.g. Mizrach and Neely, 2008; Dobрева and Schaumburg, 2023; Upper and Werner, 2007; Jappelli, Lucke, and Pelizzon, 2022). We find an even more pivotal role: bond futures are essential for the fixed-income market to function smoothly.

Yet, disentangling liquidity from information is challenging because they reinforce one another: liquidity concentrates where prices are informative, and informativeness rises with liquidity, potentially causing endogenous concentration on a single venue ([Pagano, 1989b,a](#); [Vayanos and Wang, 2007](#); [Vayanos and Weill, 2008](#)).

Intermediary Asset Pricing and Limits to Arbitrage A closely related literature studies the role of intermediaries in financial markets and the limits to arbitrage they impose (see [Long, Shleifer, Summers, and Waldmann, 1990](#); [Shleifer and Vishny, 1997](#); [Gromb and Vayanos, 2002](#); [Mitchell, Pedersen, and Pulvino, 2007](#); [Brunnermeier and Pedersen, 2008](#); [He and Krishnamurthy, 2013](#); [Allen and Wittwer, 2023b](#); [Bräuning and Stein, 2024](#)). In a recent paper on the U.S. Treasury market, e.g., [Duffie et al. \(2023\)](#) show that expected volatility explains most of the time variation in Treasury market liquidity, but when dealers exhaust their balance sheet capacity—as observed in March 2020—liquidity drops abruptly and market functionality gets impaired. The outages we study provide a clear natural experiment that corroborates this functional chain. The Eurex outages diminish dealers’ capacity to intermediate bond trades on the cash market, since they are unable to hedge their inventory risk on the futures market. Crucially, our evidence is not subject to the usual endogeneity concerns between expected volatility, market liquidity, and dealer capacity.

2 European Government Bond Market Structure

This section briefly outlines the market microstructure for euro area government bonds (EGBs) on the cash and futures market. See [Appendix A](#) for further details and [Bessembinder, Spatt, and Venkataraman \(2020\)](#) for a recent survey on fixed-income market structure. Our analysis covers the four largest euro area member states: Germany, France, Italy and Spain. They are also the only countries with corresponding bond futures.

The futures market for EGBs is highly homogeneous and centralized, as these futures and their associated options are traded exclusively on Eurex’s central limit order book

(CLOB). There are only a handful of bond futures covering selected maturity segments. A 10-year bond future exists for all four countries. For Italy, also a 2-year bond future is actively traded while for Germany, bond futures are also available for the 2-, 5-, and 30-year segment. Together, the trading volume in German bonds is roughly ten times larger on the futures than the cash market in terms of underlying value. For France and Italy, the two markets are of comparable size, while for Spain, where a future was introduced last, trading volumes in futures are only about 1% of the cash market volume. In practice, investors hedge Spanish bonds, and bonds of countries without a dedicated futures market, using whichever futures offer the best combination of liquidity and risk alignment.

In terms of market participants, the futures market is rather concentrated, with relatively few high frequency trading firms dominating trading activity ([Hautsch, Noé, and Zhang, 2017](#); [Kirilenko, Kyle, Samadi, and Tuzun, 2017](#)). The cash market, in contrast, has the typical dealer-client and core-periphery structure (see e.g. [Di Maggio, Kermani, and Song, 2017](#); [Li and Schürhoff, 2019](#); [Bessembinder et al., 2020](#); [Hendershott, Li, Livdan, and Schürhoff, 2020](#); [Allen and Wittwer, 2023a](#)).

The cash market for EGBs is also much more fragmented, segmented, and opaque compared to the futures market. There are hundreds of individual bonds outstanding at any point in time and these can be traded on very different venues. These venues differ along multiple dimensions, most importantly the trading mechanism they employ and the types of investors they grant access to. Consider concrete examples from two polar opposites. On the one side, bonds can be traded anonymously and immediately in a CLOB, just like bond futures. MTS is the dominant platform in this regard, but it is only accessible to dealers, i.e. large banks. Bonds can also be traded in a CLOB on regular stock exchanges, which are also open to small retail traders. But out of the multitude of exchanges, no single exchange captures a significant market share and the order book for most bonds is correspondingly thin. Some exchanges offer incentive programs for designated market makers to improve this poor liquidity. Importantly, on-

exchange trading is ‘lit’, i.e. all market participants can observe quotes and transaction prices and volumes.

On the other side, bonds can be traded over-the-counter (OTC). Bilateral OTC trades are neither anonymous nor immediate. Such trades are typically negotiated by voice or chat. To preserve some anonymity and to reduce search costs, market participants can also trade OTC via a broker. In this case, the initiating party communicates its trade request to a broker, who then tries to find a suitable counterparty on a ‘matched principal’ basis, see [de Roure et al. \(2025\)](#) for a detailed description of this market segment. This way, the two counterparties do not have to reveal their identity to each other. Yet another venue on the OTC market are systematic internalisers, also called single-dealer platforms. On these platforms, large dealer banks act as a central counterparty for trades initiated by their clients. Compared to on-exchange trades, all three types of OTC trades are comparatively ‘dark’, i.e. there is little pre- and post-trade transparency. In between these two extremes, another increasingly important market segment for EGB trading is electronic trading platforms, which dominate the dealer-to-client segment. Examples of such platforms are Bloomberg and Tradeweb, which use a request-for-quote (RFQ) mechanism, i.e. clients request quotes for a certain bond from multiple dealers.

Moreover, to facilitate the matching process between buyers and sellers, various companies, e.g. trading platforms themselves, disseminate indicative quotes for European government bonds. These quotes are often available only for specific bonds, e.g. ‘benchmark’ bonds of selected maturities, and the exact calculation methods behind those quotes are not disclosed.

Lastly, the market structure of EGBs differs substantially across countries. Italian bonds, e.g., are predominantly traded ‘on-exchange’ (on MTS), whereas German bonds are traded more on electronic trading platforms and OTC.⁶ To capture this heterogeneous bond market as much as possible, we combine various intraday data sources, as explained in the following subsections.

⁶Our distinction of trading venue types (on-exchange, electronic platforms, OTC) differs somewhat from the MiFID II regulation, see [Appendix A.3](#) for details.

2.1 Cash Market Data

For EGB transactions on the cash market, we start with the non-anonymous MiFID II dataset, which contains information on bond transactions since 2018. For each transaction, this dataset contains the ISIN, price, volume, time, the involved counterparties, as well as the venue on which the trade was executed. This dataset has been made available relatively recently and to the best of our knowledge, we are the first to use it for academic research.⁷

An important caveat for our purposes is that the regulatory data only covers trades in which a German security is traded or where at least one of the involved parties—the buyer or seller, the trading venue, or a central counterparty (CCP)—is domiciled in Germany. This is why trades in German bonds are overrepresented.

For the six most important days in our sample, the two Eurex outage days and the respective control days, we address this limitation by complementing the regulatory dataset with transaction data sourced directly from trading platforms (MTS, MTS BondVision, Tradeweb) and interdealer brokers (TPICAP, BGC, GFI, Aurel). This way we are able to capture also a significant share of transactions in French, Italian and Spanish government bonds. In contrast to the regulatory data, however, these datasets are anonymous, i.e. they do not contain information about the involved counterparties. Appendix A.4 describes our transaction-level data in detail.

Besides the transaction data, we also use quote data. In particular, we have executable quotes and volumes from MTS’s CLOB and indicative quotes from Bloomberg, Refinitiv and TPICAP.

2.2 Futures Market Data

For bond futures, we exploit three different datasets. First, we have the full history of transaction prices and volumes at the millisecond frequency going back to 2002. Second,

⁷While relatively new, the dataset is an extended version of the MiFID I dataset used in prior work (see e.g. [de Roure et al., 2025](#)), which was collected during 2008-2017, see [Bundesbank website](#) and [German Securities Trading Act](#).

we have the full intraday order book data, i.e. bid and ask quotes and volumes for all order book levels, going back to April 2019 from Deutsche Boerse’s A7 platform. Third and lastly, we have non-anonymous investor-level data on bond future transactions since 2019 from the European Market Infrastructure Regulation (EMIR) dataset.

2.3 Investor-Level Data

The two regulatory datasets mentioned above (MiFID II and EMIR) are non-anonymous; they contain a unique legal identifier (LEI) for the counterparties involved in each transaction. To classify investors into different sectors, we can thus use supervisory lists and the Register of Institutions and Affiliates Database (RIAD, see [Perrella and Catz, 2020](#)). We differentiate between banks, non-bank financial institutions (NBFI), investment funds, hedge funds, insurance companies and pension funds (ICPF), non-financial corporations (NFC), an official sector (including central banks), and households (retail investors). We further identify whether an investor is a primary dealer in German Bund auctions and whether the investor is usually active also on the bond futures market or only the cash market. Appendix [A.5](#) provides details.

3 Macro-Level Effects of Futures Market Outages

On 14 April 2020 and 1 July 2020, technical glitches caused outages on Eurex, the leading futures market exchange in the euro area. The first outage lasted approximately four and a half hours, from 9:25 a.m. to 2:00 p.m. while the second outage lasted less than three hours, from 8:49 a.m. to 11:31 a.m. (all in local time). In both cases, Eurex blamed ‘faulty third-party software’ as the root cause.⁸ We use these outages as natural experiments to

⁸Eurex’s operator Deutsche Boerse commented: ‘*The disruption in the T7 system in April and today’s failure had the same origin. They were due to faulty third-party software that is part of the trading system. [...] External causes can be ruled out.*’ Besides Eurex, the T7 outage also took down several stock exchanges, most notably XETRA but also local exchanges in Eastern Europe. These simultaneous outages on the spot market do not contaminate our results because they did not affect the bond market. [Appendix D](#) contains further narrative evidence on the outages. [Appendix B.1](#) exploits twelve prior outages on Eurex as a robustness check.

study the role of the futures market for the broader fixed-income market in the euro area.

We will show how these outages affect the *cash* market for EGBs, namely in terms of trading activity (Section 3.1), market liquidity (Section 3.2), and pricing (Section 3.3).

3.1 Trading Activity

To investigate the effect of the outages on cash market trading activity, we estimate the following dummy regression:

$$\log(1 + Volume_t) = \alpha + \beta \times D_t + \epsilon_t \quad (1)$$

where $Volume_t$ is the bond trading volume and D_t is a dummy that equals one during the Eurex outages and is zero otherwise. We include six days (the two outage days plus the preceding and subsequent week) and 15 intraday observations per day (from 08:30 a.m. to 4:00 p.m.). We use $\log(1 + Volume_t)$ to keep a balanced sample despite a few zero-volume intervals.⁹

Table 1 reports the results.¹⁰ Column (1) confirms that trading volumes on the cash market decrease dramatically during the futures market outages, by about 72% ($\exp(-1.28) - 1$). Column (2) looks at the maturity cross-section. Volumes drop by only 44% for bonds with less than 2.5 years to maturity, but by around 90% for longer-term bonds. Column (3) looks at the cross-section of countries and shows that German and Italian bonds are most affected by the outage, with trading volumes falling by around 85%, while volumes in French and Spanish bonds drop by around 65%.¹¹

⁹Zero-volume intervals occur because of the outages, so omitting them would bias the estimated effect toward zero. On the other hand, Chen and Roth (2023) show that the log-plus-one transform can distort percentage interpretations when treatment induces zeros. We largely avoid this problem by aggregating volumes to relatively wide time buckets, minimizing zero-volume periods.

¹⁰Throughout this paper, *, **, *** indicate statistical significance at the 10%, 5% and 1% level, respectively and robust standard errors are shown in brackets.

¹¹This country-level heterogeneity likely reflects differences in the relative size of the futures market and the cash market structure (see Appendix A.3). Appendix B.2 confirms that trading volumes catch-up when Eurex goes back online.

	(1) Aggregate	(2) Maturities	(3) Countries
Outage	-1.28*** [0.25]		
Outage $\times$ <2.5y		-0.58** [0.23]	
Outage $\times$ 2.5-5.5y		-2.38*** [0.27]	
Outage $\times$ 5.5-10.5y		-2.17*** [0.18]	
Outage $\times$ >10.5y		-2.52*** [0.32]	
Outage $\times$ DE			-2.03*** [0.31]
Outage $\times$ FR			-1.15** [0.46]
Outage $\times$ IT			-1.97*** [0.30]
Outage $\times$ ES			-0.97** [0.38]
FE Day	✓	✓	✓
FE Time	✓	✓	✓
Observations	90	360	360
Adjusted R^2	0.626	0.700	0.469

Table 1: Effect of Eurex Outages on Cash Trading Volume. Each column shows results of a different regression, see Equation 1. The dependent variable is the log of the transaction volume in bonds of a given country and maturity bucket, using 30-minute intervals between 08:30 a.m. and 4:00 p.m. Column (2) adds an interaction term between the outage dummy and the maturity buckets, column (3) does the same for countries.

3.2 Market Liquidity

3.2.1 Executable Quotes

Given that cash market trading volumes collapse during Eurex outages, the natural follow-up question is how market liquidity reacts. If liquidity remained stable, the decline in volumes would simply reflect investors sitting out the outage. If liquidity also collapses, this would indicate a broader breakdown in market functioning. To study liquidity, we start with MTS, the only major trading venue for euro area government bonds with a central limit order book, i.e. firm executable quotes.

Figure 1 shows the aggregate order book depth across all German, French, Italian, and Spanish sovereign bonds on MTS. To put the events into perspective, we compare outage days with the same day in the previous and subsequent week. Liquidity evaporates during the Eurex blackout. Within minutes after Eurex goes down on 14 April 2020, the quantity of bonds investors are willing to buy or sell at any price drops from €30bn to around €5bn. The order book depth returns to normal by 2:06 p.m. six minutes after

trading on Eurex resumed. The same is true for the second outage. While liquidity usually picks up around 9:00 a.m. on MTS, most quotes appear only at 11:43 a.m. on 1 July 2020, 12 minutes after Eurex was back online. The breakdown on MTS is market-wide. It is not confined to the handful of deliverable bonds closely linked to the futures contract traded on Eurex. Rather, liquidity drops for all bonds, suggesting that the MTS cash market platform functions properly only if the futures market is active.¹²

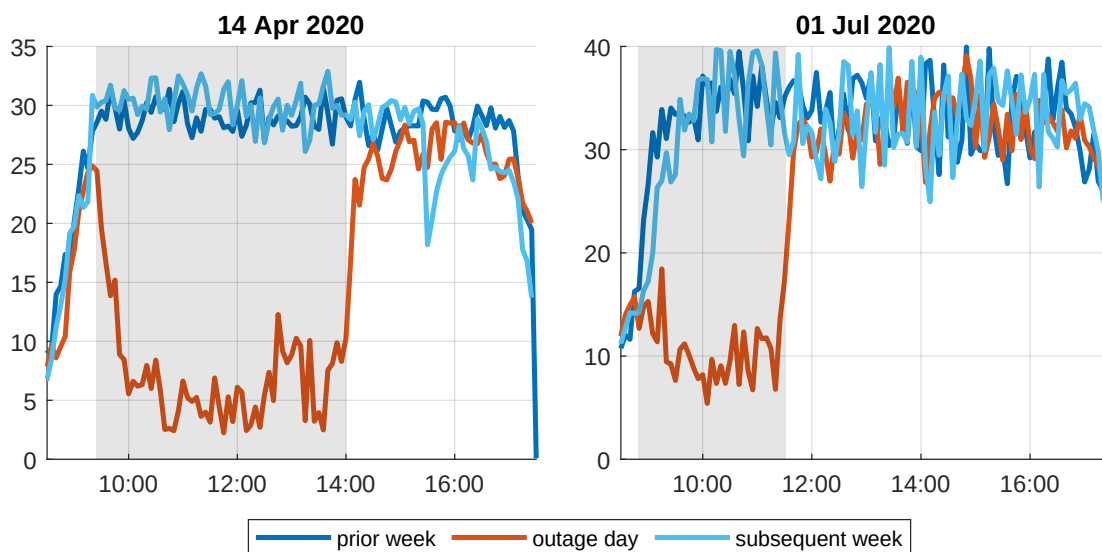


Figure 1: Order Book Depth on MTS. This figure shows the total quoted volume for all German, French, Italian and Spanish bonds (in billion €) across all three levels and both sides of the order book, at 5-minute snapshots.

Before delving into further details, recall that [Harding and Ma \(2010\)](#) report broadly similar results to what we find here: outages of the main US futures exchange (CBOT, Chicago Board of Trade) lead to a dramatic fall in trading and quoting activity on a major electronic cash market trading platform (Espeed). They attribute this finding to high-frequency trading firms that are only active on the cash market if the futures market is online and vice versa. However, we can rule out this explanation in our setting, because only dealers are allowed to trade on MTS.¹³ The fact that liquidity on MTS evaporates

¹²Appendix B.3 provides further evidence on the MTS breakdown: (i) bid-ask spreads widen, (ii) liquidity evaporates even for euro area countries that do not have a dedicated futures market, and (iii) liquidity remains slightly more robust on MTS's local market segments, where quoting obligations are enforced.

¹³The current list of members is available on the [MTS website](#).

nonetheless suggests that the forces at work are more general. We will show that the more likely explanation is simply that price discovery and liquidity provision crucially depend on an active futures market. Without bond futures as a hedging instrument and pricing signal, market functioning on the cash market deteriorates dramatically.

To study the dependency of MTS on Eurex in more detail, we run similar dummy regressions as in Equation 1 using the order book depth at 5-minute snapshots as a dependent variable. Table 2 reports the results. The available liquidity on MTS drops by about 77% when Eurex goes down, see column (1). Column (2) shows that liquidity drops by 62% for bonds with less than 2.5 years to maturity, but by around 90% for longer-term bonds, in line with the trading volume results above.

	(1) Aggregate	(2) Maturities	(3) Countries
Outage	-1.47*** [0.06]		
Outage $\times$ <2.5y		-0.97*** [0.06]	
Outage $\times$ 2.5-5.5y		-2.23*** [0.11]	
Outage $\times$ 5.5-10.5y		-2.20*** [0.10]	
Outage $\times$ >10.5y		-2.63*** [0.30]	
Outage $\times$ DE			-15.88*** [0.95]
Outage $\times$ ES			-1.86*** [0.19]
Outage $\times$ FR			-3.83*** [0.56]
Outage $\times$ IT			-1.30*** [0.18]
FE Day	✓	✓	✓
FE Time	✓	✓	✓
Observations	546	2184	2184
Adjusted R^2	0.845	0.697	0.634

Table 2: Effect of Eurex Outages on MTS Order Book Depth. Each column shows results of a different regression. Throughout, the dependent variable is the log of the quoted bid and ask volume of all bonds of a given country and maturity bucket, using 5-minute snapshots between 08:30 a.m. and 4:00 p.m. All explanatory variables are dummies, either for the maturity bucket or for different countries.

Before examining the differential effect across countries, recall that MTS is the main trading platform for Italian bonds and aggregate trading volumes in Italian bonds have a similar magnitude on the cash and futures market. This is in stark contrast to Germany e.g., where trading volumes on the futures market are roughly ten times larger than on the cash market and where MTS has only a negligible cash market share (see Appendix

A.3 for details). Hence, we would expect that Italian bonds are less affected by the Eurex blackout, which column (3) confirms. For French and especially German bonds, order book depth frequently drops to zero during outages (see Appendix B.3). These periods, when not a single dealer quotes a single bond of a given country, generate very large negative coefficients in the log-plus-one specification. Italian bonds, in contrast, continue to be quoted but quoted depth drops by 73%. Thus, market functioning on MTS is significantly impaired during Eurex blackouts, even for Italian bonds.

3.2.2 Indicative Quotes

Apart from the executable quotes on MTS, there are several providers of indicative quotes for EGBs. We obtained data from three different providers, namely Bloomberg, Refinitiv and TPICAP. Bloomberg terminals and Refinitiv’s Eikon application are the two most widely used sources for real-time financial information on the dealer-to-client segment, whereas TPICAP operates as an interdealer broker. All three disseminate quotes to facilitate trading, but the methodologies behind these quotes are proprietary and not publicly disclosed.¹⁴

Figure 2 plots the frequency of quote updates for 10-year government bonds. Across all three providers, quotes turn stale during the outages. The number of quote updates drops dramatically the moment Eurex goes down and quickly recovers once trading resumes.¹⁵ The quoting inactivity is particularly pronounced on the interdealer platform. On average across data providers and outages, the update frequency of indicative quotes drops by about 75%, mirroring the decrease in executable order book depth on MTS.

¹⁴Schestag, Schuster, and Uhrig-Homburg (2016) describe Bloomberg’s quotes as ‘a weighted average of quotes from participating dealers’. Similarly, Refinitiv’s ‘Tick History’ database contains the real-time feed updates shown on Eikon. For German and French bonds, quotes are from multiple ‘pricing contributors’, all of which are major European banks. For Italian and Spanish bonds, only a ‘composite price’, computed by Refinitiv, is available.

¹⁵Appendix B.4 confirms that these results hold across countries.

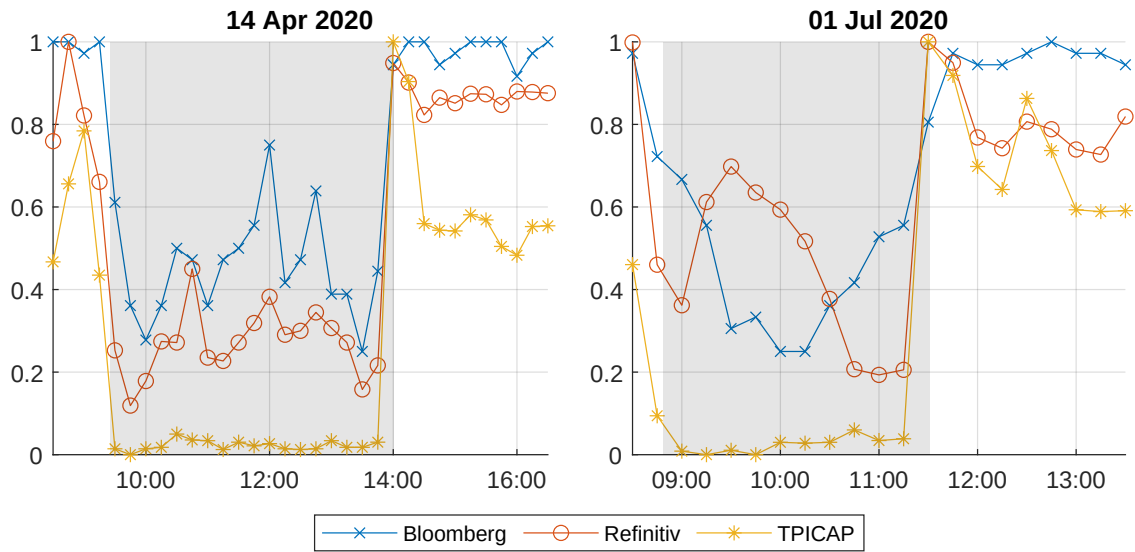


Figure 2: Quote Update Frequency for 10-Year Government Bonds on Different Platforms. Bloomberg and TPICAP data refer to quote updates in the cheapest-to-deliver bond, Refinitiv data to quote updates in the on-the-run (‘benchmark’) bond. Results for TPICAP and Refinitiv are based on the exact number of new quotes per bond. For Bloomberg, we approximate the number of new quotes as the number of tick-by-tick price changes. To show all series on a single scale, we aggregate the number of quote updates in 15-minute windows and normalize them to a 0-1 range for each data provider.

3.3 Pricing

We have documented that futures market outages lead to lower trading volumes and lower liquidity on the cash market for euro area government bonds. What we are ultimately interested in, however, is whether the price discovery process for EGBs is actually impaired due to the unavailability of bond futures.

Some *prima facie* evidence points in this direction: on 14 April 2020, the Dutch State Treasury Agency cancelled three bond auctions planned for that day, citing the Eurex outage as the reason.¹⁶ The auctions were postponed to the next day, when Eurex was back online. Dutch authorities apparently feared that their bonds, despite the AAA rating by all major rating agencies, might not be properly priced by market participants while the futures market is offline.

Was this fear justified? To find out, we study yield curve fitting errors, a popular measure for how well the bond market functions. [Hu et al. \(2013\)](#) e.g. argue that arbitrage forces usually keep the yield curve smooth. Hence, a low dispersion in bond yields along

¹⁶‘Decided to postpone today’s [...] auctions [...], due to technical issues at Eurex’ ([DSTA press release](#)).

the yield curve is a sign that bond prices are in line with fundamental values. So were the prices of bond transactions that *did* occur during the Eurex outages ‘fair’? We focus on German bonds with a maturity between one and ten years to answer this question, as they constitute the benchmark risk-free yield curve for the euro area. We convert transaction prices observed in the market into yields and then fit a term structure model to these observed yields. We do this separately for all transactions that occurred while Eurex was offline, and for all transactions that occurred during the same intraday window one week before and after.

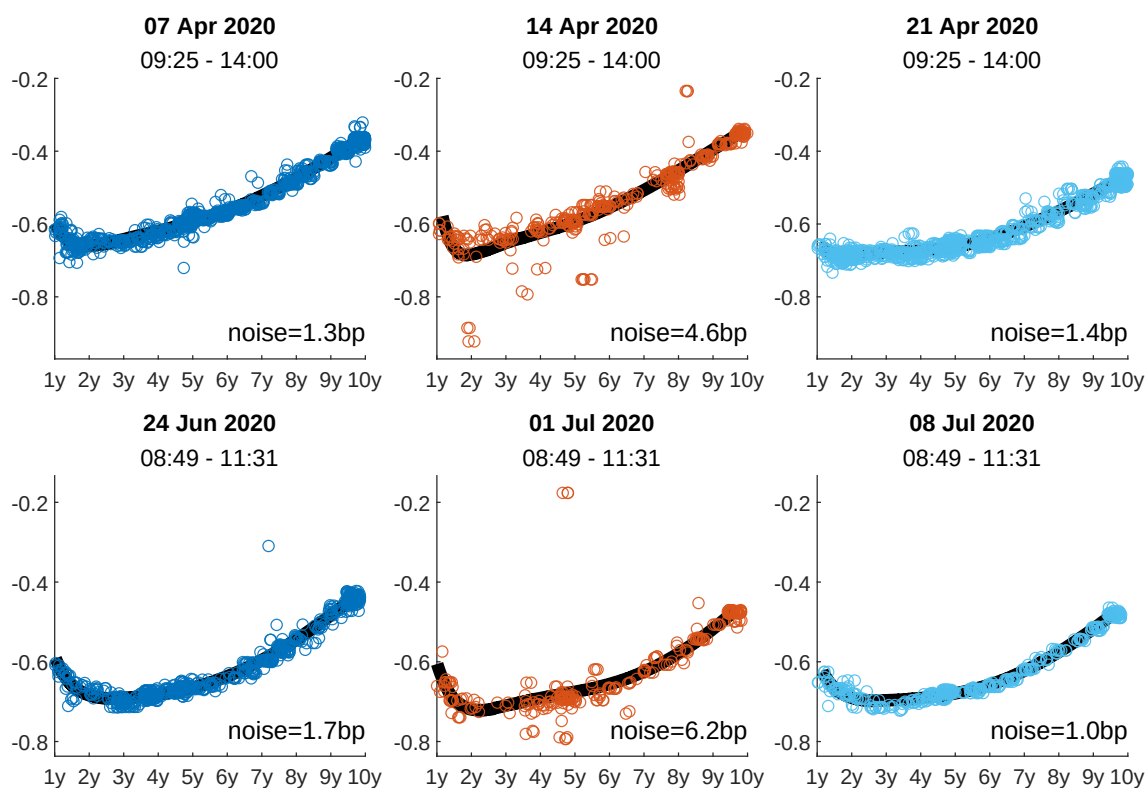


Figure 3: German Yields and Fitted Yield Curves. Each circle shows the implied yield of a transaction in German bonds during the intraday windows stated at the top of each panel. Red circles refer to the two Eurex outage days (middle column), dark and light blue circles refer to the same intraday window in the previous and subsequent week (left and right column, respectively). The vertical axis refers to the yield in percentage points, the horizontal axis to the remaining maturity in years. The black lines are fitted yield curves based on [Svensson \(1994\)](#). The value in the bottom right corner of each panel is the noise measure proposed by [Hu et al. \(2013\)](#), defined as the root mean squared pricing error (difference between actual and fitted yields).

Figure 3 shows the results. The observed market yields usually lie on a smooth yield curve, but not when Eurex is offline. During the outages, many transactions deviate

strongly from model-implied ‘fair’ yields. In fact, the dispersion in bond yields is so large that the exact method to compute ‘fair’ yields is secondary. We fit yield curves based on [Svensson \(1994\)](#), but we would obtain virtually the same increase in ‘pricing errors’ using other methods. During the outages, the noise measure proposed by [Hu et al. \(2013\)](#), defined as the root mean squared pricing error, is three to six times higher than during the same intraday window one week before or after. Some market participants apparently struggle to price risk-free German sovereign bonds without bond futures as a guidepost. [Section 4](#) will show who these investors are.

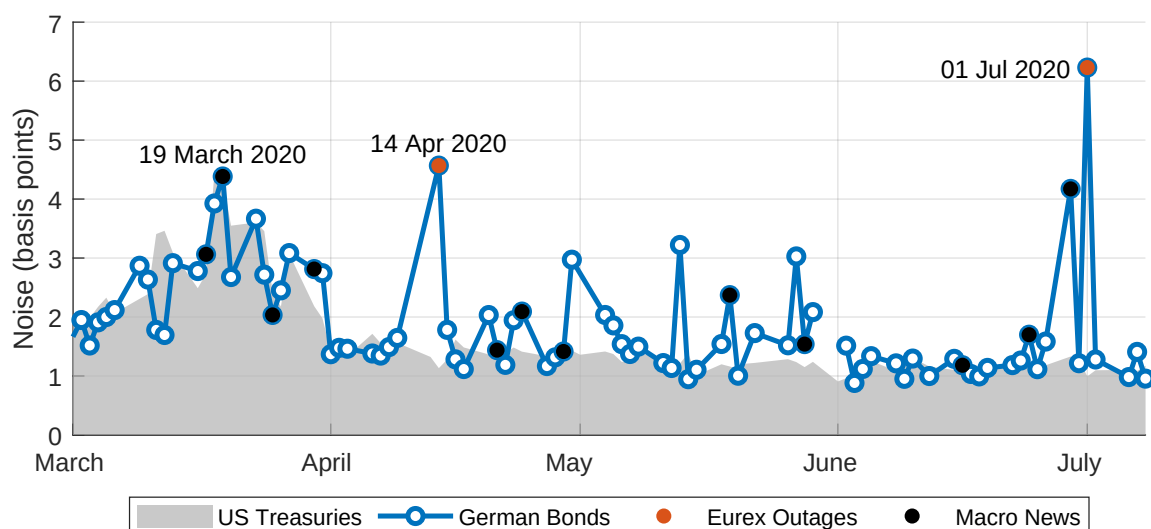


Figure 4: Yield Curve Noise during 2020. The grey area is the US Treasury noise measure published by [Hu et al. \(2013\)](#). It refers to the root mean squared pricing error, based on the cross-section of end-of-day bond prices from the CRSP database and using the [Svensson \(1994\)](#) method to fit yield curves. The blue line applies the same methodology to German government bonds, as in [Figure 3](#). Before (after) 1 June 2020, the measure is based on all transactions between 9:25 a.m. and 2:00 p.m. (8:49 a.m. and 11:39 a.m.) each day, which corresponds to the intraday times of the first (second) Eurex outage. Red dots mark the Eurex outage windows. Black dots mark windows with major macroeconomic data releases (ifo survey, ZEW survey, German CPI, or US mortgage applications). 19 March 2020 indicates the peak of the Covid-19 market turmoil. The ECB announced its €750 billion Pandemic Emergency Purchase Programme shortly before midnight on 18 March.

[Figure 4](#) puts the Eurex outages into perspective, repeating the above exercise for all trading days between 1 March 2020 and 8 July 2020. The mispricing of German government bonds during the two Eurex outages was severe, comparable to or even higher than at the peak of the Covid-19 market turmoil in March 2020. For reference, the figure also shows the yield curve noise measure for US Treasuries published by [Hu et al. \(2013\)](#).

Due to our use of intraday transaction prices, rather than end-of-day prices, our measure of noise is itself more noisy. This is because within our intraday windows, surprising macro news might hit the market, moving fair yields and hence mechanically increasing pricing errors. Nonetheless, the mispricing we compute for German bonds is remarkably close to US Treasuries. In both countries, the yield curve noise hovers around 1.5 basis points during normal times, but shoots up to more than 4 basis points during the Covid-19 market turmoil. On the two Eurex outage days, lastly, we observe a spike in mispricing only for German bonds.¹⁷

	(1)	(2)	(3)	(4)
Outage	1.27*** [0.18]	1.25*** [0.17]	1.07*** [0.16]	0.72*** [0.18]
Constant	0.98*** [0.02]	0.98*** [0.02]	0.99*** [0.03]	1.00*** [0.03]
FE Minute	✓	✓	✓	✓
FE ISIN		✓	✓	✓
FE Venue			✓	✓
FE Buyer/seller ID				✓
Observations	3352	3352	3055	2404
Adjusted R^2	0.072	0.115	0.127	0.374

Table 3: Effect of Eurex Outages on Mispricing. Each column shows results of a different regression. Throughout, the dependent variable is the absolute pricing error in basis points, i.e. the difference between the observed and fitted yield based on [Svensson \(1994\)](#). The sample spans all trades shown in [Figure 3](#), i.e. all trades in one to ten year German bonds during the Eurex outages and during the same intraday window on the same day in the previous and subsequent week. We sequentially add fixed effects: (1) for the time-of-day, using 15-minute buckets, (2) for the bond ISIN, (3) for the venue type the transaction is executed on, (4) for the buyer and seller ID.

[Table 3](#) confirms that cash bond transactions become significantly more mispriced. The average pricing error more than doubles during outages. By sequentially adding fixed effects, we show that the mispricing spike is not due to different bonds being traded (column 2). Instead, a sizable share of the pricing errors is caused by a shift in trading activity toward investors and venues with systematically higher mispricing. The outage coefficient drops by 14% when we add venue-level fixed effects (column 3), and by

¹⁷Appendix [B.5](#) zooms in on the two outage days and confirms that the mispricing in German bonds spikes the moment Eurex goes down and quickly recovers once trading resumes.

another 33% with investor-level fixed effects (column 4). Nevertheless, the spike in mispricing remains economically large and highly significant even after controlling for all these compositional effects.

To summarize our macro-level findings, the euro area government bond market becomes dysfunctional when the futures market goes down. Next, we use granular transaction-level data to identify the mechanisms behind this breakdown.

4 Micro-Level Mechanisms of Futures Market Outages

This section provides micro-level evidence on how outages in the futures market affect the cash market for Bunds. The data point to two main mechanisms. First, by removing a hedging instrument, outages reduce the intermediation capacity of dealers. Second, by removing a benchmark price, outages exacerbate information asymmetries for clients. We present two stylized facts for these mechanisms before analyzing each one in more detail in [Section 4.1](#) and [Section 4.2](#).

First, [Table 4](#) shows that outages reduce volumes most for dealer-to-client trades (-92%) and least for client-to-client trades (-67%), consistent with dealers cutting back intermediation when they cannot hedge. Mispricing, in contrast, increases by 3 basis points for client-to-client trades, 0.72 basis points for dealer-to-client trades and not at all for dealer-to-dealer trades, consistent with clients relying on futures prices as a benchmark.

Second, [Table 5](#) documents that dealers' trading profits increase during outages, both in relative and absolute terms, at the expense of clients. This is exactly in line with [Duffie et al. \(2017\)](#): *introducing* a benchmark reduces the informational advantage of dealers relative to clients, and hence lowers dealers' profit margins. The outages are natural experiments of this effect in reverse: *removing* the benchmark exacerbates information asymmetries and hence raises dealers' profit margins.

	(1) Volume	(2) Mispricing
Outage $\times$ C2C	-1.10*** [0.23]	3.30*** [0.61]
Outage $\times$ D2C	-2.54*** [0.23]	0.72*** [0.19]
Outage $\times$ D2D	-1.45*** [0.23]	0.36 [0.22]
FE Time	✓	✓
FE ISIN		✓
Observations	252	3231
Adjusted R^2	0.656	0.163

Table 4: Effect of Eurex Outages on Trading Volumes and Mispricing across Investor Segments. Column (1) refers to a balanced panel regression using the log of the trading volume in Bunds in Euro plus one, in 15-minute intervals, as the dependent variable. Column (2) refers to a transaction-level regression with absolute pricing errors as the dependent variable. Throughout, the sample covers all trades in 1-10y Bunds during the two outages and the same intraday windows on the same day in the previous and subsequent week. We differentiate trades across three investor segments: client-to-client (C2C), dealer-to-client (D2C) and dealer-to-dealer (D2D). For brevity, the table shows results only for the outage dummy interaction terms.

	Profit (€ million)		Profit (% of volume)		% profitable trades	
	No Outage	Outage	No Outage	Outage	No Outage	Outage
Client	-27.79	-48.85	-0.18	-3.73	48.97	39.38
Dealer	28.40	48.85	0.19	3.73	51.19	60.62

Table 5: Distribution of Daily Trading Profits across Dealers and Clients. To obtain a profit measure, we follow [Allen and Wittwer \(2023a\)](#) and compute markups as the deviation of a trade’s yield from the fair yield based on a fitted curve ([Figure 3](#)), appropriately signed for buys and sells such that markups are positive when favorable for the investor. We then multiply markups by trade volume and maturity to measure the euro value of gains for one counterparty and losses for the other, reflecting the zero-sum nature of trading (equivalently, we could multiply volumes with the deviation between actual and fair *prices*). The sample covers all dealer-to-client trades during the outages and non-outage control windows. To obtain a daily measure, we scale the profits from 7.28 hours (the combined duration of both outages) to 10 hours (the typical trading hours for Bunds), and divide the sum during control windows by two (since we use two control windows for each outage). Profits do not necessarily sum to zero due to a few transactions where only one counterparty can be identified.

4.1 Bond Futures as Hedging Instruments

This section provides direct evidence that the reduced trading volumes of dealers is indeed driven by their reduced intermediation capacity. To do so, we first establish that dealers usually use bond futures to hedge their temporary inventory risk that arises from intermediating trades on the cash market, as already shown by [Naik and Yadav \(2003\)](#) for UK Gilts and [Fleming, Nguyen, and Rosenberg \(2024\)](#) for US Treasuries. We confirm

this for German Bunds by running the following regression:

$$\Delta Duration_{it}^f = \alpha - \beta \times \Delta Duration_{it}^c \times Dealer_i + \epsilon_{it} \quad (2)$$

where $\Delta Duration^f$ is the duration-weighted net trading volume (buy minus sell) of investor i on day t in German government bond futures, based on EMIR data ([Section 2.2](#)). For each of the four futures, we compute this measure as the net number of contracts traded multiplied by the contract notional of €100,000 and the modified duration of the cheapest-to-deliver bond on day t , divided by that bond’s conversion factor. $\Delta Duration^c$ analogously is the duration-weighted net trading volume in German bonds on the cash market. $Dealer_i$ is a dummy variable equal to one if investor i is a dealer. We are interested in the β coefficient, the contemporaneous hedge ratio.

[Table 6](#) shows a strong and highly significant negative relationship between the cash and bond future trading volumes for dealers. On average, dealers offset 16.3% of the duration risk arising from their cash market trades by taking opposite positions on the futures market on the same day (column 1). This hedge ratio increases to 30.3% when we exclude days where we expect the trading behavior to be dominated by exogenous factors, namely primary auctions, quarter ends, and futures expiries (column 2). Restricting the sample to days and investors with non-zero inventory changes in both markets makes little difference (column 3). Equally importantly, we find a precisely estimated null effect for clients across all three specifications, confirming that clients do not use bond futures for hedging purposes. For many clients, this is obvious because they are not active on the futures market at all (see [Table A9](#) in [Appendix A.5](#)). But our results confirm that it is also true for the subset of clients that is active on both the cash and futures market.

[Table 7](#) shows that the average hedging coefficient of roughly 31% for the dealer sector masks substantial variation *across* dealers. To see this, we re-estimate [Equation 2](#) individually for each dealer. The β coefficients range from 0% to 75%, meaning that some dealers do not hedge their cash market trades at all while others hedge the bulk of their

	(1) all days	(2) excl. auctions, EoQ, fut.exp.	(3) excl. zeros
Client	0.43 [0.95]	0.34 [0.93]	0.39 [1.00]
Dealer	16.30*** [3.92]	30.33*** [5.39]	31.03*** [5.51]
FE Day	✓	✓	✓
FE LEI	✓	✓	✓
Observations	45812	36870	16418
Adjusted R^2	-0.006	-0.001	-0.024

Table 6: Duration-weighted Net Trading Volumes in German Bunds on the Cash and Futures Market. This table shows regression results of Equation 2, predicting daily changes in the duration exposure on the futures market with those on the cash market, at the investor-level, distinguishing clients from dealers. In column (2), we drop roughly 50 trading days related to primary auctions, quarter ends, and days around the expiration of bond futures. In column (3), we further drop all observations with zero inventory change at the day-LEI level on either the futures or cash market. Coefficients are multiplied by 100 such that they refer to percentage effects.

	min	p50	mean	max
no hedging (<5%)	0.0	0.0	1.5	4.9
moderate hedging	7.9	22.8	24.6	49.8
intense hedging (>50%)	59.4	68.2	67.5	75.1
All Dealers	0.0	22.8	27.7	75.1

Table 7: Dealer-Heterogeneity in Bond Futures Use for Hedging. This table refers to the cross-section of β coefficients when estimating Equation 2 at the dealer-level. The estimation sample is identical to Table 6 column (3). β measures the percent of daily cash duration exposure changes offset via bond futures. For the two dealers that are not active on the futures market at all (see Appendix A.5), we set $\beta = 0$. The final classification includes five dealers in the ‘no hedging’, five in the ‘intense hedging’, and 22 dealers in the ‘moderate hedging’ group.

exposure. This heterogeneity lets outages bite different dealers with different strength. We exploit this heterogeneity by grouping dealers into a ‘no hedging’ group if the dealer hedges less than 5% of the daily cash market exposure via futures, an ‘intense hedging’ group if they hedge more than 50%, and a ‘moderate hedging’ group with the remaining dealers.

	(1) volume	(2) mispricing	(3) profit
Outage $\times$ Dealer not hedging	0.54 [2.17]	0.35 [0.30]	-0.46 [2.71]
Outage $\times$ Dealer moderately hedging	-2.36*** [0.71]	0.83*** [0.21]	2.97*** [0.75]
Outage $\times$ Dealer intensely hedging	-4.42** [2.19]	0.44* [0.26]	1.74 [1.88]
FE Minute	✓	✓	✓
FE ISIN		✓	✓
Observations	162	1920	1920
Adjusted R^2	0.360	0.064	0.029

Table 8: Effect of Eurex Outages on D2C Trades across Dealer Groups. The sample includes all D2C transactions during the two outage and four non-outage control windows. In column (1), we construct a balanced panel of trading volumes (in logs plus 1) at the 30-minute frequency, to avoid compositional effects. Columns (2) and (3) are at the transaction-level. The dependent variables are calculated from the dealer perspective. For brevity, the table shows results only for the outage dummy interaction terms.

Table 8 shows how the different hedging strategies of dealers relate to their D2C trading behavior during Eurex outages. Besides trading volumes and absolute mispricing, we also look at the profitability of trades (in Euros, used in signed-log form in the regressions; see Table 5 for definitions). The ‘no hedging’ group shows no obvious reaction. These dealers keep trading with their clients as usual, both in terms of volumes and prices. That means they neither increase their activity to compensate for the withdrawal of other dealers nor do they exploit the situation by increasing markups. D2C volumes drop significantly for the ‘moderate hedging’ group and even more for the ‘intense hedging’ group, reflecting these dealers’ progressively greater reliance on the futures market. Regarding prices, we see that both groups cause mispricing but only dealers in the ‘moderate hedging’ group trade more profitably.¹⁸

¹⁸Naik and Yadav (2003) document similarly large differences in hedging ratios across UK Gilt dealers. Appendix B.6 shows that (i) dealers’ hedging intensity is barely related to other characteristics like connectedness, centrality and size and (ii) our regression results hold when controlling for the additional

One interpretation of these results is that for dealers in the ‘intense hedging’ group, intermediation capacity constraints become binding when Eurex goes down, forcing them to forego even highly profitable trades. Dealers in the ‘moderate hedging’ group, in contrast, retain some intermediation capacity, allowing them to selectively deploy capacity when it is profitable.

Overall, this section provides quasi-experimental evidence that dealers are important for market functionality (see e.g. [Duffie et al., 2023](#)). Outages on the futures market reduce dealers’ capacity to intermediate bond trades on the cash market, as they are unable to hedge additional inventory risk. Consequently, clients either accept unfavorable quotes by dealers, or they have to trade directly with other clients, causing large mispricing.

4.2 Bond Futures As Benchmark Prices

Outages cause severe mispricing in client-to-client trades and reallocate profits toward dealers ([Table 4](#); [Table 5](#)). This suggests that dealers have an informational advantage over clients and rely less on Bund futures as pricing benchmarks (in line with [Hortaçsu and Kastl, 2012](#); [Paiardini, 2015](#)). This section tests two further predictions of the benchmark channel.

First, less sophisticated clients should fare worse when futures go dark. We test this by separating clients into different types: non-dealer banks, NBFIs, investment funds, hedge funds, ICPFs, NFCs, the official sector (including central banks) and households (retail investors).

[Table 9](#) column (1) shows that absolute pricing errors rise broadly across client types, with the largest increase for households and hedge funds. The increase in mispricing is significant for all but three client types (ICPFs, NFCs and the official sector), probably because they execute few trades during the outages (see [Appendix A.5](#)). Looking at markups in column (2), the largest effect is on households, whose markups decrease by

effect of dealer inventory: dealers with larger pre-outage inventories raise markups and extract profits less during the outage.

4.3 basis points while Eurex is down, i.e. they buy Bunds at excessively low yields (high prices) and sell at excessively high yields (low prices). Hedge funds, in contrast, increase their markups significantly. Column (3), lastly, confirms that dealer banks' trades become significantly more profitable when the futures market is down, despite a non-significant increase in markups. This is because their markups are concentrated in long-term bonds and in larger trades. The opposite is true for households, which mitigates the negative effect of markups on their profits. Similarly, the significant spike in markups for hedge funds and the drop in markups for investment funds turns insignificant once we take into account the traded volume and maturity.

	(1) mispricing	(2) markup	(3) profit
Outage $\times$ Bank Dealer	0.62*** [0.13]	0.25 [0.22]	2.27*** [0.65]
Outage $\times$ Bank Non-Dealer	1.86*** [0.43]	0.66 [0.51]	-1.22 [0.85]
Outage $\times$ NBF	1.25*** [0.44]	0.48 [0.49]	0.95 [1.17]
Outage $\times$ Investment Fund	0.44*** [0.12]	-0.67*** [0.21]	-2.94** [1.17]
Outage $\times$ Hedge Fund	2.63*** [0.68]	2.75*** [0.80]	1.57 [1.51]
Outage $\times$ ICPF	0.27 [0.17]	-0.51* [0.27]	-6.18*** [2.34]
Outage $\times$ NFC	1.78 [1.25]	-1.28 [1.59]	-1.14 [3.49]
Outage $\times$ Official	0.20 [0.21]	-0.31 [0.36]	-3.10 [1.91]
Outage $\times$ Household	4.30*** [0.95]	-4.04*** [1.11]	-2.45*** [0.91]
FE Minute	✓	✓	✓
FE ISIN	✓	✓	✓
Observations	6117	4795	4795
Adjusted R^2	0.165	0.054	-0.009

Table 9: Effect of Eurex Outages on Absolute Pricing Errors, Markups, and Profits across Investor Types. The dependent variable is the transaction-level pricing error, in basis points of yield, in absolute value (column 1), in net terms (i.e. as markups, column 2), and volume- and duration-weighted (i.e. as profits, column 3). In columns (2)-(3), we drop trades between investors of the same type and the variables are calculated from the perspective of the investor type listed in the first column. Positive coefficients hence imply a favorable increase in markups or profits for that investor type. For brevity, the table shows results only for the outage dummy interaction terms.

To sum up, the client cross-section broadly fits the information asymmetry channel. Households, typically thought of as the least sophisticated client type, fare worst during outages, particularly compared to sophisticated clients like hedge funds.

A second prediction of the benchmark channel is that when the public benchmark dis-

appears, private signals and relationships should become more important. This prediction moves us from sectors to networks. We study three established network proxies, namely the number of clients’ connections (Kondor and Pintér, 2022), the intensity of client-dealer relationships (Di Maggio et al., 2017), and dealer centrality (Li and Schürhoff, 2019).¹⁹

In Table 10, we use all three measures to explain the dysfunction on the dealer-to-client segment. Note that this analysis does not cover the most mispriced trades. That is because we cannot compute network measures for trades by households (since individuals remain anonymous in our dataset), or for trades on exchanges (as these occur anonymously in a CLOB). Nonetheless, we find that better connected clients fare better during outages. They cause less mispricing, they pay less markup, and they incur less trading losses. This is exactly in line with client’s connections measuring informational advantages in decentralized markets. Similarly important is the dealer-client relationship. The stronger the relationship, the less trades become mispriced, and the less dealers extract profits from their clients. The centrality of the dealer, in contrast, has no significant impact on trading outcomes.

To conclude, the dysfunction observed on the D2C segment during Eurex outages is not entirely explicable by the reduced intermediation capacity of dealers. Instead, outages also exacerbate information asymmetries and increase the importance of client-dealer relationships. Who you trade with matters more when the common price benchmark is gone.

4.3 Further Evidence

Swap Market Dysfunction If bond futures merely served as hedging instruments, price discovery for the risk-free yield curve could instead take place in the swap market. Instead, Figure 5 shows that interest rate swaps become severely mispriced during Eurex

¹⁹Following Kondor and Pintér (2022) we measure the number of different dealers a client has traded with. Following Di Maggio et al. (2017), we measure the intensity of each client-dealer pair as the fraction of sales by client j to dealer k , normalized by j ’s total selling volume, and we compute the analogous measure for buys. Following Li and Schürhoff (2019), lastly, we capture each dealer’s centrality as the volume-weighted eigenvector centrality. All measures are based on the universe of Bund trades in 2020.

	(1) mispricing	(2) markup	(3) profit
Outage	0.40 [0.52]	-1.99*** [0.73]	-11.54*** [3.92]
Outage $\times$ # Connections	-0.02* [0.01]	0.04*** [0.02]	0.34*** [0.09]
Outage $\times$ Relationship w/ Dealer	-1.79*** [0.66]	0.48 [0.77]	5.95* [3.60]
Outage $\times$ $\ln(\text{Dealer Centrality})$	-0.20 [0.13]	-0.21 [0.18]	-0.21 [0.76]
FE Minute	✓	✓	✓
FE ISIN	✓	✓	✓
Observations	1700	1700	1700
Adjusted R^2	0.119	0.036	0.030

Table 10: Role of Dealer-Client Network Characteristics during Outages. The dependent variables are calculated from the client perspective. # Connections refers to the number of different dealers a client has traded with. Dealer Relationship measures the fraction of buying/selling volume of each client-dealer pair as a share of the total volume of each client. Dealer Centrality is the volume-weighted eigenvector centrality of the dealer involved in the transaction. The last two regressors are dummy variables equal to one if the dealer hedges between 5-50% or more than 50% of its cash market duration risk via bond futures (see Table 7). The sample includes all D2C transactions during the two outage and four non-outage control windows. We drop trades by households, since the various network measures cannot be computed, and trades on exchanges, as these occur anonymously in a CLOB. This leads to a slightly smaller sample compared to Table 8.

outages, just like Bunds on the cash market.²⁰

Why do Eurex outages spill over to the swap market, even though swaps are not directly linked to any Eurex-traded derivative? An unsatisfactory explanation is that just like on the cash bond market, dealers on the swap market—which are mostly the same large banks—hedge their trades with bond futures. Hence, when the futures market suffers an outage, these dealers stop intermediating on the swap market, causing the illiquidity and mispricing we observe.

In our view, a more satisfactory answer is that bond future prices are the most informative asset price on the fixed-income market. Bund futures provide the benchmark that anchors pricing not only in the bond but also the swap market. When that benchmark disappears, price discovery in both markets collapses, in line with cross-asset learning

²⁰Appendix B.7 provides further evidence of a swap market breakdown. Indicative quotes become stale the moment bond futures become unavailable, particularly for longer-dated swaps. On Bloomberg, bid-ask spreads spike during outages, if quotes are available at all. Lastly, on one particular interdealer broker platform (for which we were able to obtain the full order book of executable quotes), Eurex outages cause a complete evaporation of liquidity, which is reversed within half an hour after Eurex goes back online.

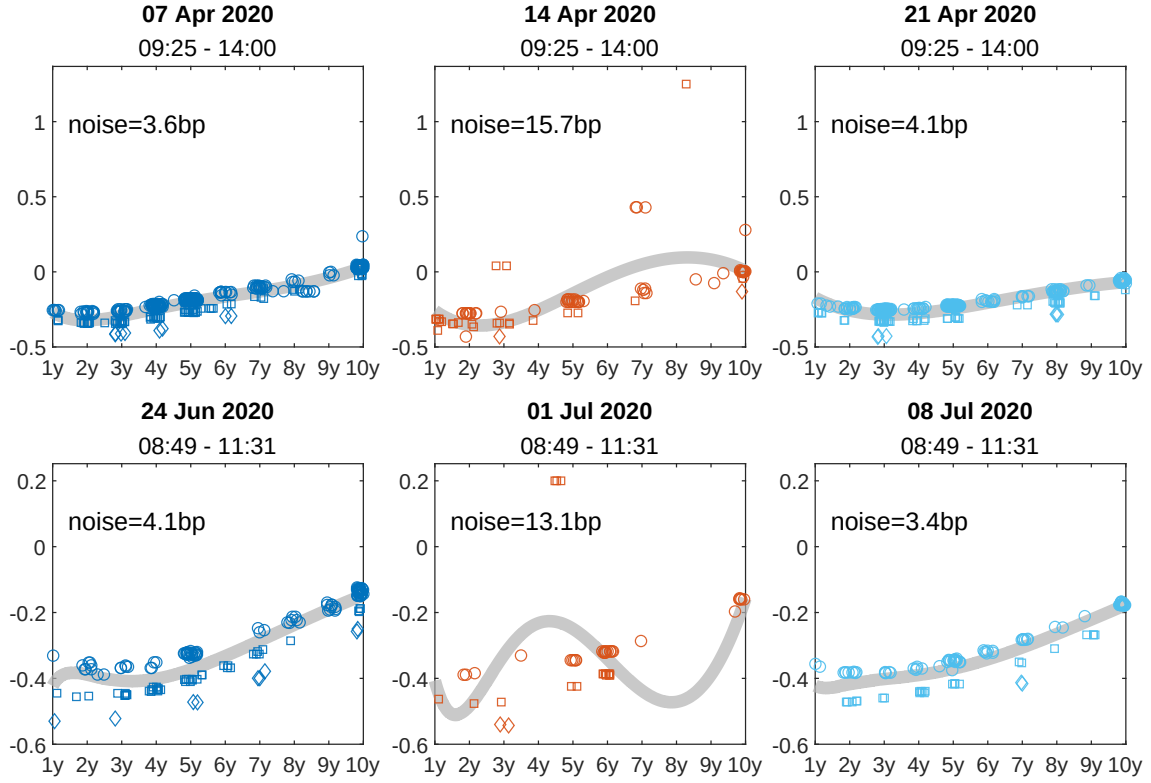


Figure 5: Euribor Swap Rates and Fitted Yield Curves. This figure replicates Figure 3 for the swap market. Each marker refers to an interest rate swap transaction during an outage (middle column) or the same intraday window in the previous and subsequent week (left and right column, respectively). The vertical axis refers to the fixed-leg interest rate (in percentage points), the horizontal axis to the contract horizon (in years). The floating rate is the Euro Interbank Offered Rate (Euribor) with a maturity of one, three, or six months, marked as diamonds, squares and circles, respectively. Grey lines are fitted yield curves based on Svensson (1994). The value in the top left corner of each panel is the noise measure proposed by Hu et al. (2013), defined as the root mean squared deviation of actual rates from the fitted curve. Based on EMIR data, see Section 2.2.

models (Cespa and Foucault, 2014).²¹

Venue-Level Heterogeneity Table 11 shows that outages affect trading venues very differently. On electronic platforms, volumes collapse and mispricing rises, consistent with dealers pulling back intermediation when they cannot hedge. Trading on systematic internaliser platforms also declines but prices remain fair, consistent with dealers favoring loyal clients. On regular exchanges, volumes barely move, yet mispricing spikes, consis-

²¹An ideal test of the impaired price formation hypothesis would check if important fundamental news is properly incorporated into bond prices while Eurex is offline. Unfortunately, there were no such news. According to the extensive intraday news database of Kerssenfischer and Schmeling (2024), the most significant news during either outage was the Swedish central bank's announcement to extend its QE program on 1 July 2020 at 9:00 a.m., see Riksbank press release. However, this decision was largely anticipated and caused only minor market reactions, even in Sweden, see Riksbank staff memo.

tent with unsophisticated retail investors continuing to trade as usual, likely unaware of the Eurex outage. These venue-level discrepancies reinforce both mechanisms via which outages operate: they impair dealer capacity and exacerbate information asymmetries.²²

	(1) Volume	(2) Mispricing
Outage $\times$ OTC bilateral	-1.51*** [0.31]	1.97*** [0.50]
Outage $\times$ OTC via IDB	-1.74*** [0.32]	0.14 [0.17]
Outage $\times$ OTC via SI	-1.64*** [0.28]	0.17 [0.26]
Outage $\times$ electronic platforms	-2.30*** [0.23]	0.91*** [0.19]
Outage $\times$ MTS	-0.39*** [0.15]	
Outage $\times$ regular exchange	-0.23** [0.11]	4.46*** [1.44]
FE Time	✓	✓
FE ISIN		✓
Observations	504	3055
Adjusted R^2	0.639	0.166

Table 11: Effect of Eurex Outages on Trading Volumes and Mispricing across Venues. See Table 4 for details. In column (2), the differential effect for MTS cannot be estimated, since there was not a single trade in 1-10y German bonds on MTS during the outages.

5 Evidence from Cash Venue Outages

The previous sections show that when the centralized futures market suffers an outage, the decentralized cash market for EGBs is severely impaired. What about the other way around?

An ideal experiment to answer this question would require a random incident that halts trading on the entire cash market, i.e. on all exchanges, electronic trading platforms, as well as the over-the-counter segment. Fortunately for the public at large, but unfortunately for us researchers, such a comprehensive and uniform shock has never occurred. Hence, the best we can do is to study outages on individual cash trading platforms.

²²Appendix B.8 provides additional evidence based on the effect of outages on trade sizes. Dealer-to-client trades become much smaller while dealer-to-dealer sizes hold up, in line with dealers cutting back intermediation. Mispricing increases disproportionately in these small trades. Appendix B.9, lastly, shows that our findings are not driven by liquidity demand, market panic, or changes in trading relationships.

We found five such outages affecting four different platforms:²³ MTS, Bloomberg, the Frankfurt stock exchange (Frankfurter Wertpapier Börse), and Brokertec.

To study the effect of these cash venue outages on the aggregate cash and futures market, we run dummy regressions like below:

$$Y_{it} = \alpha + \beta \times D_t + \epsilon_{it} \quad (3)$$

where D_t is a dummy variable equal to one during the outage. We include the outage period and the same intraday window during the preceding and subsequent week (as controls).

For the cash market, Y_{it} measures the transaction volume in € or the number of transactions (both in logs plus one), separately for German, French, Italian and Spanish government bonds, and t refers to a 30-minute frequency. A major caveat is that three of the five outages occurred before 2019, i.e. before the start of the MiFID II dataset. Hence, the only available transaction-level dataset for the cash market is the less exhaustive and less granular MiFID I dataset, see Appendix A.1.

For the futures market, Y_{it} also covers the trading volume (in number of contracts or the number of trades, both in logs plus one), separately for each of the eight bond futures traded on Eurex, and t refers to a 15-minute frequency. Since we have the entire history of transactions on Eurex, we also look at more direct measures of market functioning. In particular, Y_{it} also covers volatility (the sum of absolute log returns), an intraday Amihud (2002) measure (ratio of volatility over volume), and the Roll (1984) bid-ask spread estimate. For compatibility, these last three measures are normalized to mean zero and unit variance for each future.

Table 12 contains results. The top panel shows that outages on individual cash venues reduce the aggregate trading activity on the cash market. This is at odds with the common

²³Appendix D confirms all outages through newspaper reports, except the most recent one on MTS. We could not confirm this outage independently, but the episode we identified has the same characteristics as an earlier confirmed outage: no intraday transactions or quotes for any bond for a prolonged period during trading hours and in the absence of any major news.

claim that bond trading seamlessly moves to alternative platforms in case of outages, see the narrative evidence in Appendix D.4.1. So given that individual cash venue outages affect the aggregate cash market, do they also affect the futures market?

	MTS 12 Jan 2010 8:00-10:35	Bloomberg 17 Apr 2015 9:20-10:10	FWB 27 May 2015 8:00-11:00	Brokertec 11 Jan 2019 19:43-21:35	MTS 26 Jul 2019 12:30-13:20	Pooled
<i>Cash market</i>						
Volume	-0.20 [1.05]	-1.97*** [0.59]	-1.71 [1.03]		-1.42** [0.49]	-1.19* [0.71]
#Trades	-0.22 [0.15]	-0.59** [0.23]	-0.32 [0.19]		-0.10 [0.18]	-0.31* [0.17]
<i>Futures market</i>						
Volume	0.15 [0.17]	-0.32 [0.21]	-0.65*** [0.12]	-0.57*** [0.18]	-0.53** [0.21]	-0.42*** [0.13]
#Trades	0.18 [0.12]	-0.32** [0.15]	-0.36*** [0.07]	-0.45*** [0.10]	-0.57*** [0.12]	-0.29*** [0.10]
Volatility	0.24** [0.10]	-0.08 [0.07]	-0.55*** [0.12]	-0.04*** [0.01]	-0.23 [0.15]	-0.22*** [0.07]
Amihud	0.14 [0.24]	0.03 [0.09]	0.16* [0.08]	-0.46 [0.32]	0.08 [0.08]	0.04 [0.08]
Roll	0.01 [0.19]	-0.02 [0.23]	0.19* [0.11]	0.19 [0.29]	0.55 [0.35]	0.16* [0.09]

Table 12: Effect of Cash Venue Outages. Each column refers to a different outage, each row to a different dependent variable. Each sample includes the outage period and the same intraday window on the preceding and subsequent week (as controls). The last column refers to a pooled regression of all outages. For the cash market, variables are at the country-level and 30-minute frequency and Volume refers to the transaction volume in €. For the futures market, variables are at the future-level and 15-minute frequency and Volume refers to the number of traded contracts. Volume and #Trades are in logs plus one, Volatility is the sum of absolute log returns, Amihud (2002) is the ratio Volatility/Volume, and Roll (1984) is a bid-ask spread estimate. The last three measures are normalized to mean zero and unit variance. Since these three measures are undefined when there are zero trades in a given period, estimates may suffer from selection bias. All regressions include fixed effects for the minute within a day and for each country or bond future. The pooled regressions further includes fixed effects for the month in which the outage occurred.

The lower panel in Table 12 suggests they do, but mainly in terms of trading activity,

not market functioning. The trading volume on Eurex drops significantly during cash market outages, but not nearly as much as in the other direction: the pooled estimate is roughly one-quarter as large as the equivalent estimate in Table 1. More direct measures of market functioning show even less of an impact. Focusing on the pooled estimates, bond future prices become *less* volatile and the illiquidity measure of Amihud (2002) does not react significantly. Only the bid-ask spread estimate of Roll (1984) increases significantly, but the effect is economically small at 0.16 standard deviations. These effects pale in comparison to the complete evaporation of liquidity on the cash market during Eurex outages (see Section 3.2).

A natural concern is that this asymmetry might merely reflect the smaller scale of any individual cash venue outage. To address this concern, we proxy for the scale by interacting the outage dummy with the affected venue’s market share in the respective country:

$$Y_{it} = \alpha + \beta \left(D_t \times \text{MktShare}_{vc} \right) + \varepsilon_{it}, \quad (4)$$

where MktShare_{vc} denotes the market share of cash venue v in country c (expressed as a fraction, e.g. 0.1 for 10%), and Y_{it} covers the same future-level outcomes as before.²⁴

	Futures Market				
	Volume	#Trades	Volatility	Amihud	Roll
Cash Venue Outage $\times$ MktShare	-1.95***	-1.63***	-0.24	0.48*	-0.12
	[0.74]	[0.56]	[0.39]	[0.27]	[0.40]
Adjusted R^2	0.763	0.704	0.244	0.122	0.111

Table 13: Effect of cash venue outages on the futures market, scaled by venue importance. The table reports β estimates from regression (4), where the outage dummy is interacted with the affected cash venue’s market share in the respective country. The sample construction and dependent-variable definitions follow Table 12. All regressions include fixed effects for the month in which the outage occurred, for the minute within a day, and for the bond future.

Table 13 confirms that scaling by venue importance matters for quantities. Both the

²⁴Given the lack of a consolidated tape, venue-level market shares are unavoidably imprecise, especially at the country level and for historical episodes. Table A21 in the Appendix summarizes how we arrived at our estimates.

trading volume as well as the number of trades drop more when the affected cash venue has a larger market share. Importantly, however, the market functioning measures do not exhibit comparable scaling with venue importance. Even when the affected venue commands a large share, only the Amihud measure points to a slight deterioration in liquidity. Taken at face value, the estimate implies a 0.24 standard deviation increase in bond future illiquidity when a cash venue with a 50% market share in the underlying country’s cash bond market suffers an outage.

Taken together, our results point to a sharp asymmetry in how outages transmit across markets. Cash venue outages propagate to the futures market mainly via the hedging channel: with fewer cash trades executed, hedging demand and hence trading activity in bond futures falls, while liquidity remains largely unchanged. Futures market outages, by contrast, disrupt the cash market through two channels: they remove access to a key hedging instrument and they remove the centralized benchmark that anchors price formation. This difference helps explain why a fragmented cash market can tolerate outages of individual venues with only limited effects on futures market functioning, whereas an outage of the centralized futures market has first-order effects on cash market functioning. This interpretation is consistent with evidence from equity markets, where price discovery is concentrated on the primary (lit) venue rather than on secondary trading (dark) venues (see e.g. [Hasbrouck, 1995](#); [Hagströmer and Menkveld, 2023](#); [Degryse, Hagströmer, and Landsberg, 2025](#)).

Two limitations are worth emphasizing when interpreting [Table 13](#). First, the interaction estimates are identified from variation in venue-country market shares that are generally small, given the cash market’s high fragmentation. The linear interaction coefficient should therefore be interpreted locally: it speaks to how incremental increases in venue importance map into futures trading activity given the existing fragmented market structure, not to what would happen if a fully centralized cash market experienced an outage. Second, our scaling proxy abstracts from cross-country hedging and benchmarking externalities. For example, market participants may use Bund futures to hedge the

risk-free component of French bonds, or use a combination of German and Italian futures to approximate the risk profile of Spanish bonds, rather than trade the relatively illiquid local bond futures directly. This cross-country substitution weakens the mapping from a disruption in a given country’s cash trading venue to market functioning in that country’s futures contract. Importantly, even with this limitation, we find that cash venue outages scale with futures activity but not with futures market functioning, consistent with the view that the price discovery and liquidity provision originates on the centralized futures market.

[Appendix C](#) zooms in on three country-outage pairs where the affected cash market venue had a large market share: Bloomberg in German Bunds, MTS in Italian bonds, and Brokertec in US Treasuries. In all cases, cash venue outages barely affect market functioning on the futures market. The benign effect of the Brokertec outage is particularly noteworthy, suggesting that one of our main findings — that price formation and liquidity provision is more of a one-way street from the futures to the cash market than previously thought — also applies to the US.

6 Conclusion

Sovereign bond markets build the core of the financial system. We show that this market becomes dysfunctional without bond futures in the euro area. When the Eurex futures exchange suffers an outage, trading activity on the cash market declines, liquidity dries up, and the remaining transactions occur at prices far from fundamental values.

Our non-anonymous transaction-level dataset reveals two mechanisms behind this breakdown: outages remove a hedging instrument, limiting dealers’ capacity to intermediate, and they remove a benchmark price, amplifying information asymmetries among clients. Dealers that rely heavily on futures retreat most from the cash market, while dealers with a moderate use deploy their remaining intermediation capacity selectively. In fact, dealers’ profit margins rise during the outage, as the lack of a public bench-

mark price widens the informational advantage of dealers over clients (in line with [Duffie et al., 2017](#)). With private signals and relationships becoming more important, clients fare very differently: sophisticated clients and those with strong dealer relationships are least affected by the outage, whereas households and less connected clients suffer most.

Lastly, we show that outages on individual cash trading platforms barely affect bond futures, suggesting that price formation and liquidity provision is more of a one-way street from the futures to the cash market than previously thought. This asymmetry suggests that for sovereign bonds, cross-asset learning models (e.g. [Cespa and Foucault, 2014](#)) offer a more realistic framework than symmetric arbitrage models (e.g. [Gromb and Vayanos, 2010](#)).

Our results have important implications for a number of ongoing policy debates, e.g. regarding the pros and cons of centralization (see e.g. [Duffie, 2025](#)). Notably, the U.S. Securities and Exchange Commission’s mandate to centrally clear U.S. Treasury cash transactions, currently scheduled to take effect in 2027 ([SEC press release](#)), promotes all-to-all trading. This would make the cash market, which is currently fragmented across multiple trading platforms, many of which lack central clearing, more similar to the futures market, which is fully centralized and clearly dominant in terms of price discovery. The downside is that an outage of the futures market becomes a systemic risk. In contrast, the decentralized cash market is less liquid, provides little price discovery, heavily relies on dealers as financial intermediaries, but it is robust to an outage of any individual trading platform.

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INTERNET APPENDIX

for

OUTAGES IN SOVEREIGN BOND MARKETS

by MARK KERSSENFISCHER and CASPAR HELMUS

(not for publication)

[Appendix A](#) details the data sources, accessibility, coverage, and data cleaning procedures.

[Appendix B](#) provides further evidence on futures market outages.

[Appendix C](#) provides further evidence on cash venue outages.

[Appendix D](#) lists the outages we study and provides narrative evidence.

Appendix A Data Details

This section contains background information on our data. Appendix A.1 provides an overview of our datasets and Appendix A.2 explains how to get access to the data. Appendix A.3 outlines the market structure on the European cash and futures market for bonds. Appendix A.4 provides details on our transaction-level data, and Appendix A.5 explains how we classify investors into different types and segments.

A.1 Overview of Data Sources

Dataset / Source	Content	Anony- mous	Sample period
<i>Cash Market</i>			
MiFID I	transactions	no	2008-2017
MiFID II	transactions	no	since 2018
MTS	transactions and order book updates	yes	since 2008
MTS BondVision, Tradeweb, BGC, TPICAP, GFI, Aurel	transactions	yes	six days in 2020 (two Eurex outage days and four control days)
Bloomberg, Refinitiv, TPICAP	indicative quotes	yes	
<i>Futures Market</i>			
EMIR	transactions	no	since 2008 (prior to 2018, data comes from the MiFID I dataset)
Eurex	transactions	yes	since 2002
Eurex (A7)	order book updates	yes	since April 2019 (data since January 2019 comes from Bloomberg)
CME (A7)	order book updates	yes	since February 2019

Table A1: Overview of Data Sources Used in This Paper.

Table A1 lists all datasets we use. We focus most of our analysis on the two most recent Eurex outages in 2020 for data availability and quality reasons. In particular, the MiFID I dataset captures less than half as much daily trading volume as the MiFID II dataset and covers particularly few trades by clients such as investment funds. MiFID II has better data coverage and much more granular information on trades, e.g. a flag identifying trades by retail investors. See Appendix A.4 for a detailed description. However, MiFID II data quality improved gradually over time and remains relatively poor for 2018 and 2019. Moreover, we obtained transaction-level data from various trading venues only for the six relevant days in 2020. For these reasons, we study the micro-level mechanisms only for the outages in 2020.

A.2 Access and Availability

Access to the non-anonymous regulatory MiFID I, MiFID II and EMIR datasets can be requested via the [Research Data and Service Centre \(RDSC\)](#) at Bundesbank and is subject to strict requirements, see [rules for visiting researchers at the RDSC](#).

The [application process](#) requires researchers to write a short proposal outlining the research question and its policy relevance. The data request is then evaluated by the RDSC and economists in the statistics department providing the data. Once the evaluation has been successfully completed, researchers need to sign a contract with the RDSC, before being granted secure on-site access to the data.

All other data is available directly from data vendors.

A.3 Future and Cash Market Structure

This section provides further details on the market structure for European government bonds briefly outlined in [Section 2](#) of the main text.

[Table A2](#) gives an overview of the available bond futures and their trading volumes in 2022. Because of their benchmark status in the euro area fixed income markets, futures on German government bonds dominate trading with roughly 80% of the total in 2022. The remaining activity is distributed among Italian (10%), French (8%) and Spanish futures (<1%). Importantly, the futures market is fully centralized on Eurex. For each future, three contracts with different expiration horizons can be traded, one for each of the three nearest months in the March, June, September and December cycle. We focus on the nearest-to-maturity contracts at any point in time. These account for over 90% of trading volume, as most investors ‘roll over’ to the next-nearest contract a few days before expiration. That means for the Eurex outage (and control days) in April 2020, we use future contracts that expire in June 2020. For the July outage, we use contracts that expire in September 2020.

[Table A3](#) gives a rough comparison of the relative size of the future and cash markets. For Germany, the trading volume on the futures market is roughly ten times as large as on the cash market. For France and Italy, the two markets are of comparable size. For Spain, where a future was introduced last, namely in October 2015, trading volumes in futures are only 1% of the cash market volume. Within the cash market, the trading volume in German and Italian bonds is similar, whereas the volume is only 75% as large for French and 30% as large for Spanish bonds.

[Figure A1](#) visualizes EGB trading volumes, not only across but also within the futures and cash market. The figure highlights that trading volumes on the futures market are concentrated into a handful of bond futures whereas on the cash market, the trading volume is dispersed across hundreds of individual bonds, many with tiny trading volumes. For Germany, France and Italy, not even the most-traded cash bond comes close to any of the bond futures in terms of trading volume. For Spain, in contrast, the trading volume in the bond future is roughly equal to the median volume across cash bonds.

[Table A4](#) gives a stylized overview of the different cash market venues for EGBs, differentiating between exchanges, electronic platforms, and the over-the-counter segment. We classify venues based on the employed trading protocol, which leads to some discrepancies with the MiFID II regulation. MTS, e.g., is regulated as a multilateral trading facility, whereas we classify it as an exchange, since MTS employs a central limit order book just like stock exchanges.²⁵

²⁵MiFID II Art. 4(1)(20)-(24) differentiates between regulated markets (exchanges in our terminology) and multilateral and organised trading facilities (MTF and OTF; referring to electronic platforms and IDBs in our terminology). Single-dealer platforms are called ‘systematic internalisers’ (SI) and are defined as ‘an investment firm which, on an organised, frequent systematic and substantial basis, deals on own account when executing client orders outside a regulated market, an MTF or an OTF without operating a multilateral system’. We follow the MiFID II regulation and consider these trades a particular form of OTC trading. In practice, systematic internalisers are usually large banks and the ‘SI’ status comes with extra regulatory responsibilities, e.g. an

The market shares of different EGB trading venues are notoriously difficult to measure, due to the poor post-trade transparency and the lack of a consolidated tape. The last row in [Table A4](#) provides a rough estimate based on public sources, which suggest that roughly half of the trading volume occurs OTC, with the remainder split 2:1 between electronic platforms and exchanges (mostly MTS).²⁶ In fact, our transaction-level dataset roughly confirms these market shares, see [Appendix A.4](#).

Name	Code	Country	Maturity (years)	Contracts (million)	%	Volume (billion €)	%
Bund	FGBL	DE	8.5-10.5	216	33%	32,835	37%
Bobl	FGBM	DE	4.5-5.5	158	24%	19,873	23%
Schatz	FGBS	DE	1.75-2.25	142	21%	15,543	18%
OAT	FOAT	FR	8.5-10.5	54	8%	7,745	9%
BTP	FBTP	IT	8.5-11	42	6%	5,390	6%
Short Term BTP	FBTS	IT	2-3.25	27	4%	2,977	3%
Buxl	FGBX	DE	24-35	22	3%	3,793	4%
Bono	FBON	ES	8.5-10.5	0	0%	21	0%
*Mid-Term BTP	FBTM	IT	4.5-6	0		0	
*Mid-Term OAT	FOAM	FR	4.5-5.5	0		0	
Sum				662		88,177	

Table A2: Euro Area Government Bond Futures on Eurex. Trading volumes refer to 2022 ([source: Eurex](#)). The maturity column refers to the remaining maturity a cash bond must have in order to be deliverable into the respective futures contract. *Since their trading volume is virtually zero, we exclude the FBTM and FOAM futures from our analysis.

Country	Future Volume	Cash Volume	Ratio Future/Cash
Germany	72,044	7,404	9.7
France	7,745	5,435	1.4
Italy	8,367	7,197	1.2
Spain	21	2,002	0.01
Sum	88,177	22,038	4.0

Table A3: Comparison of Euro Area Government Bond Trading Volumes on the Futures and Cash Market in 2022 in billion €. The bond future trading volumes correspond to the aggregate volume of all futures of a given country, see [Table A2](#). The cash market trading volumes are based on the [European Secondary Bond Market Data Report](#) by the International Capital Market Association (ICMA). These numbers are roughly in line with data from the [Government Bond Data Report](#) from the Association for Financial Markets in Europe (AFME), whose numbers for Germany are based on a survey of the debt management agency among dealer banks. This survey puts the cash trading volume in German bonds in 2022 roughly 10% lower at 6,636 billion €, see [Finanzagentur website](#).

obligation to report trades through an Approved Publication Arrangement (APA). A recent [ESMA Consultation Paper](#) provides a concise overview of European trading venues and the [ESMA Register](#) contains a list of regulated markets, MTFs, OTFs, SIs, and APAs. Lastly, a [2021 ICMA Directory](#) catalogues cash trading venues including eligible participants and trading protocols.

²⁶ A 2021 [European Commission Report](#) about the proposed merger between the London Stock Exchange Group (owner of the MTS platform at that time) and Refinitiv (owner of the Tradeweb platform) provides a detailed overview of the European government bond market and the different trading platforms. The merger was approved only after the London Stock Exchange Group sold off its MTS platform to Euronext (see [European Commission press release](#)).

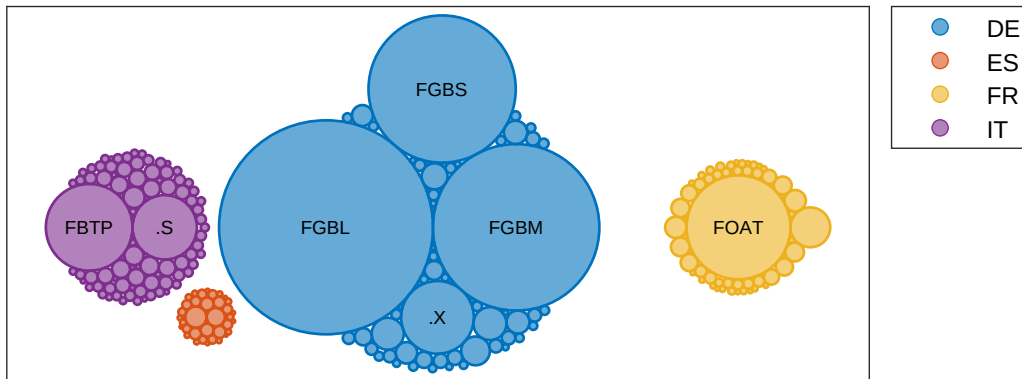


Figure A1: EGB Trading Volumes at the Instrument-Level. Each circle refers to an individual bond or bond future. Bond futures are labelled, e.g. ‘FGBL’ for the 10-year Bund future. The size of each circle is proportional to the trading volume in that instrument. In particular, the relative trading volume across countries and across the futures and cash market matches [Table A3](#). Within the futures market, the relative volumes match [Table A2](#). For the cash market, the number of bonds per country and the distribution of trading volumes across those bonds are based on our transaction dataset for a particular day (7 April 2020), see [Appendix A.4](#). Since this dataset is restricted to plain vanilla bonds, the above figure slightly understates the true number of bonds per country. Lastly, note that the Spanish bond future is not labelled due to its low trading volume (roughly equal to the median volume across Spanish cash bonds).

	On-Exchange	Electronic Platforms	Over-the-Counter
Trading Protocol	Central Limit Order Book	Request for Quote	Voice, Chat
Immediacy, Centralization, Pre-Trade Transparency	High	Moderate	Low
Post-Trade Transparency	High	Low	Low
Anonymity	High	Low	Low if bilateral, moderate if via broker
Examples	MTS, stock exchanges	Tradeweb, Bloomberg	interdealer brokers: TPICAP, BGC, ...
Market Share*	16.6%	33.3%	50%

Table A4: Stylized Differences between Cash Trading Venues for European Government Bonds. *The market shares are rough estimates based on public trading volume data from different sources covering different time periods. In particular, the above-mentioned [2022 ICMA report](#) puts the OTC market share at roughly 50% (labelled ‘systematic internaliser’ trades, which are a major segment of the OTC market, see also [Appendix A.4](#)). To decompose the remaining “on-venue” volume into exchanges and electronic platforms, we use the [2019 EC Report](#) cited in the main text. This report estimates that roughly 25% of “on-venue” volume occurs “on-exchange” (0-5% on regular stock exchanges and 15-30% on MTS, see [Table 3](#) on p.168 of the report).

A.4 Cash Market Transaction Dataset

As explained in [Section 2.1](#), we exploit multiple datasets to capture as many cash transactions in EGBs as possible. For the entire sample since 2008, we use data from the MiFID I/II datasets and from MTS. For the six most important days—the two Eurex outage days and the respective control days in 2020—we exploit further sources, as explained below.

Data Sources

In a nutshell, we start with a non-anonymous regulatory dataset and add non-duplicate anonymous transactions sourced directly from trading venues.

To give more context, note that the MiFID II regulation differentiates between *trade* and *transaction* reporting. Trade reports are anonymous, need to be sent in near real-time, and must be made publicly available, e.g. via an Approved Publication Arrangement (APA).²⁷ These reports usually contain the intraday timestamp, ISIN, price and volume of a traded bond, with some exceptions.²⁸ Transaction reports, in contrast, contain additional information, most importantly about the counterparties involved in each trade.²⁹ These reports need to be sent on a $T + 1$ basis, either to the National Competent Authority (NCA) or ESMA, possibly via an Approved Reporting Mechanism (ARM).

What we refer to as ‘MiFID II’ data refers to the *transaction* reports sent to Bafin, Germany’s NCA. This data only covers trades in which a German security is traded or where at least one of the involved parties—the buyer/seller, the trading venue, or a central counterparty—is domiciled in Germany. We augment this data with non-duplicate *trade* reports we obtained directly from trading platforms (MTS, MTS Bondvision, Tradeweb) and interdealer brokers (TPICAP, BGC, GFI, Aurel).³⁰ See below for further details on how we define duplicates and clean the data.

Cash Bond Universe

To make life easier, we restrict our analysis to ‘plain vanilla’ European government bonds whenever we study the cash market. That means we ignore inflation-indexed bonds, bonds with variable

²⁷However, the regulation does allow to ‘charge fees for the use and redistribution of historic data’, see [ESMA Q&A on MiFID II](#). Hence, in practice, trade reports are only available for the current and previous business day, see e.g. [Tradeweb website](#), [Bloomberg website](#) and [Brokertec website](#). In June 2023, the European Commission has announced to improve this state of affairs by creating a ‘*consolidated tape [that] will bring together the prices and volumes of financial instruments, such as shares and bonds, from hundreds of execution venues across all Member states into a single stream of information, equally accessible for everybody*’, see [EC press release](#).

²⁸Under MiFID II, transactions above specific ‘large in scale’ (LIS) or ‘size specific to the instrument’ (SSTI) thresholds or transactions in ‘illiquid’ bonds benefit from deferred publication. The thresholds vary across ISINs and are regularly updated by ESMA, as is the list of bonds deemed liquid. At the time of the two Eurex outages in 2020, the LIS threshold for sovereign bond transactions was €15 million, the SSTI threshold was €6.5 million (the latter only applies to trades where a party is dealing on own account) and ESMA deemed a total of 520 bonds liquid (see [ESMA website](#)). In some cases the deferral period can be indefinite, i.e. these transactions are never published individually. Instead, only a weekly aggregate volume is published at the ISIN-level. Lastly, MiFID II allows ‘volume masking’, i.e. some transactions can be reported without volumes. Regarding our dataset, these reporting exemptions apply only to the Tradeweb dataset, which covers only few trades above €6.5 million, none above €15 million, and roughly 5% of transactions omit the volume information. For details on the post-trade reporting requirements, see this [AFME report](#).

²⁹For the full list of reported fields, see [EU regulation 2017/590 Annex 1 Table 2](#).

³⁰For the sake of completeness, note that we unsuccessfully tried to obtain intraday transaction data from further data sources. Tradition, another large interdealer broker for European government bonds, see [Tradition website](#), did not retain transaction data for the period we study. Similarly, Bloomberg provides the transaction data it collects under MiFID II only with a limited history. Lastly, various sources provide data on ISIN-level trading volumes, but only at a daily frequency, which is too coarse for our purposes (e.g. Clearstream and Euroclear, the two principal clearing houses in Europe, MarketAxess, a trading venue, and Markit, an independent data provider).

coupon, ‘strips’, ‘green’ bonds, and any other exotic bonds. For each bond ISIN, we obtain a number of bond characteristics from Bloomberg, including the issuance and maturity date, the coupon rate, coupon frequency and day-count basis. [Table A5a](#) provides summary statistics regarding the time to maturity, time since issuance, original maturity, and the coupon rate of the bonds we study. [Table A5b](#) provides a breakdown of our bond universe by country and maturity bucket.

Table A5: Characteristics and Composition of the Cash Bond Universe.

	min	p50	mean	max
Years to Maturity	0.0	2.9	6.3	46.9
Years since Issuance	0.0	2.5	4.7	28.4
Original Maturity in Years	0.2	10.0	11.0	50.4
Coupon Rate	0.0	0.9	1.7	9.0

(a) Bond characteristics.

Maturity Bucket	DE	ES	FR	IT	Total
< 2.5 years	37	24	46	50	157
2.5 – 5.5 years	14	12	11	25	62
5.5 – 10.5 years	15	16	14	18	63
> 10.5 years	11	12	16	21	60
Total	77	64	87	114	342

(b) Number of bonds by country and maturity bucket.

We use maturity buckets of less than 2.5 years to maturity, 2.5 to 5.5 years, 5.5 to 10.5 years, and more than 10.5 years. These thresholds ensure that we have an ‘on-the-run’ (OTR) bond in each of the first three buckets and a cheapest-to-deliver bond in all buckets (if a corresponding future exists, see [Table A2](#)).

[Table A6a](#) shows the ISINs of the bonds that were cheapest-to-deliver into futures contracts at the time of the Eurex outages in 2020. Similarly, [Table A6b](#) shows the ISINs of on-the-run bonds, i.e. the most recently issued bond with approximately two, five or ten year original maturity. For France and Spain, we identify OTR bonds for the 1-year rather than the 2-year maturity, since the latter maturity is uncommon in these countries.

Final Sample

[Table A7](#) panel (a) shows the final number of transactions we use from each data source. As expected, our MiFID II data has the best coverage for trades in German bonds. In theory, the dataset should cover all trades, and we are close to this ideal in practice. For instance, adding transactions from Tradeweb to the MiFID II dataset gives roughly 3% more observations for Germany, but almost doubles the number of observations for France and Spain. Similarly, adding transactions from MTS and MTS Bondvision is crucial for Italy, but rather negligible for Germany.

[Table A7](#) panel (b) contains average daily trading volumes at the country-level. To put these numbers into perspective, we capture a volume of 24 billion Euro in German government bonds, more than the German debt management agency estimates based on a survey among dealer banks (17 billion Euro daily trading volume in 2020, see [Finanzagentur website](#)). This is reassuring, as the survey covers a smaller universe of trades, namely only trades in which one counterparty is a primary dealer.

The last row in [Table A7](#) contains the relative shares of the trading volume we capture across countries. German bonds account for 51% of volume in our dataset compared to a ‘true’ cash

Table A6: Overview of Futures Contracts, Cheapest-to-Deliver Bonds, and On-the-Run Bonds.

Maturity	Country	Future	April 2020	July 2020
2y	Germany	FGBS	DE0001104792	DE0001104800
	Italy	FBTS	IT0005367492	IT0005282527
5y	Germany	FGBM	DE0001102374	DE0001141810
10y	Germany	FGBL	DE0001102465	DE0001102473
	France	FOAT	FR0013407236	FR0013407236
	Italy	FBTP	IT0005365165	IT0005365165
	Spain	FBON	ES0000012E51	ES0000012E51
30y	Germany	FGBX	DE0001135481	DE0001102341

(a) Futures contracts and cheapest-to-deliver (CTD) bonds.

Country	Time	1y/2y	5y	10y
Germany	April 2020	DE0001104792 [#]	DE0001141810	DE0001102499
	July 2020	DE0001104800 [#]	DE0001141810 [#]	DE0001102507
Italy	April 2020	IT0005388928	IT0005344335	IT0005403396
	July 2020	IT0005412348	IT0005408502	IT0005413171
France	April 2020	FR0125848699	FR0013157096	FR0013451507
	July 2020	FR0126001801	FR0013157096	FR0013516549
Spain	April 2020	ES0L02103056 [*]	ES0000012F92	ES0000012F76
	July 2020	ES0L02106117	ES0000012F92	ES0000012F76

(b) On-the-run (OTR) bonds. For Germany and Italy, we identify 2-year OTR bonds; for France and Spain, 1-year OTR bonds. [#]For Germany, some bonds are OTR and CTD at the same time, see Panel A. ^{*}For Spain, the 1-year OTR bond changes between the first Eurex outage on 14 April 2020 and the control day on 21 April (to ES0L02104161 on 17 April 2020).

(a) Number Of Transactions					
	DE	FR	IT	ES	Total
Mifir	16891	1821	13172	1950	33834
Tradeweb	550	1738	2365	1379	6032
MTS	36	431	5278	461	6206
BondVision	125	206	2711	218	3260
Interdealer Broker	601	660	516	103	1880
Total	18203	4856	24042	4111	51212

(b) Average Daily Trading Volume (in billion €)					
	DE	FR	IT	ES	Total
in billion €	24.0	6.0	13.5	4.0	47.5
in % of total	50.5	12.6	28.5	8.4	100.0

Table A7: Number and Volume of Cash Transactions by Data Source and Country. All figures refer to transactions in ‘plain vanilla’ European government bonds, see Appendix A.4 for details. Panel (a) refers to the total number of transactions on all six days in our sample (the two Eurex outage days and the respective control days), whereas panel (b) refers to the average across the four control days.

market share of only 34% in 2022 based on Table A3. The analogous figures are 13% vs. 25% for France, 29% vs. 33% for Italy, and 8% vs. 9% for Spain. That means even after augmenting the regulatory data, trades in German bonds are still overrepresented. This non-representativeness is not too concerning for our aggregate volume analysis, which relies on differential effects (outage days versus matched control days, and outage windows versus adjacent intraday windows). In that diff-in-diff setting, level biases in country coverage matter little for the estimated treatment effects. It is problematic, however, for the mechanism analysis, which requires a near-complete view of trading across venues and investor types. For this reason, we focus on German Bunds in Section 3.3 and Section 4 of the main text.

Figure A2 breaks down trading volumes by venue type and documents substantial country-level heterogeneity. German bonds, e.g., are traded predominantly OTC or on electronic platforms and rarely on-exchange.³¹ Italian and Spanish bonds, in contrast, are frequently traded on-exchange. Note, however, that the venue market shares for France, Italy and Spain should be interpreted with caution. Our non-anonymous MiFID II data misses many trades in these bonds and thus a proper identification of trade chains (affiliate trades) is less reliable, even for the trades covered, leaving the ultimate trading venue of many transactions unidentified.

MiFID II data cleaning procedure

From the raw MiFID II dataset, we first delete transactions i) with missing or implausible prices or quantities, ii) that occurred outside of trading hours (prior to 8 a.m. or after 7 p.m.), and iii) that occurred more than two days prior to the bond’s issuance (so-called ‘when-issued’ transactions). We round prices to one decimal because we observed that multiple reports of the same transaction sometimes deviate slightly in prices. Since timestamps are vital for our deduplication procedure and the linkage of plausible trade chains, we correct those where a misreporting is highly likely. E.g. differences of exactly one or two hours for trades with identical bond, price, and quantity

³¹de Roure et al. (2025) study dealer-to-dealer trades in German bonds (using the MiFID I dataset mentioned in Section 2.1) where they can identify both counterparties. In this subsample, they find that interdealer brokers account for roughly 80% of trading volume. Our data puts the IDB share at roughly 25% of the aggregate trading volume.

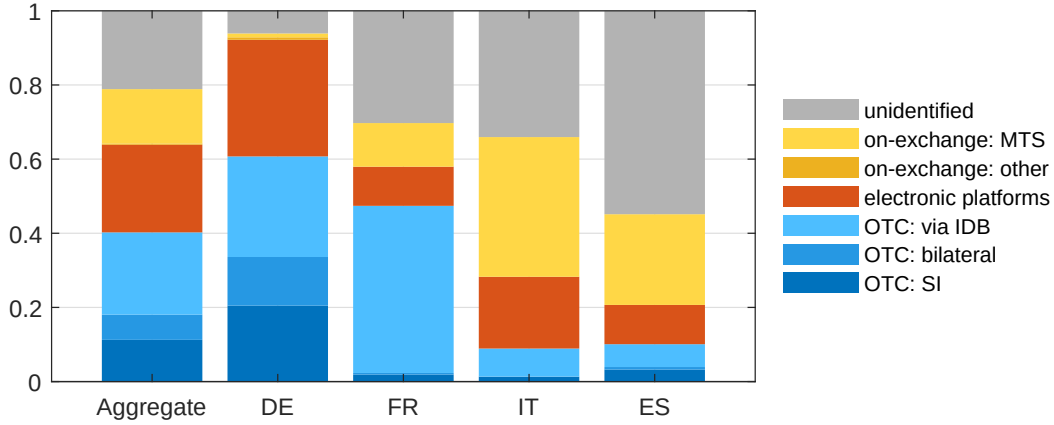


Figure A2: Market Shares of Cash Trading Venues. This figure breaks down the aggregate and country-level trading volume in EGBs into different cash trading venues. Results are based on our transaction dataset for all four non-outage days (i.e. April 7 and 21, June 30 and July 8, 2020), see Appendix A.4. ‘IDB’ refers to interdealer brokers, ‘SI’ refers to systematic internalisers and ‘other exchanges’ refer to regular stock exchanges.

likely reflect misreported time zones. We convert everything to CET. Timestamp deviations by up to two seconds are also accounted for when everything else points towards a trade chain.

For a probable trade chain with multiple different reported trade venues, we identify the ultimate trading venue by placing actual venues over systemic internalisers and OTC flags to identify the ultimate trading venue. Our final goal as regards trades’ counterparties is to identify the ultimate buyers and sellers, i.e. those that ultimately exchange ownership in the bond. First we delete duplicates in timestamp, bond, buyer ID, seller ID, price and quantity but different submitting IDs. Trade chains are then identified by timestamp, bond, price and quantity. In the large majority of cases, netting over all involved buyers and sellers results in only two remaining counterparties of which one is a buyer and the other a seller with identical quantities. Only if this is not the case, we deploy a carefully designed algorithm. To give a basic example, when one buyer but two sellers remain after netting and both of these sellers show identical quantities, i.e. this is not a cumulative trade, we place the selling counterparty less likely to be an intermediary first (e.g. an investment fund) and drop the other seller (e.g. a dealer bank). A set of these measures results in the identification of two (or in the case of cumulative trades three or more) ultimate counterparties for the bulk of identified trade chains. For the remainder, we drop the trade side that remains ambiguous. We also drop intragroup trades, i.e. trades between different LEIs belonging to the same investor, see Appendix A.5

Additional Filters for Yield Curve Fitting

In Section 3.3, where we compute yield curve fitting errors for German government bonds, we apply some additional filters to the ones mentioned above. In particular, we exclude intragroup trades (see next section) and drop trades in bonds with less than one or more than ten years to maturity or that have implausible prices. For the latter filter, we convert each transaction price to an implied yield³² and compute the difference to the daily maturity-matched yield from the Bundesbank’s term structure model (see Bundesbank website). We drop a few transactions where the absolute yield difference is larger than 75 basis points, or larger than 25 basis points and the transaction price is exactly 100. These transaction prices most likely reflect reporting errors. These filters leave us with over 3,000 transactions during the six selected intraday windows.

³²We use Matlab’s built-in *bndyield* function, taking into account the price of the bond, its maturity and issuance date, the coupon rate, coupon frequency, first and last coupon date, and assuming $T+2$ settlement.

Table A8 provides some descriptive statistics on these transactions.

	min	p50	mean	max
No Outage				
Volume (million Euro)	0.00	1.22	9.96	1292.00
Years to Maturity	1.00	5.36	5.77	9.86
Years since Issuance	0.11	2.27	3.27	26.47
Coupon Rate	0.00	0.00	0.53	6.50
Yield	-0.73	-0.62	-0.59	-0.31
Yield Curve Fitting Error	-12.23	-0.23	-0.18	27.94
Outage				
Volume (million Euro)	0.00	0.90	2.37	200.00
Years to Maturity	1.01	5.84	6.11	9.84
Years since Issuance	0.09	2.25	3.59	26.28
Coupon Rate	0.00	0.25	0.62	6.50
Yield	-0.92	-0.57	-0.56	-0.18
Yield Curve Fitting Error	-23.80	-0.12	-0.20	50.53

Table A8: Descriptive Transaction Statistics for Yield Curve Fitting Sample. These statistics refer to the over 3,000 transactions in German sovereign bonds with 1-10 year maturity shown in Figure 3 and underlying the regression results in Table 3.

A.5 Investor-Level Info

A key feature of our regulatory transaction data is that it contains a unique legal identifier (LEI) for at least one of the two counterparties involved in each cash transaction.³³ We can thus use Eurosystem supervisory lists and the Register of Institutions and Affiliates Database (RIAD) database, see [Perrella and Catz, 2020](#)) to group LEIs into unique investors.³⁴ Then, we classify investors into the following categories: banks, non-bank financial institutions (NBFI), investment funds, hedge funds, insurance companies and pensions funds (ICPF), non-financial corporations (NFC), an official sector (including central banks) and households (retail investors).³⁵

Table A9 gives an overview of the number of investors of each type and Table A10 gives an overview the number of transactions across investor segments and types.

For banks, it is important to differentiate between (non)dealers. We identify dealers using the list of primary dealers participating in Bund auctions, see [Finanzagentur website](#). In total, our sample covers cash market transactions by 32 different dealer banks. With this classification at hand, we categorize each transaction into one of three investor segments: client-to-client (C2C), dealer-to-client (D2C), and dealer-to-dealer (D2D).

³³This is in contrast to the Canadian transaction data used e.g. by [Allen and Wittwer \(2023a\)](#), which contains a LEI only for dealers, brokers, and the largest investors, whereas other investors remain anonymous. The US transaction data used e.g. by [Duffie et al. \(2023\)](#) only covers trades by registered broker-dealers, identified via Market Participant Identifier (MPID) codes. The UK transaction data used e.g. by [Czech, Huang, Lou, and Wang \(2021\)](#), lastly, contains a Business Identifier Code (BIC, issued by SWIFT) or a Firm Reference Number (FRN, issued by the UK's Financial Conduct Authority).

³⁴The 32 dealers in our sample, e.g., comprise over 100 different LEIs (similar to [Duffie et al., 2023](#)).

³⁵We start classifying investors using the Eurosystem lists of financial institutions (covering banks, investment funds and ICPFs, see [ECB website](#)) and use RIAD for the remaining investors. In some cases, we manually adjust the investor classification. Some of the largest global banks trade in EGBs via European subsidiaries, for instance, and RIAD identifies these as NBFIs rather than banks, probably because they are no depository institutions in the euro area.

	Eurex-Active	Eurex-Inactive	Total
Bank Dealer	30	2	32
Bank Non-Dealer	13	68	81
NBFI	6	39	45
Investment Fund	5	220	225
Hedge Fund	6	61	67
ICPF	1	55	56
NFC	0	20	20
Official	0	16	16
Household	0	1	1
Total	61	482	543

Table A9: Number of Investors by Type. Data refers to investors with Bund trades on the cash market on the six selected days in 2020. Investors are aggregated to group-level. The category ‘household’ comprises an unknown number of retail investors, as these are not uniquely identifiable via a LEI.

(a) Investor Segments			
	No Outage	Outage	Total
C2C	879	261	1140
D2C	3195	640	3835
D2D	978	164	1142
Total	5052	1065	6117

(b) Investor Types			
	No Outage	Outage	Total
Bank Dealer	2578	484	3062
Bank Non-Dealer	590	180	770
NBFI	474	117	591
Investment Fund	489	99	588
Hedge Fund	256	64	320
ICPF	175	19	194
NFC	30	12	42
Official	323	25	348
Household	137	65	202
Total	5052	1065	6117

Table A10: Number of Cash Transactions by Investor Segment and Type. All figures refer to transactions in German sovereign bonds with 1-10 year maturity during the two Eurex outages in 2020 and the four non-outage control windows. With two sides to each trade, the totals sum to roughly twice the number of observations shown in Figure 3 and underlying the regression results in Table 3 (abstracting from unknown investor types due to anonymous transaction data).

We also separate investors into those that are usually active on the bond futures market and those that are not, see last column of Table A9. In particular, we exploit the European Market Infrastructure Regulation (EMIR) dataset, see Section 2.2. We deem any investor with more than 20 trades in any German bond future on the six selected days in 2020 (outage and control days) as ‘active on Eurex’. This way we can differentiate between cash market trades in which both, only one, or none of the two counterparties are usually active on Eurex.

For our trade report data, the investor classification is less straightforward, because this data is anonymous. However, the data source itself is informative about the venue type and investor segment and we use this information to the extent possible, see [Table A11](#). For instance, we attribute transactions on MTS to the dealer-to-dealer segment, based on the fact that MTS is restricted to dealers. The classification is equally uncontroversial for data we obtained from interdealer brokers. The MTS Bondvision and the Tradeweb platform, in contrast, are open to both dealers and clients, and hence we leave the investor segment of these trades unclassified. However, besides transactions executed directly on its platforms, Tradeweb also publishes data on OTC trades (for which Tradeweb acted as an APA). Within these trades, systematic internaliser trades are explicitly flagged and we classify the remaining trades as bilateral OTC trades.

Data Source	Venue Type	Venue Subtype	Segment
MTS	On Exchange	MTS	D2D
TPICAP, BGC, GFI, Aurel	OTC	OTC via IDB	D2D
MTS Bondvision	Electronic Platforms		-
Tradeweb MTF [*]	Electronic Platforms		-
Tradeweb SI [†]	OTC	OTC via SI	-
Tradeweb other [†]	OTC	Bilateral OTC	-

Table A11: Venue and Investor Segment Classification for Anonymous Transaction Data. This table shows how we classify anonymous transactions, sourced directly from trading venues, into venues and investor segments. Empty cells mean we leave the segment unidentified. ^{*}Due to Brexit, Tradeweb runs two separate multilateral trading facilities (MTF) in the UK and the EU (MIC Codes ‘TREU’ for London and ‘TWEM’ for Amsterdam). [†]Tradeweb also offers an APA service, reporting trades from the OTC market, explicitly flagging trades by systematic internalisers (SI).

Appendix B Further Evidence on Eurex Outages

This section provides further evidence on the effect of futures market outages.

Appendix B.1 uses prior outages on Eurex as a robustness check of our main results. Appendix B.2 confirms that trading volumes on the EGB cash market catch up after the outages on the futures market are over. Appendix B.3 documents how widespread the liquidity dry-up on MTS is. Liquidity drops slightly less on the separately run local market segments, where ‘market making obligations’ are enforced. Appendix B.4 provides robustness checks regarding the effect of Eurex outages on indicative EGB quotes on the cash market. Appendix B.5 shows that the mispricing on the Bund market is restricted to the intraday periods of the two Eurex outages. Appendix B.6 provides further micro-level on dealers’ hedging strategies. Appendix B.7 shows that long-maturity interest rate swaps are severely affected by outages on Eurex. Short-term swaps, in contrast, continue to provide a reliable indicator of ‘fair’ short-term rates during the outages. Appendix B.8 shows that outages reshape the size of trades differentially across investors segments and trading venues. Appendix B.9 rules out alternative mechanisms via which outages on the futures market might disrupt the cash market.

B.1 Previous Eurex Outages as Robustness

Section 3 in the main text suggests that market functioning for euro area government bonds depends heavily on the futures market, but that evidence is based on just two Eurex outages in 2020. We put particular emphasis on these two most recent outages because we have the best data for this period, in particular regarding EGB transactions on the cash market, see Appendix A.4. The two Eurex outages in 2020 have not been without precedent, however. Hence, as a robustness check, this section studies nine previous outages on Eurex between 2009 and 2018, see Appendix D.1.³⁶

Table A12 shows that these previous outages also cause trading on the cash market to decline. The only difference is that trading volumes in bonds with more than 10.5 years to maturity do not drop significantly.³⁷

How about MTS? Did Eurex outages always lead to an evaporation of liquidity on MTS? Yes, as shown in Table A13. As in our main results, liquidity dries up least in short-term Italian bonds.

One last piece of evidence on the interaction between Eurex and MTS comes from different non-trading days between the two platforms. While MTS usually has the same holiday schedule as Eurex, 25 May 2015 was an exception. On that day, Eurex was closed while MTS remained open.³⁸ Figure A3 shows that MTS was basically inactive that day. There were very few quotes for any bonds and barely any transactions. Of course, this non-trading day does not constitute a true shock, since it was not exogenous and since it could be well anticipated by market participants. But in a sense, this makes our point stronger: even when market participants have enough time to prepare, they are not willing to trade or quote EGBs without an active futures market.

³⁶In the following analysis, we omit three of the twelve previous outages. The one on 23 December 2009, when the start of futures trading was delayed from 8:00 a.m. to 8:20 a.m. because this outage is too short and too early in the day to observe any effect on the cash market. And the two partial outages on 21 November 2016 and 26 May 2014, as these are not directly comparable to the other system-wide outages.

³⁷We cannot study country-level trading volumes because the regulatory transaction-level MiFID I dataset we use (see Section 2.1) mainly captures trades in German bonds and we were not able to obtain intraday transaction data directly from trading platforms like we did for the 2020 outages.

³⁸Our results suggest an obvious explanation for why the two trading calendars coincide: since MTS is dependent on Eurex, it adopts their trading calendar. Regarding the 25 May 2015 exception, see [MTS trading calendar 2015](#) and [Eurex press release](#). The latter explains that ‘Eurex [...] decided, as an exception, not to offer any trading [...] on 25 May 2015. On that day there will be national public holidays in the U.S. (‘Memorial Day’), Great Britain (‘Late May bank holiday’), Germany, Austria and Switzerland (‘Whit Monday’), as well as in South Korea (‘Buddha’s Birthday’). Therefore, essential markets are not available.’

	(1) Aggregate	(2) Maturities
Outage	-2.13** [1.03]	
Outage $\times$ <2.5y		-1.14 [0.96]
Outage $\times$ 2.5-5.5y		-2.71*** [0.99]
Outage $\times$ 5.5-10.5y		-3.63*** [0.76]
Outage $\times$ >10.5y		0.97 [1.37]
FE Day	✓	✓
FE Time	✓	✓
Observations	405	1620
Adjusted R^2	0.724	0.502

Table A12: Effect of Previous Eurex Outages on Cash Trading Volumes. This table replicates [Table 1](#) from the main text, using previous Eurex outages as a robustness check. The sample covers nine outage days plus the same day in the preceding and subsequent week, i.e. 27 days in total. Each day, we have 15 intraday observations (from 08:00 a.m. to 3:00 p.m.)

	(1) Aggregate	(2) Maturities	(3) Countries
Outage	-3.33*** [0.37]		
Outage $\times$ <2.5y		-2.43*** [0.33]	
Outage $\times$ 2.5-5.5y		-7.52*** [0.61]	
Outage $\times$ 5.5-10.5y		-8.31*** [0.64]	
Outage $\times$ >10.5y		-9.89*** [0.67]	
Outage $\times$ DE			-7.15*** [0.58]
Outage $\times$ ES			-7.18*** [0.60]
Outage $\times$ FR			-4.67*** [0.50]
Outage $\times$ IT			-2.96*** [0.34]
FE Day	✓	✓	✓
FE Time	✓	✓	✓
Observations	2943	11772	11772
Adjusted R^2	0.546	0.601	0.563

Table A13: Effect of Previous Eurex Outages on MTS Order Book Depth. This table replicates [Table 2](#) from the main text, using previous Eurex outages as a robustness check. Throughout, the dependent variable is the log of the quoted bid and ask volume of bonds of a given country and/or maturity bucket, at 5-minute snapshots.

B.2 Trading Volumes

[Figure A4](#) shows cumulative trading volumes in 10-year government bond futures, comparing the two outage days with the same day in the previous and subsequent week. The figure confirms that during both outages, trading in bond futures came to a complete halt.

[Figure A5](#) reproduces the equivalent figure for the cash market, based on our extensive transaction-level dataset. Trading volumes drop while Eurex is offline and surge once it comes back online.

[Table A14](#) confirms the catch-up. Trading volumes are significantly higher roughly two hours after Eurex goes back online, suggesting that some investors are either forced or choose to delay their trades.

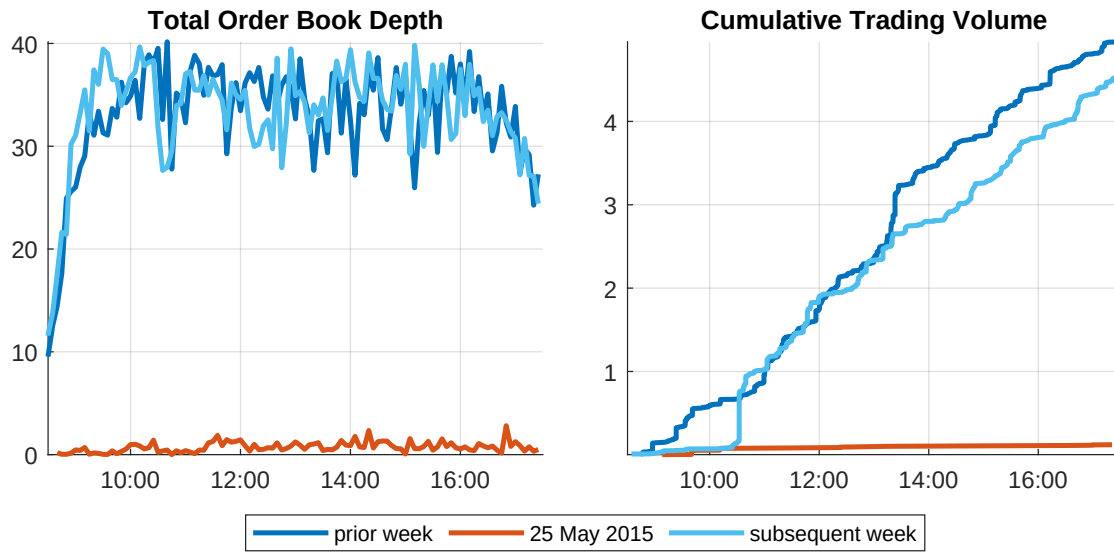


Figure A3: MTS Inactivity on Eurex Holiday. The left panel shows the total order book depth on MTS (in billion €), across all German, French, Italian and Spanish bonds and all market segments, in 5-minute snapshots. The right panel shows the total cumulative trading volume, in billion €. Red lines refer to 25 May 2015 (when Eurex was closed due to a holiday), dark and light blue lines refer to the same day in the previous and subsequent week.

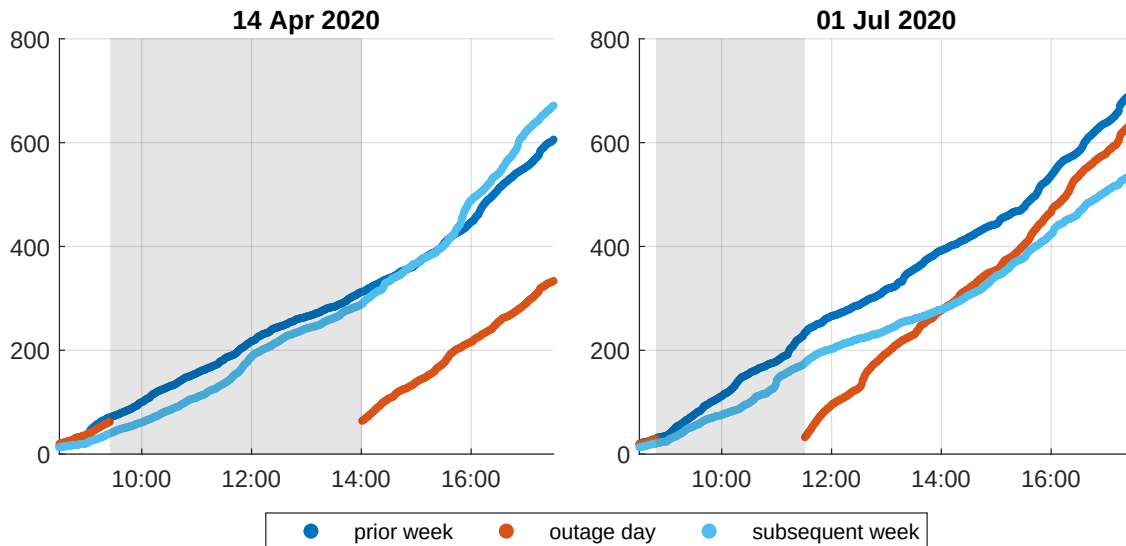


Figure A4: Trading Volume in 10-year Bond Futures. This figure shows the cumulative number of traded contracts (in thousands). Red dots refer to the outage day, dark and light blue dots refer to the same day in the previous and subsequent week.

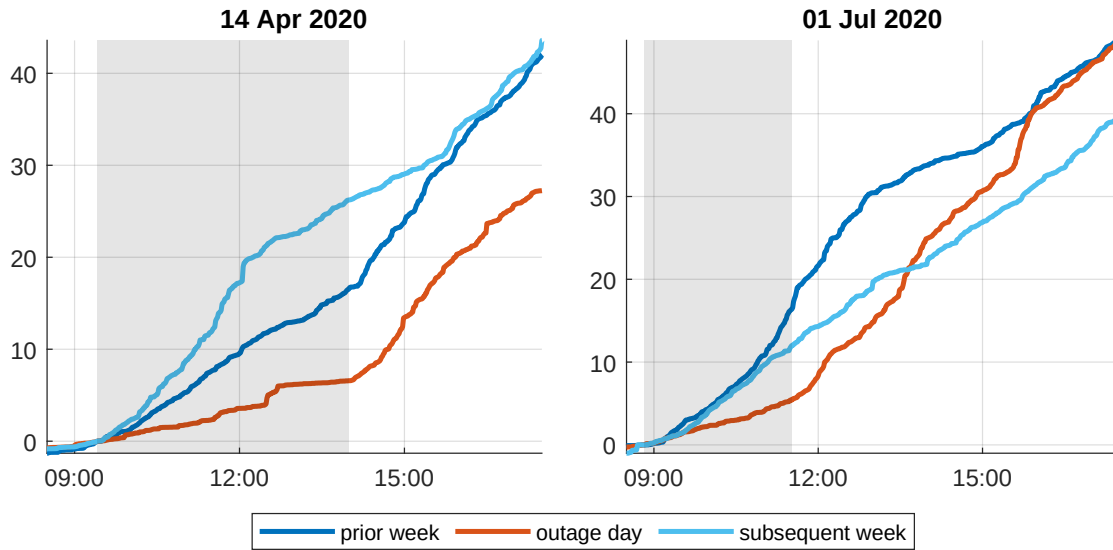


Figure A5: Trading Volume in EGBs on the Cash Market. This figure shows the cumulative trading volume in all German, French, Italian and Spanish sovereign bonds (in billions of Euro, normalized to zero at the intraday time of the outage). Red dots refer to outage days, dark and light blue dots to the previous and subsequent week.

	(1) All Countries	(2) Germany
Outage Window	-1.46*** [0.38]	-1.14*** [0.35]
+ [0,30) min	-0.38 [0.38]	0.06 [0.58]
+ [30,60) min	0.30 [0.42]	0.83 [0.53]
+ [60,90) min	-0.00 [0.45]	0.73 [0.66]
+ [90,120) min	0.21 [0.37]	0.87** [0.36]
+ [120,150) min	0.81* [0.42]	1.99*** [0.35]
FE Day	✓	✓
FE Time	✓	✓
FE Country	✓	
Observations	360	90
Adjusted R^2	0.459	0.491

Table A14: Catch-up in Cash Trading Volumes after Eurex Outages. The dependent variable is the log of the transaction volume in German, French, Italian and Spanish bonds in 30-minute intervals, analogous to [Equation 1](#).

B.3 MTS Liquidity

This section expands on the results presented in [Section 3.2.1](#) in the main text.

[Figure A6](#) shows MTS bid-ask spreads during Eurex outages. If bonds are quoted at all, then at much wider spreads than usual.

[Figure A7](#) replicates [Figure 1](#) at the country-level. The order book depth on MTS drops for all countries when Eurex goes down. [Figure A8](#) shows that the liquidity evaporates also for other euro area countries, which do not have an own futures market. [Figure A9](#), lastly, looks at non-euro countries whose bonds are traded on MTS. Except for Denmark, whose currency is pegged to the Euro, market liquidity seems unaffected by the outage on Eurex. In a sense, these results serve as a placebo test: the breakdown of MTS was indeed caused by the outage on Eurex, there was no simultaneous independent incident on MTS.

Lastly, we study the relevance of so-called ‘market making commitments’. While MTS’s euro area wide ‘EBM’ market has no such commitments, MTS also runs parallel local bond markets (labelled ‘GEM’ for Germany, ‘FRF’ for France, ‘MTS’ for Italy, and ‘ESP’ for Spain). On these local markets, each local issuer, i.e. usually the debt management office, can set obligations and requirements for their own country. For instance, market makers might have to provide bid and ask quotes for specific bonds, sometimes at pre-defined maximum spreads and minimum volumes and for a specified duration, e.g. at least five hours a day or 50% of trading time. See [Cheung, Rindi, and De Jong \(2005\)](#) for a detailed overview of the microstructure of the MTS trading platforms and the [MTS website](#) for current rules.

Do these quoting obligations matter? [Figure A10](#) compares the order book depth on the EBM and local markets for each country. We can clearly see liquidity drops on both segments for all countries, but it is hard to eyeball whether the relative decline differed across countries.

Hence, [Table A15](#) reports results from another dummy regression, this time run separately for each country, namely:

$$\log(1 + OBdepth_{st}) = \alpha + \beta_1 \times D_t + \beta_2 \times D_t \times local + \epsilon_t \quad (A1)$$

where $OBdepth_{st}$ is the order book depth (in €) of all bonds at time t , measured at 5-minute snapshots, quoted on market segment s . D_t is a dummy that equals one during the Eurex outages and $local$ is a dummy variable that equals one for MTS’s local market segment and is zero for the EBM segment. We include six days (the two outage days plus the preceding and subsequent week) and 91 intraday observations per day (5-minute snapshots from 08:30 a.m. to 4:00 p.m.).

We see that the order book depth is higher on the local market for all countries, especially so for Italy and Spain. And precisely for these two countries, we see a differential effect during the Eurex outage. The liquidity of Italian and Spanish bonds drops markedly less on the local market segment compared to the EBM market. This is in line with the stricter quoting obligations on the local Italian and Spanish market compared to the German and French market.

Note that these results are analogous to [Clark-Joseph, Ye, and Zi \(2017\)](#), who show that even mild quoting obligations significantly improve liquidity on US stock exchanges.

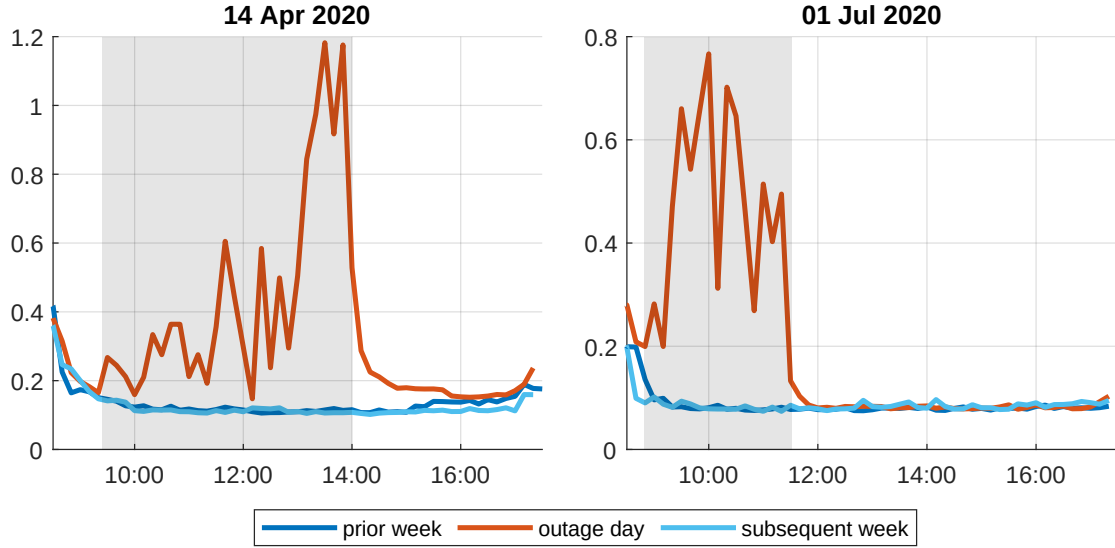


Figure A6: Bid-Ask Spreads on MTS. This figure shows the median top-level bid-ask spread, in percent of the mid price, across all German, French, Italian and Spanish bonds, at 10-minute snapshots.

	(1) DE	(2) ES	(3) FR	(4) IT
Outage	-15.42*** [0.91]	-2.52*** [0.21]	-3.73*** [0.53]	-0.94*** [0.06]
local	0.16 [0.11]	0.71*** [0.11]	0.10** [0.05]	0.51*** [0.01]
Outage $\times$ local	1.04 [1.20]	0.12 [0.18]	0.50 [0.70]	0.14* [0.08]
FE Day	✓	✓	✓	✓
FE Time	✓	✓	✓	✓
Observations	1092	1092	1092	1092
Adjusted R^2	0.719	0.628	0.314	0.802

Table A15: Differential Effect of Eurex Outages on MTS Market Segments. Each column shows results of a separate regression run at the country-level, see Equation A1. Throughout, the dependent variable is the log of the order book depth plus one of all bonds of a given country. We differentiate between the common EBM and local market segment.

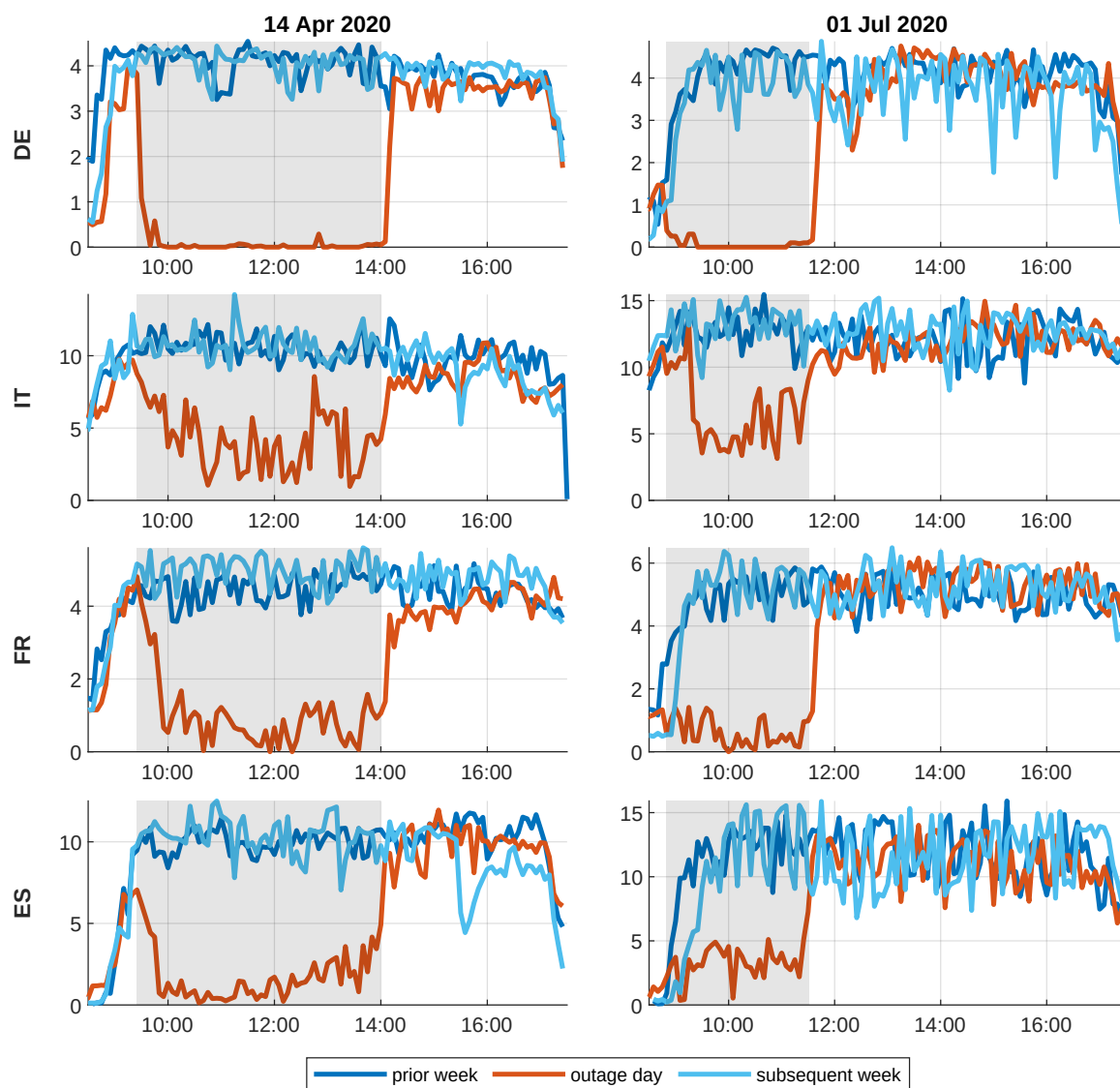


Figure A7: Total Order Book Depth on MTS across Countries. This figure shows the total quoted volume (in billion €) for German, French, Italian and Spanish bonds across all three levels and both sides of the order book, at 5-minute snapshots.

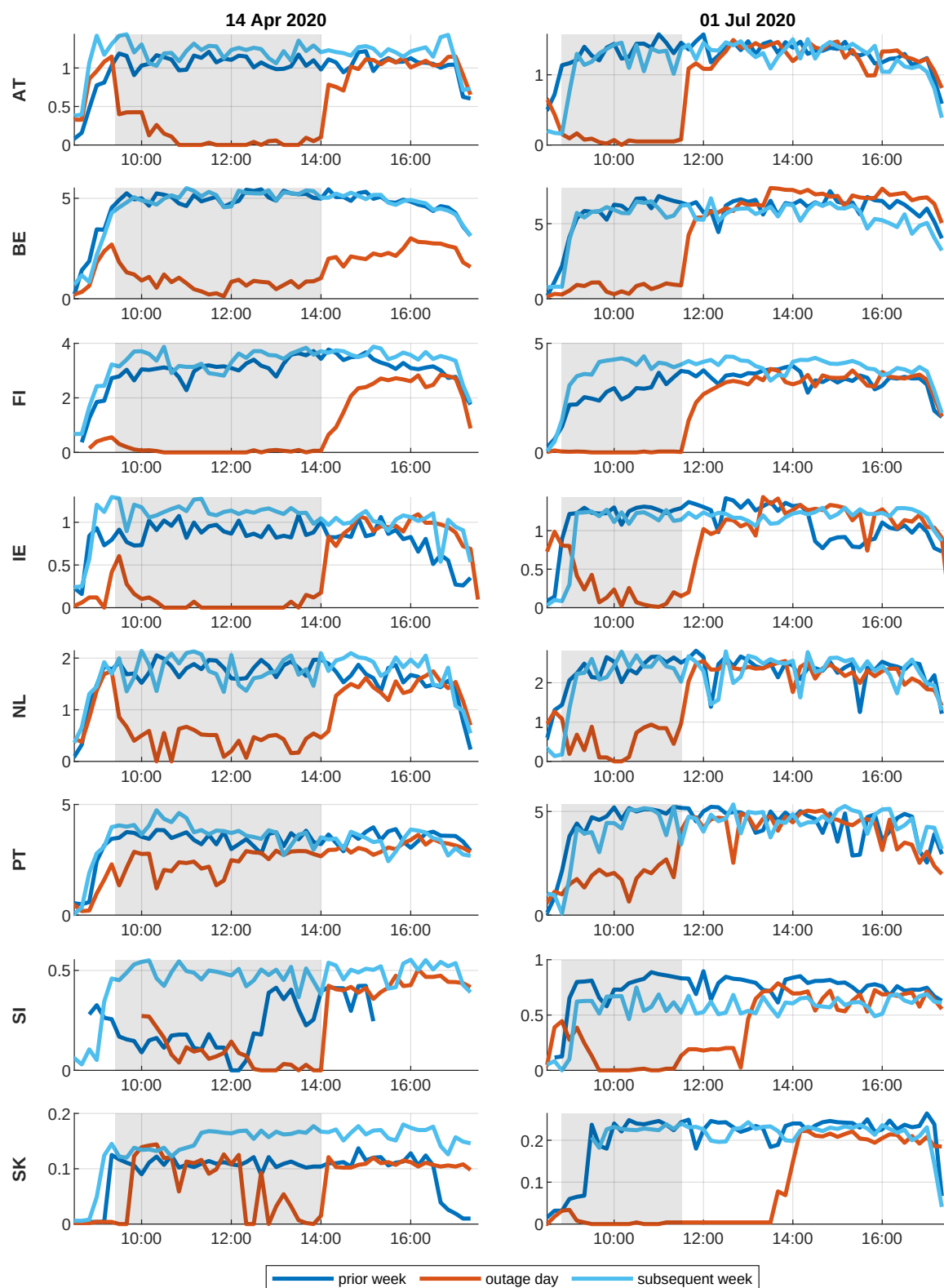


Figure A8: Total Order Book Depth on MTS for Other Euro Area Countries. This figure shows the total quoted volume (in billion €), across all three levels and both sides of the order book, at 10-minute snapshots, for other euro area member states whose bonds are traded on MTS.

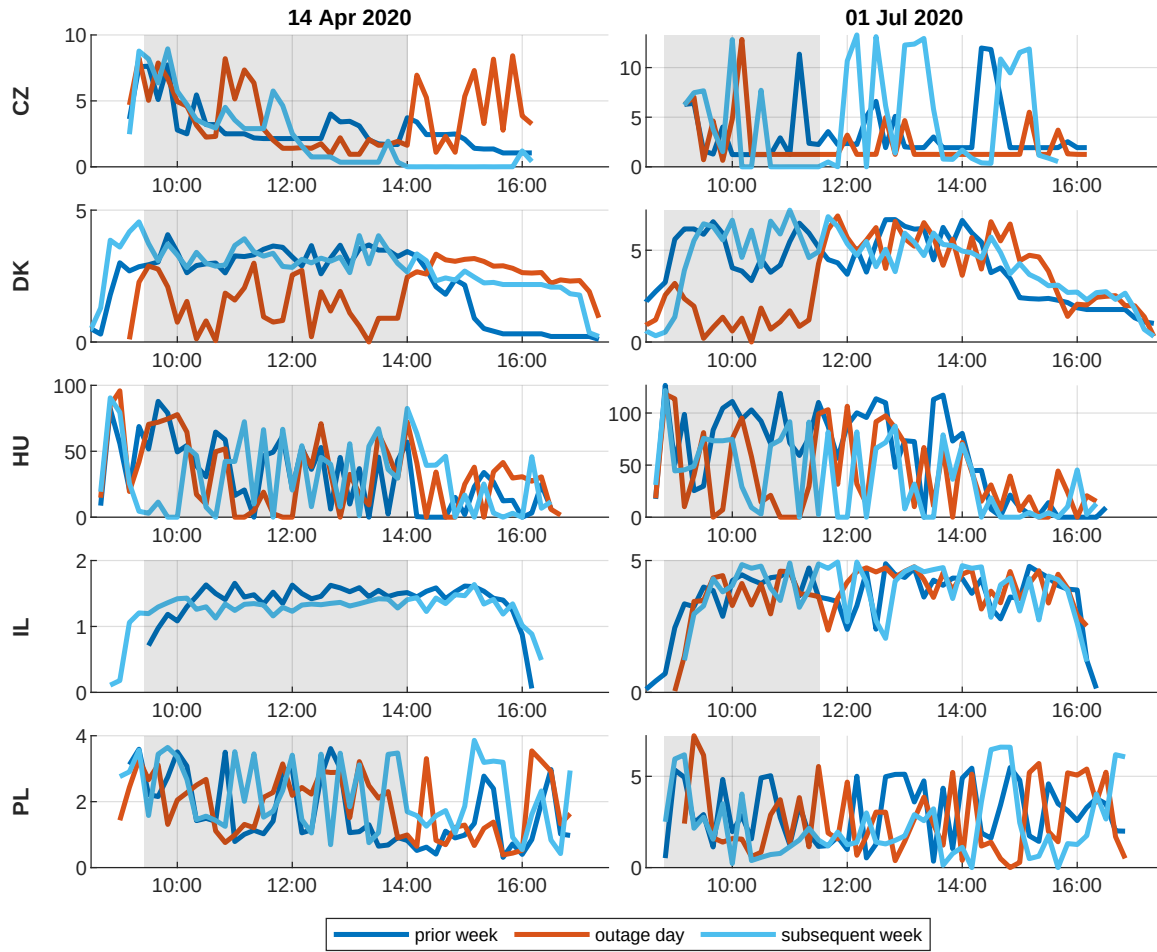


Figure A9: Total Order Book Depth on MTS for Non-Euro Area Countries. This figure shows the total quoted volume (in billion of local currency), across all three levels and both sides of the order book, at 10-minute snapshots, for non-euro countries (CZ for Czech Republic, DK for Denmark, HU for Hungary, IL for Israel, PL for Poland). The lack of liquidity on 14 April 2020 for Israeli bonds was due to a public holiday, not the outage on Eurex.

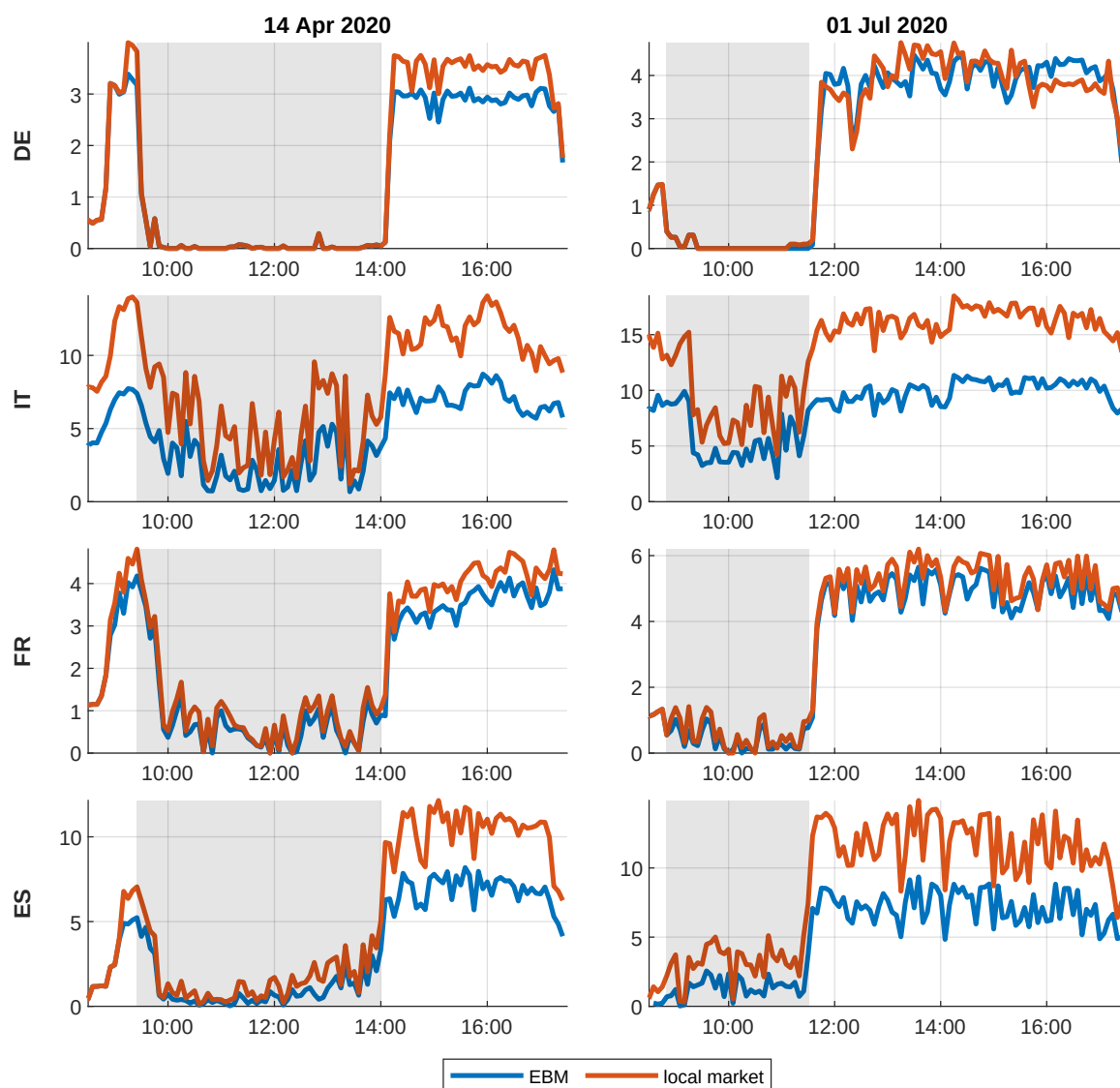


Figure A10: Order Book Depth across different Market Segments on MTS. This figure shows the total quoted volume (in million €) across all three levels and both sides of the order book, at 5-minute snapshots, for all bonds of a given country. Blue lines refer to the ‘European Bond Market’ segment (as in the main text), red lines refers to the respective domestic market segment.

B.4 Indicative Quotes

This section reproduces the results presented in [Section 3.2.2](#) of the main text at the country-level. Recall that [Figure 2](#) shows across three different data sources that indicative quotes for EGB cash bonds become stale during the Eurex outage.

[Figure A11](#) confirms that the result holds for Germany, Italy, France, and Spain: the number of quote updates drops dramatically during Eurex outages. [Figure A12](#) plots the bid yield of 10-year bonds quoted for each country. Yields stay virtually constant while Eurex is offline. By all appearances, these prices are stale.

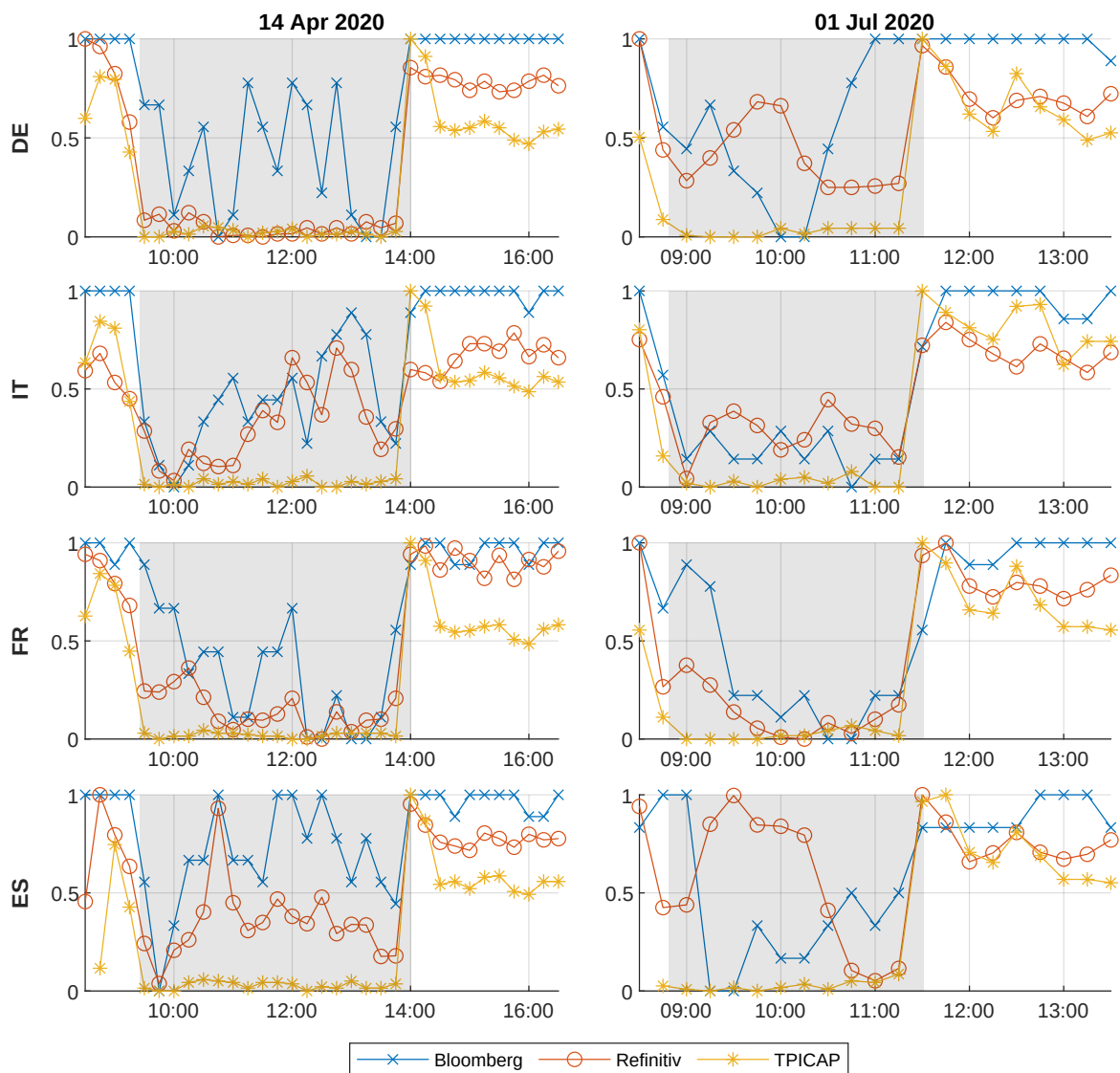


Figure A11: Quote Update Frequency for 10-Year Government Bonds on Different Platforms. Bloomberg and TPICAP data refer to quote updates in the cheapest-to-deliver bond, Refinitiv data to the on-the-run bond. For Bloomberg, we approximate the number of new quotes as the number of tick-by-tick price changes. To show all series on a single scale, we sum the number of quotes updates in 15-minute windows and normalize them to a 0-1 range for each data provider.

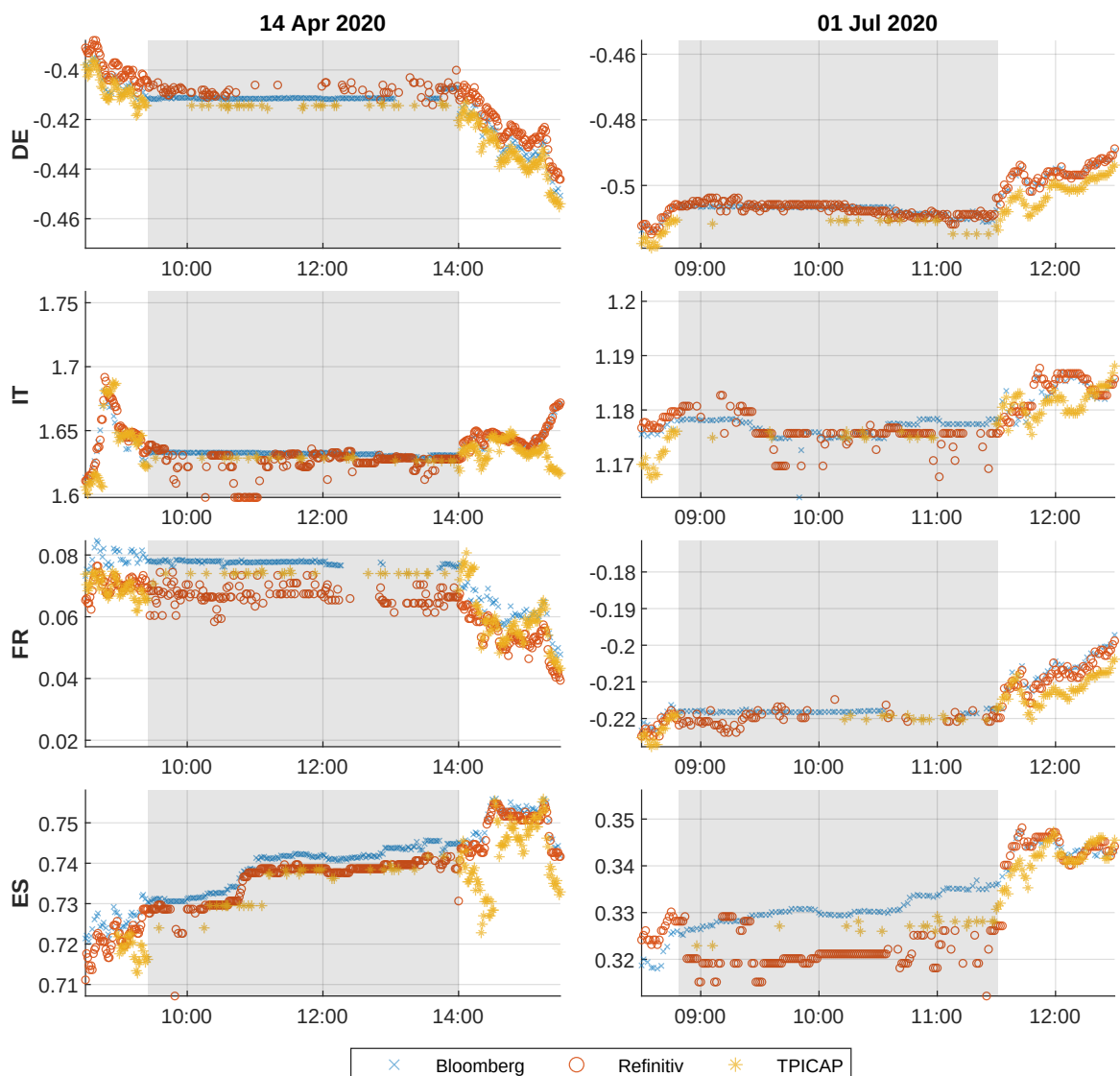


Figure A12: Quoted Bid Yield of 10-Year Government Bonds on Different Platforms. Bloomberg and TPICAP data refer to the cheapest-to-deliver bond, at minutely snapshots. Refinitiv data refers to the on-the-run ('benchmark') bond, at minutely snapshots and as the median value across all available 'pricing contributors'. To show all series on a single scale, we apply a level adjustment to the Refinitiv yield, such that the daily median yield matches the daily median yield of Bloomberg and TPICAP.

B.5 Mispricing Before and After Eurex Outages

Figure A13 zooms in on the two outage days to rule out the possibility that pricing errors in German bonds are higher throughout the two outage days, for reasons other than the Eurex outages. We again fit yield curves and compute root mean squared pricing errors, but this time for hourly windows from one hour before till four hours after the outages. These hourly windows allow a sharp identification, while simultaneously ensuring that we have sufficiently many observations along the yield curve for the Svensson (1994) methodology to work. The figure shows that the huge spike in pricing errors is indeed restricted to the Eurex outage periods. The noise measure jumps up immediately during the first hour of the outage and then quickly recedes, reverting back to normal once Eurex is back online.

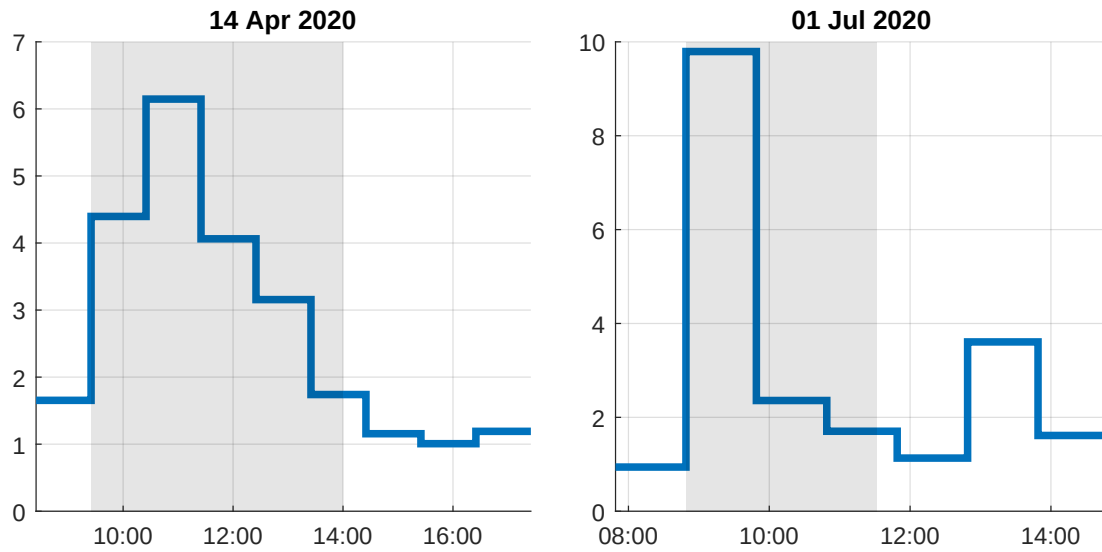


Figure A13: German Yield Curve Noise on Eurex Outage Days. This figure shows the noise measure proposed by [Hu et al. \(2013\)](#) on the two Eurex outage days, from one hour before the outage till four hours after the outage. For each one hour window, we fit a [Svensson \(1994\)](#) curve to the observed market yields and compute the root mean squared pricing errors. Grey areas mark the outage periods on Eurex.

B.6 Hedging Mechanisms

This section expands on the results presented in [Section 4.1](#) of the main text.

[Table A16](#) shows that the different hedging strategies across bond market dealers are barely correlated with other important characteristics. Neither the connectedness, the centrality, nor the size of the dealer reliably predicts how much of its cash market duration risk she offsets on the futures market. In a kitchen sink regression, the only statistically significant relationship is that more connected dealers use bond futures slightly more to hedge their inventory risk.

	(1)	(2)	(3)	(4)	(5)
# Connections	1.81 [1.18]				4.25** [2.02]
ln(Centrality)		0.62 [2.65]			-2.06 [22.54]
ln(Yearly # Trades)			1.32 [2.85]		0.13 [6.71]
ln(Yearly Trading Volume)				0.59 [2.55]	-4.57 [22.11]
Observations	32	32	32	32	32
Adjusted R^2	0.042	-0.031	-0.026	-0.031	0.027

Table A16: Relation between Dealers' Hedging Intensity and other Characteristics. The dependent variable is the dealer-level β coefficient from [Equation 2](#), which measures how much of the cash market duration risk the dealer offsets via bond futures. See [Table 7](#) for details. # Connections refers to the number of different dealers each dealer has traded with and Centrality is the volume-weighted eigenvector centrality, based on all trades in 2020. As a proxy for size, we use the number of trades and total trading volume of each dealer in 2020. See [Section 4.2](#) for details.

[Table A17](#) confirms that the results from [Table 8](#) are virtually unchanged when we include

dealer inventory as an additional explanatory variable. For each dealer, we compute the net trading volume in 1-10y German Bunds from the start of the prior trading day until the intraday time of the respective outage. Since the first outage on 14 April 2020 occurred on the Tuesday following the Easter holidays, we compute the net trading volume from the prior Thursday till 9:25 a.m. on Tuesday. For all other outage and control days, the previous day was a trading day. Since the inventory can be positive or negative, we use its inverse hyperbolic sine as a regressor. The results show that dealers with larger inventories cause have lower markups and trade less profitably.

	(1) mispricing	(2) markup	(3) profit
Outage $\times$ Dealer not hedging	0.36 [0.30]	-0.24 [0.40]	-0.54 [2.67]
Outage $\times$ Dealer moderately hedging	0.86*** [0.24]	0.22 [0.26]	2.76*** [0.77]
Outage $\times$ Dealer intensely hedging	0.45* [0.26]	0.29 [0.38]	1.91 [1.94]
Outage $\times$ asinh(Inventory)	0.01 [0.01]	-0.02* [0.01]	-0.11*** [0.04]
FE Minute	✓	✓	✓
FE ISIN	✓	✓	✓
Observations	1920	1920	1920
Adjusted R^2	0.064	0.033	0.034

Table A17: Effect of Eurex Outages on D2C Trades across Dealer Groups. This table replicates [Table 8](#) from the main text, adding dealer inventory as an additional explanatory variable.

B.7 Effect on Swap Market

Interest rate swaps are a key segment of the fixed-income market. These swaps exchange fixed-rate interest payments for floating-rate interest payments over a specified period.³⁹ [Figure A14](#) shows that quotes on €STR interest rate swaps become stale the moment bond futures become unavailable, particularly at longer maturities.⁴⁰

[Figure A15](#) confirms that bid-ask spreads for longer-dated swaps spike during outages, if quotes are available at all.

[Figure A16](#) lastly, confirms these results with executable quotes from an interdealer broker. In particular, Compagnie Financière Tradition, a listed company on the Swiss stock exchange, runs the Trad-X platform. This platform is based on a central limit order book, i.e. immediately executable quotes, in contrast to the indicative quotes on Bloomberg. According to their own statements, Trad-X is the market-leading platform for interest rate derivatives and is used by market participants from over 29 countries. For ease of exposition, we focus on the most liquid instrument on Trad-X, namely 10-year/6-month Euro Interbank Offered Rate (Euribor) swaps, which exchange 10-year fixed-rate interest payments for six-month floating-rate interest payments.

Eurex outages cause a complete evaporation of the order book for these swaps. The number of available order book levels declines and spreads widen immediately. After Eurex goes back online, the order book recovers within half an hour.

³⁹See [Dalla Fontana, Holz auf der Heide, Pelizzon, and Scheicher \(2019\)](#) and [Boudiaf, Frieden, and Scheicher \(2024\)](#) for a detailed anatomy of the euro area interest rate swap market.

⁴⁰Recall that the €STR is closely linked to the ECB's policy rates, and that the ECB provided extensive forward guidance at the time. Between April and July 2020, the €STR rate fluctuated between -.53 and -.56 percent.

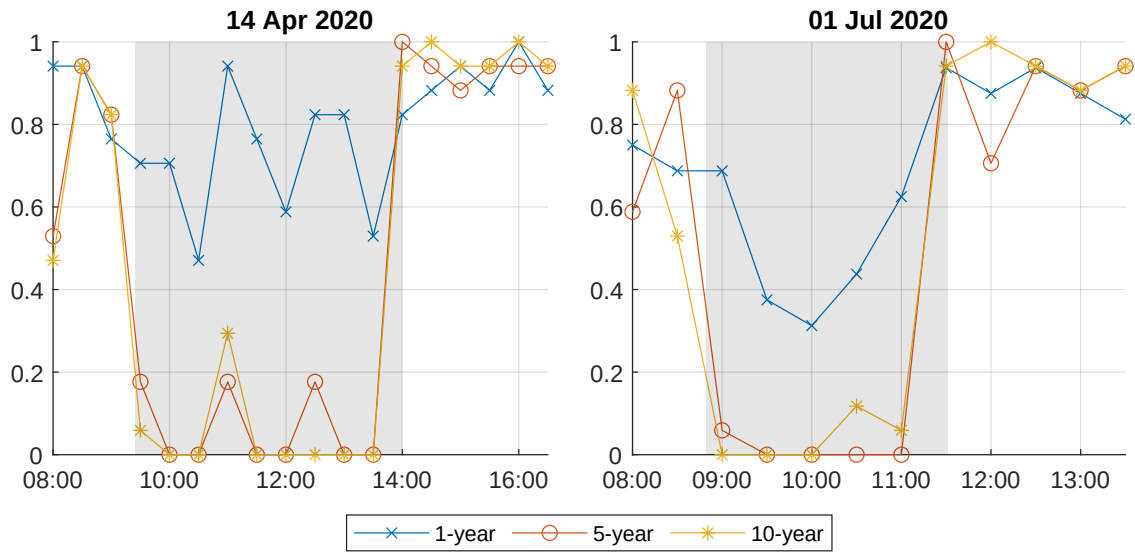


Figure A14: Quote Update Frequency for €STR Swap Rates on Bloomberg. We approximate the number of new quotes as the number of tick-by-tick price changes. To show all series on a single scale, we aggregate the number of quote updates in 30-minute windows and normalize them to a 0-1 range for each maturity. All series refer to overnight index swaps of different maturity based on the euro short-term rate (€STR)

B.8 Transaction Sizes

Table A18 shows that outages also reshape the size of trades. Dealer-to-client trades shrink by 65%, pointing to reduced willingness of dealers to take on inventory when they cannot hedge. Client-to-client trades also contract, though less strongly. On the dealer-to-dealer segment, trade sizes marginally *increase*, consistent with dealers postponing rather than scaling down their own trades. Across venues, the sharpest decline occurs on electronic platforms, the main D2C venue. On systematic internaliser platforms, by contrast, trade sizes hold up, consistent with dealers filling full orders for loyal clients. On regular exchanges, trade sizes also remain stable, consistent with unsophisticated retail investors continuing to execute small orders, likely unaware of the Eurex outage.

	(1) Segment	(2) Venue
Outage × C2C	-0.86*** [0.25]	
Outage × D2C	-1.05*** [0.13]	
Outage × D2D	0.34* [0.19]	
Outage × OTC bilateral		-0.13 [0.36]
Outage × OTC via IDB		-0.69*** [0.13]
Outage × OTC via SI		-0.31 [0.25]
Outage × electronic platforms		-1.02*** [0.17]
Outage × regular exchange		-0.22 [0.30]
FE Time	✓	✓
FE ISIN		✓
Observations	3231	3033
Adjusted R^2	0.264	0.310

Table A18: Effect of Eurex Outages on Transaction Sizes. See Table 4 for details.

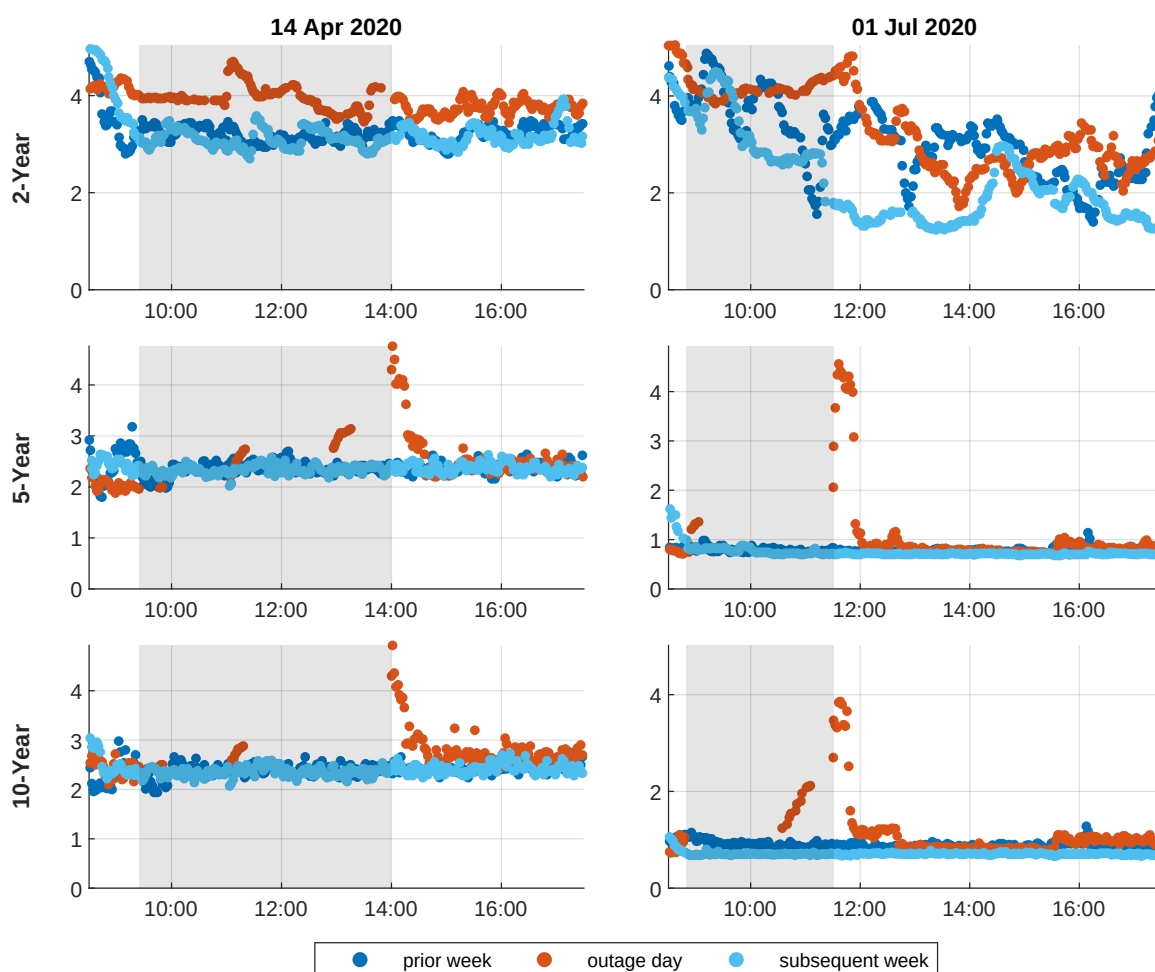


Figure A15: Bid-Ask Spread of €STR Swap Rates on Bloomberg. This figure shows the bid-ask spread (in basis points) for overnight index swaps of different tenors based on the euro short-term rate (€STR). Red lines refer to outage days, dark and light blue lines refer to the same day in the previous and subsequent week.

The consequences are visible in pricing. [Table A19](#) shows that mispricing during outages is disproportionately concentrated in small trades: the outage effect declines with $\log(\text{Volume})$ across segments and venues, implying that the smallest transactions experience the largest deterioration in execution quality. This size-dependence is strongest in the client-facing segments (C2C and D2C) and remains present, though much weaker, in interdealer trading (D2D). Taken together with the documented drop in D2C ticket sizes, this points to a reduced-trade-size channel—clients are pushed into smaller trades, and these smaller trades execute at worse prices—consistent with both constrained dealer capacity and heightened information asymmetries during outages.

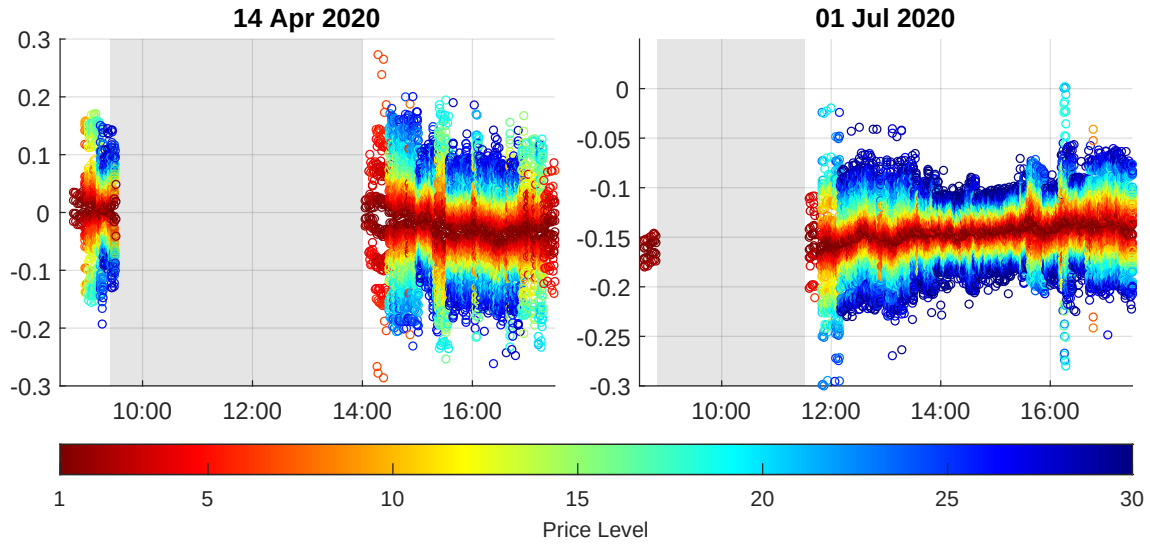


Figure A16: Bid and Ask Quotes for 10y/6m Euribor Swaps on Trad-X. The figure shows bid and ask quotes across different order book levels.

	(1) Segment	(2) Venue
Outage $\times$ C2C $\times$ log(Volume)	-1.15*** [0.20]	
Outage $\times$ D2C $\times$ log(Volume)	-0.49*** [0.18]	
Outage $\times$ D2D $\times$ log(Volume)	-0.17** [0.07]	
Outage $\times$ OTC bilateral $\times$ log(Volume)		-1.00*** [0.24]
Outage $\times$ OTC via IDB $\times$ log(Volume)		-0.41*** [0.10]
Outage $\times$ OTC via SI $\times$ log(Volume)		-0.47*** [0.16]
Outage $\times$ electronic platforms $\times$ log(Volume)		-0.46*** [0.13]
Outage $\times$ regular exchange $\times$ log(Volume)		-1.10*** [0.39]
FE Minute	✓	✓
FE ISIN	✓	✓
Observations	3231	3033
Adjusted R^2	0.239	0.213

Table A19: Reduced-trade-size channel of Eurex Outages on Mispricings. This table replicates [Table 4](#) column (2) and [Table 11](#) column (2), adding another interaction term with the log transaction volume.

B.9 Ruling out alternative explanations

Urgency or Liquidity Demand The venue-level trading volume results in Table 11 in the main text are closely aligned with Menkveld, Yueshen, and Zhu (2017)’s ‘pecking order hypothesis’. According to this hypothesis, investors rank low-cost-low-immediacy venues (like the OTC segment) first and high-cost-high-immediacy venues (like exchanges) last. But when investors’ trading needs become more urgent, they reverse this ranking. This is exactly what we find for cash bonds during Eurex outages. So one might think that a higher urgency to trade, or equivalently a higher demand for liquidity, is driving the results. However, in Menkveld et al. (2017)’s framework, a pure ‘urgency shock’ would push more flow to high-immediacy venues and raise total volume. We see the venue shift but not the volume rise. Outages merely change the composition of investors. Those with a higher urgency to trade withdraw less from the market.

Another way to rule out that higher liquidity demand plays a major role is to look at the comovement of mispricing and dealer inventory. As Goldberg (2020) and Goldberg and Nozawa (2021) explain, higher liquidity demand increases both Hu et al. (2013)’s noise measure as well as dealer’s inventory. Lower liquidity supply, in contrast, increases noise but reduces dealer’s inventory. This latter channel is clearly more in line with our evidence that dealers withdraw from the market, particularly those dealers that usually use bond futures as a hedging instrument.

Market panic Is the dysfunction we observe on the cash bond market simply the result of futures market outages causing a market panic? To rule out this possibility, we first look at other asset classes. Figure A17 shows that stock prices do not crash and the USD/EUR exchange rate barely moves during the outage, at odds with a market-wide panic.

Next, we rule out the possibility that outages cause a ‘flight to safety’ or ‘dash for cash’. In a ‘flight to safety’ event, Bunds of all maturities should appreciate in value, i.e. the entire yield curve should shift downwards, given that Bunds are deemed to have zero default risk (see e.g. Baele, Bekaert, Inghelbrecht, and Wei, 2020). A ‘dash for cash’ scenario predicts the exact opposite: as investors try to sell Bunds in order to get cash, bond prices should fall and the yield curve should shift upward.⁴¹

While the dispersion of transaction yields *around* the curve spikes during outages, Figure A18 shows that the yield curve *itself* barely changes. This null effect is at odds with both a ‘flight to safety’ and ‘dash for cash’ explanation.

Dealer Quotes and New Trade Pairs Table 4 in the main text shows that mispricing spikes most strongly on the client-to-client market, where dealers do not act as intermediaries by taking bonds on inventory. This points to an impaired price formation process. One caveat, however, is that some C2C trades might occur only because dealers stop providing quotes during outages, forcing clients to search for alternative counterparties. While we cannot observe individual quote provision directly, Table A20 provides suggestive evidence: Eurex outages lead to a significant increase in newly formed buyer-seller pairs on the cash market. The probability that a trade is the first between a given pair rises by 2 percentage points in the C2C segment and 9 percentage points in the D2C segment. These numbers might appear small, but when compared to the low baseline probabilities, they imply a 60% and 160% increase in new trade pair formations, respectively. These new connections arise mostly on electronic trading platforms, where the likelihood of a trade being the first between two investors more than doubles during outages. The increase on the OTC market is smaller, unsurprising given higher search frictions and the short duration of the disruptions.

Do these new pairings explain the dislocations we observe? Column (3) suggests not. Mispricing rises sharply during outages regardless of whether trades occur between established partners

⁴¹Eisenbach and Phelan (2025) show in a model how this ‘dash for cash’ effect can outweigh the ‘flight to safety’ effect, rationalizing the temporary spike of Treasury yields in mid-March 2020.

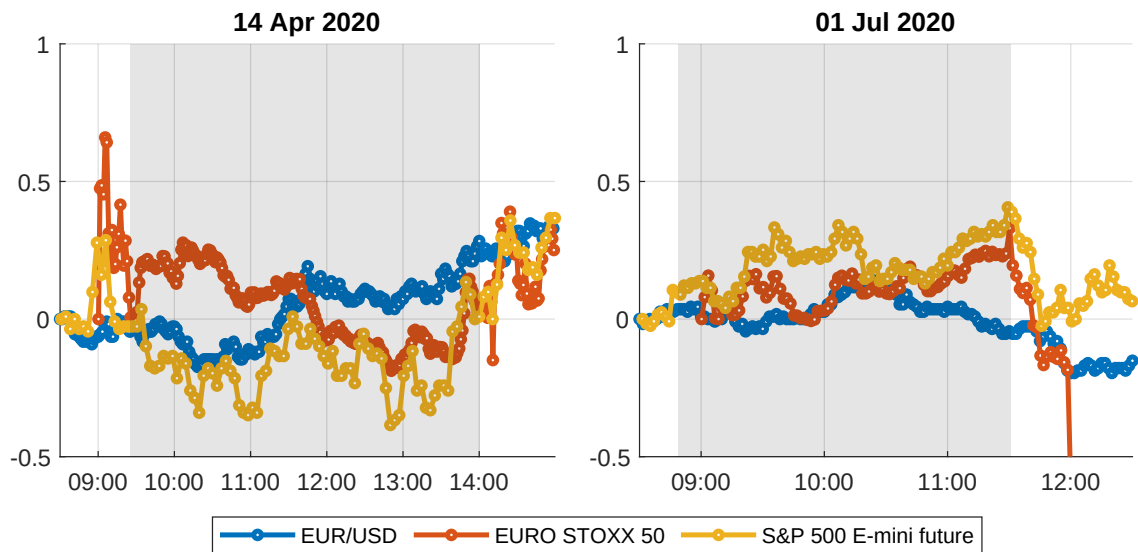


Figure A17: EUR/USD Exchange Rate and Stock Prices During Eurex Outages. All price changes are in percentage points since 08:30 a.m. the S&P 500 future refers to the front-end contract. Data is from Bloomberg. The grey area refers to the outage period on Eurex.

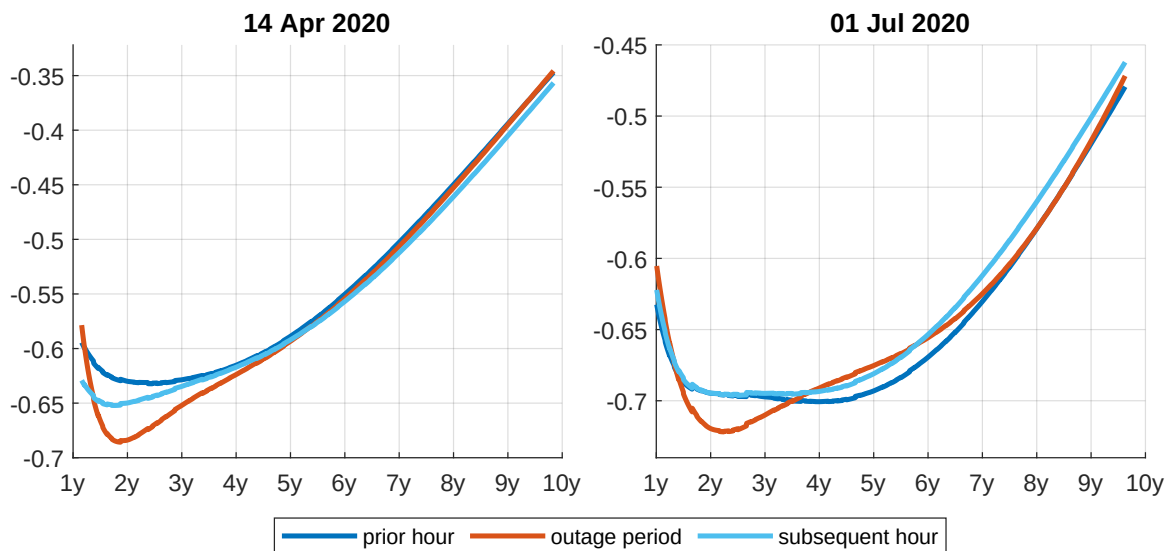


Figure A18: German Yield Curve before, during, and after Eurex Outages. This figure shows yield curves based on the [Svensson \(1994\)](#) method, using all transactions in German sovereign bonds with 1-10y maturity one hour before the outage, during the outage, and one hour after the outage. The fitted yield curves during the outages are identical to the ones shown in [Figure 3](#)

or first-time counterparties. In other words, while outages push clients into new trading relationships, the broader dysfunction in the cash market is not driven by these first-time trades.

	New Buyer-Seller Pair		Mispricing
	(1) Venue	(2) Segment	(3) Venue
Outage $\times$ C2C	0.02 [0.02]		3.40*** [0.61]
Outage $\times$ D2C	0.09*** [0.02]		0.66*** [0.20]
Outage $\times$ OTC bilateral		0.06 [0.04]	
Outage $\times$ OTC via SI		0.03 [0.03]	
Outage $\times$ electronic platforms		0.10*** [0.03]	
Outage $\times$ C2C $\times$ New Pair			-2.20 [1.50]
Outage $\times$ D2C $\times$ New Pair			-0.17 [0.91]
FE Minute	✓	✓	✓
FE ISIN	✓	✓	✓
Observations	2516	2248	2516
Adjusted R^2	0.024	0.033	0.180

Table A20: Effect of Eurex Outages on the Formation of New Buyer-Seller Pairs and their Mispricing. The dependent variable in columns (1)-(2) equals one if a trade is the first between a given investor pair and is zero otherwise (using all Bund trades in 2020). In column (3), the dependent variable is the absolute pricing errors (in basis points of yield), and we add an interaction term between the explanatory variables and the aforementioned dummy indicator. For brevity, the table shows results only for the outage dummy interaction terms. The sample covers all Bund trades, except D2D trades (due to the lack of new buyer-seller pairs) and trades on regular exchanges, as these occur anonymously in a CLOB.

Appendix C Further Evidence from Cash Venue Outages

This appendix complements the analysis in [Section 5](#) by documenting the cash-venue market share assumptions used in the scaling regressions and providing event-level evidence for the three outages in which the disrupted cash venue plausibly accounted for a large share of trading in the underlying segment (Bloomberg for German Bunds, MTS for Italian bonds, and BrokerTec for U.S. Treasuries).

C.1 Cash Venue Market Share Assumptions

[Table A21](#) reports the venue–country market share inputs used to construct MktShare_{vc} in regression (4). In the absence of a consolidated tape for euro area government bonds, venue-level market shares are inherently difficult to measure consistently across time and countries. We therefore treat these values as best-guess proxies for relative venue importance based on our transaction data ([Appendix A.3](#)), public data sources and bilateral discussions with market participants. In the regressions, we convert the reported percentages into fractions.

Also note that our set of historical outages provides limited variation to isolate the role of trading protocols: all but one episode affect venues operating a central limit order book (CLOB), with Bloomberg (RFQ) being the exception. We therefore refrain from drawing strong conclusions about how protocol differences shape the transmission of cash venue outages to the futures market.

Cash venue	DE	FR	IT	ES
MTS	5	20	70	60
Bloomberg	30	20	10	10
FWB (Frankfurt Stock Exchange)	1	0	0	0
BrokerTec	2	2	1	1

Table A21: Assumed cash-venue market shares used to scale cash-venue outages. The [2019 European Commission Report](#) estimates that MTS and Bloomberg each had a 20-30% market share in *electronic* EGB trading, compared to 0-5% for Brokertec and FWB (see Table 3 on p.168 of the report). We verified the MTS country-level shares with our MTS transaction-level dataset ([Table A1](#)). The 5% MTS share for Germany is also roughly in line with [de Roure et al. \(2025\)](#), who report a 3.7% market share, after applying some filter to the MiFID I dataset.

C.2 Event-Level Evidence for High-Share Outages

Bloomberg Outage (Bunds). On 17 April 2015, Bloomberg terminals experienced a widespread outage beginning around 8:20 a.m. London time and lasting at least until 9:10 a.m. with some newspaper reports suggesting a longer duration, probably because Bloomberg terminals came back online gradually. In any case, traders worldwide were not able to access their terminals temporarily. Since Bloomberg is a major trading platform for German Bunds, one might expect large effects of this outage on the four Bund futures. [Figure A19](#) instead documents a null effect. Neither the Amihud nor the Roll illiquidity proxy spike, suggesting that market functioning in bond futures is not impaired. This is also in line with [Figure A20](#), which shows the country-level liquidity on MTS during the Bloomberg outage. Liquidity is slightly depressed in Italian bonds but not in German, French or Spanish bonds.

MTS Outages (Italy). On January 12, 2010, MTS experienced an outage early in the trading day, with the first order book quotes appearing only at 10:35 a.m. instead of 8:00 a.m. Despite MTS being the dominant cash venue for Italian sovereign bonds, this outage has no obvious effect on the two illiquidity proxies for the 10-year FBTP future, the only available Italian bond future at that time, see [Figure A21](#).

The unconfirmed but suspected outage on 26 July 2019 (see [Figure A27](#)) yields the same conclusion. Since this event took place quite recently, we can directly observe the order book liquidity on the futures market and do not have to resort to illiquidity proxies. [Figure A22](#) shows that there is no discernible drop in market liquidity for the 10-year bond future for any country when MTS turns inactive. Similarly, indicative quotes on Bloomberg do not show any obvious reaction to the suspected MTS outage, see [Figure A23](#). Both the frequency of quote updates as well as bid-ask spreads stay virtually constant throughout the outage.

BrokerTec Outage (U.S. Treasuries). Finally, we examine the outage on BrokerTec, the dominant electronic interdealer platform for U.S. Treasuries (see [Fleming, Nguyen, and Ruela, 2024](#)). On 11 January 2019, BrokerTec experienced an outage from 1:43 p.m. to 3:35 p.m. U.S. Eastern Time. Because this event occurred late in the European day, this outage has no effect on the euro area *cash* market, which was effectively inactive at that time. Even so, [Table 12](#) shows a reduction in euro area futures trading activity during the outage, while market functioning remains largely unchanged. Since this outage also took place quite recently, we can again directly observe the liquidity on the futures market. [Figure A24](#) plots the order book depth in 10-year German Bund futures and 10-year U.S. Treasury futures. Bund futures depth appears unaffected. In U.S. Treasury futures, depth slightly *increases* during the outage and reverts thereafter, consistent with the notion that disruptions can temporarily weaken latency arbitrage between cash Treasuries and Treasury futures, similar in spirit to the weather-based identification in [Shkilko and Sokolov \(2020\)](#). Overall, this episode reinforces our conclusion that cash market disruptions have limited effects on futures market functioning, and that the asymmetry between futures-to-cash and cash-to-futures transmission is not unique to the euro area.

[Figure A24](#) shows that the order book depth in German Bund futures seems unaffected by the outage on Brokertec. For US Treasury futures, liquidity slightly *improves* during the outage and falls back to normal once Brokertec goes back online. This seemingly counterintuitive effect suggests that outages reduce latency arbitrage between Treasuries and Treasury futures, similar to the effect of adverse weather between NYC and Chicago on the equity market. In that setting, [Shkilko and Sokolov \(2020\)](#) show that adverse weather interrupts the early access to futures price movements in Chicago for some high-frequency trading firms, which improves liquidity for cash equities in NYC. The overall benign effects of the Brokertec outage suggest that one of our main findings — that price formation and liquidity provision is more of a one-way street from the futures to the cash market than previously thought — also applies to the US.

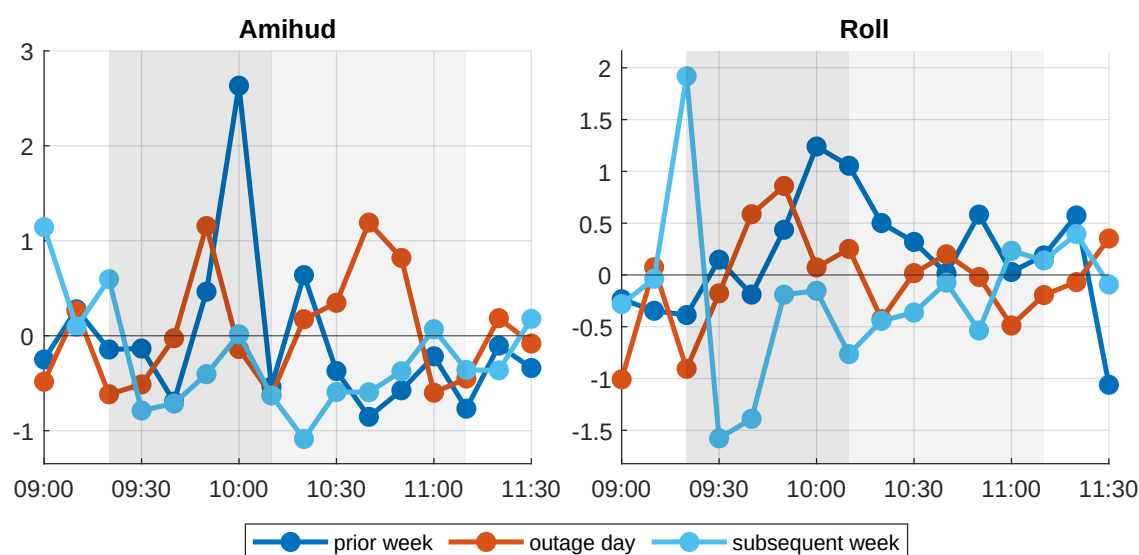


Figure A19: Estimated Illiquidity of German Bund Futures during Bloomberg Outage on 17 April 2015. This figure shows the illiquidity proxies of Amihud (2002) and Roll (1984) for German Bund futures. The proxies are computed separately for each of the four futures at 10-minute frequency, then normalized to mean zero and unit variance, and finally aggregated to a weighted mean, using the number of traded contracts. Red dots refer to the outage day on 17 April 2015, dark and light blue dots refer to the same day in the previous and subsequent week. The dark shaded area refers to the confirmed outage period, the light shaded covers an additional hour as trading resumed gradually.

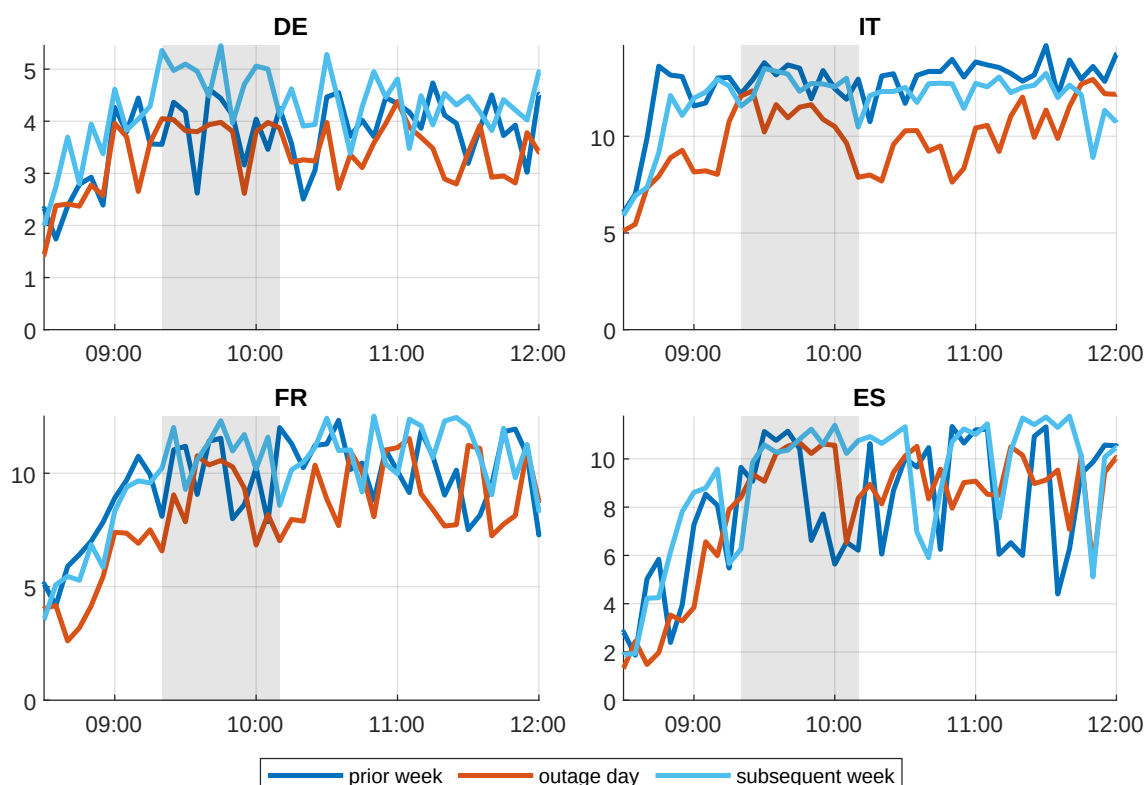


Figure A20: MTS Order Book Depth during Bloomberg Outage on 17 April 2015. This figure shows the total order book depth for German, French, Italian and Spanish sovereign bonds on MTS, at 5-minute snapshots (in billion €). Red lines refer to the outage day on 17 April 2015, dark and light blue lines refer to the same day in the previous and subsequent week.

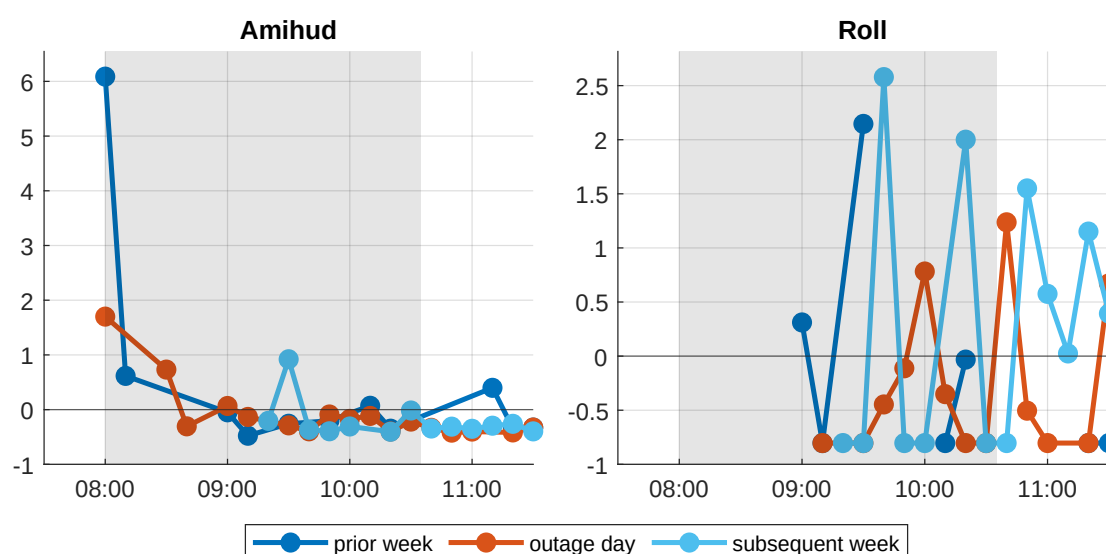


Figure A21: Estimated Illiquidity of Italian Bond Futures during MTS Outage on 12 January 2010. This figure shows the illiquidity proxies of Amihud (2002) and Roll (1984) for the 10-year FBTP future, the only available Italian bond future at that time. The proxy is computed at 10-minute frequency and normalized to mean zero and unit variance. Red dots refer to the outage day on 12 January 2010, dark and light blue dots refer to the same day in the previous and subsequent week. The dark shaded area refers to the confirmed outage period.

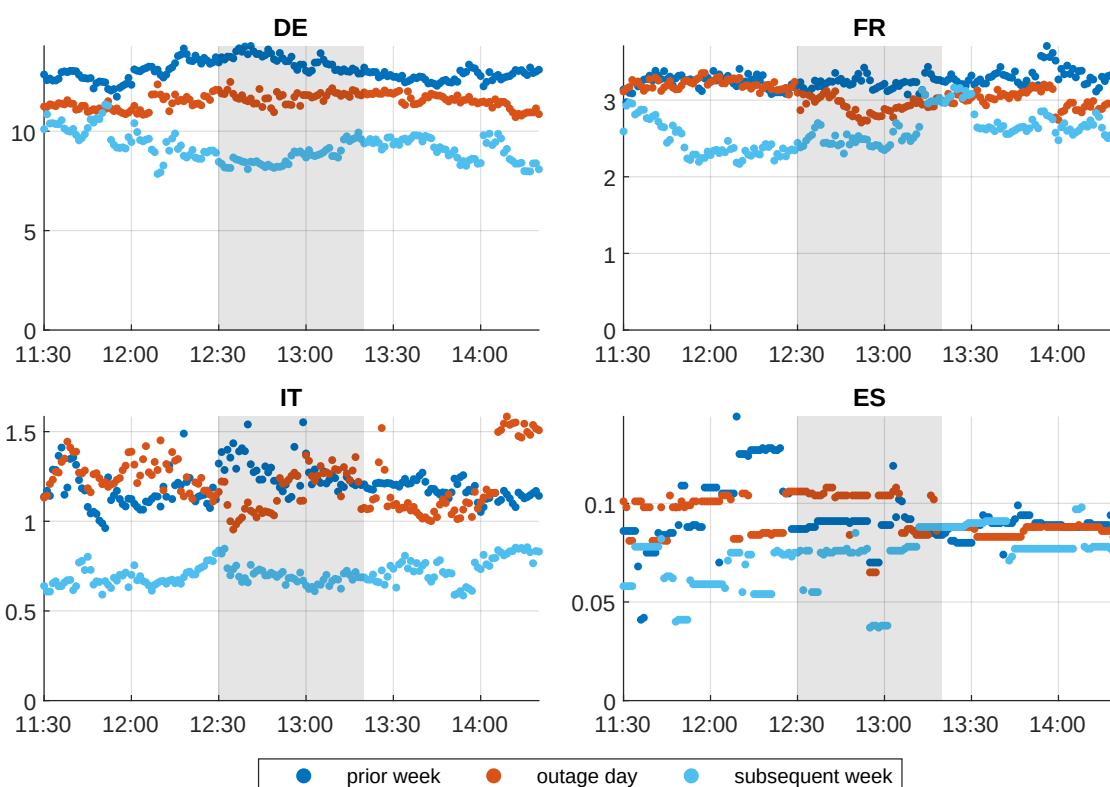


Figure A22: Order Book Depth of 10-year Bond Futures during suspected MTS Outage on 26 July 2019. This figure shows the total number of contracts (in thousands) quoted at the first fifteen levels on both sides of the order book, at minutely snapshots. Red dots refer to the suspected MTS outage day on 26 July 2019, dark and light blue dots refer to the same day in the previous and subsequent week.

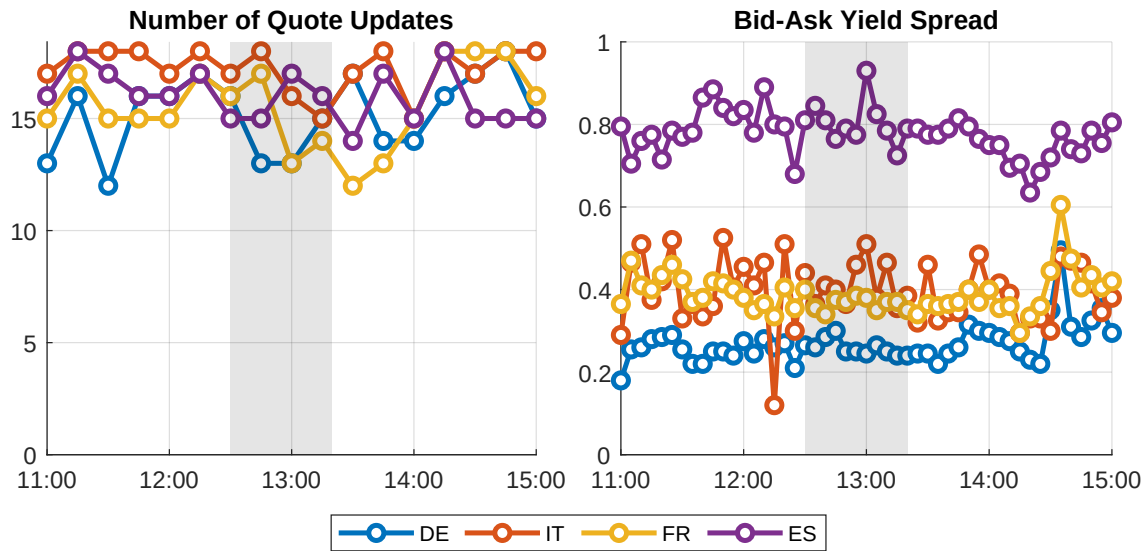


Figure A23: Quote Update Frequency and Bid-Ask Spreads on Bloomberg during suspected MTS Outage on 26 July 2019. The left panel shows number of quote updates, the right panel shows bid-ask spreads (in basis points of yield). All data refers to 10-year cheapest-to-deliver bonds. The grey areas mark the times of the suspected MTS outage.

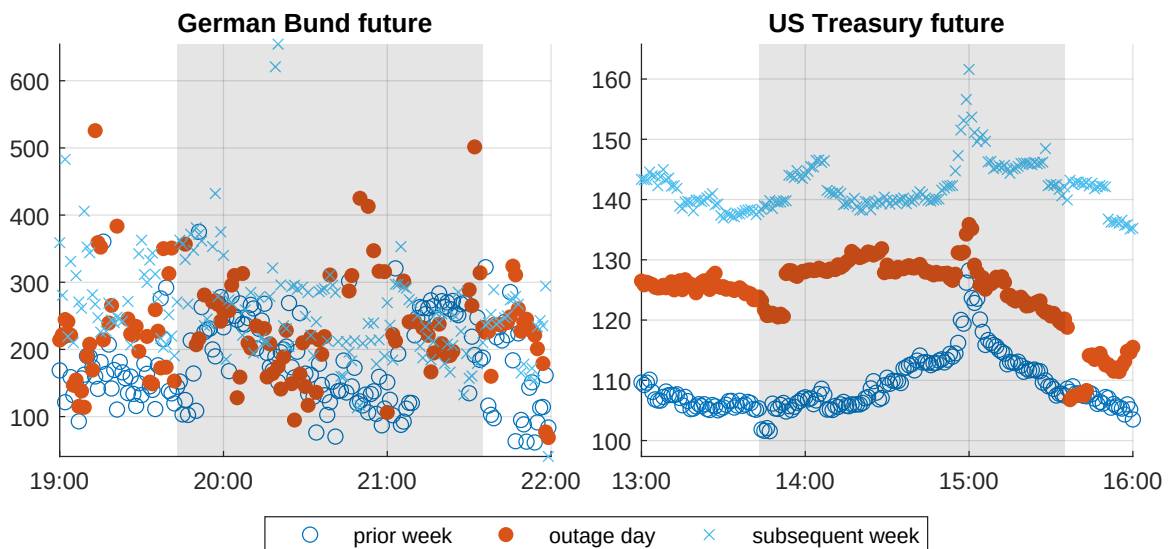


Figure A24: Order Book Depth of Bond Futures during Brokertec Outage on 11 January 2019. This figure shows the number of quoted contracts in 10-year German Bund futures on Eurex (left panel; covering both sides of the first three levels of the order book; data comes from Bloomberg, since our Eurex order book data starts in April 2019) and 10-year US Treasury futures on CME (right panel; in thousands; covering both sides of the first 15 levels of the order book). The grey area refers to the outage period on the Brokertec cash trading platform. Red markers refer to the outage day, dark and light blue markers refer to the same day in the previous and subsequent week. Timestamps are in local time.

Appendix D Overview of Market Outages and Narrative Evidence

This section provides an overview of the outages we study. Appendix D.1 lists all outages and affected assets, Appendix D.2 recounts newspaper reports on all Eurex outages and Appendix D.3 on all cash venue outages. Many articles are from the [Factiva news archive](#), which is how we identified some of the outages in the first place. Appendix D.4, lastly, summarizes market participants' views on the effect of outages and the role of different trading venues on the euro area bond market.

D.1 Outage List

Table A22 contains a list of the market outages we study. Figure A25 shows the exact intraday times of each system-wide Eurex outage. We focus on outages since 2008 because we do not have cash market data prior to that. Figure A26 confirms the two partial outages on Eurex. The figure shows that only 5-year and 10-year German bond futures were unaffected by the outages on 26 May 2014 and 21 November 2016. On the first of these days, there were two separate outages. Figure A27 confirms the known outage of the MTS platform on 12 January 2010 and shows another suspected outage on 26 July 2019.

Exchange/Platform	Affected fixed-income assets	Dates
Eurex	European Government Bond (EGB) futures	1 July 2020
		14 April 2020
		16 March 2018
		13 December 2017
		22 February 2016
		20 July 2015
		17 February 2015
		26 August 2013
		11 October 2011
		23 December 2009
		18 November 2009
		4 February 2009
	all EGB futures except 5/10-year German Bund futures	21 November 2016 26 May 2014
MTS (unconfirmed)	cash bonds	26 July 2019
Brokertec		11 January 2019
Frankfurt stock exchange (FWB)		27 May 2015
Bloomberg		17 April 2015
MTS		12 January 2010

Table A22: Overview of Market Outages Studied in This Paper.

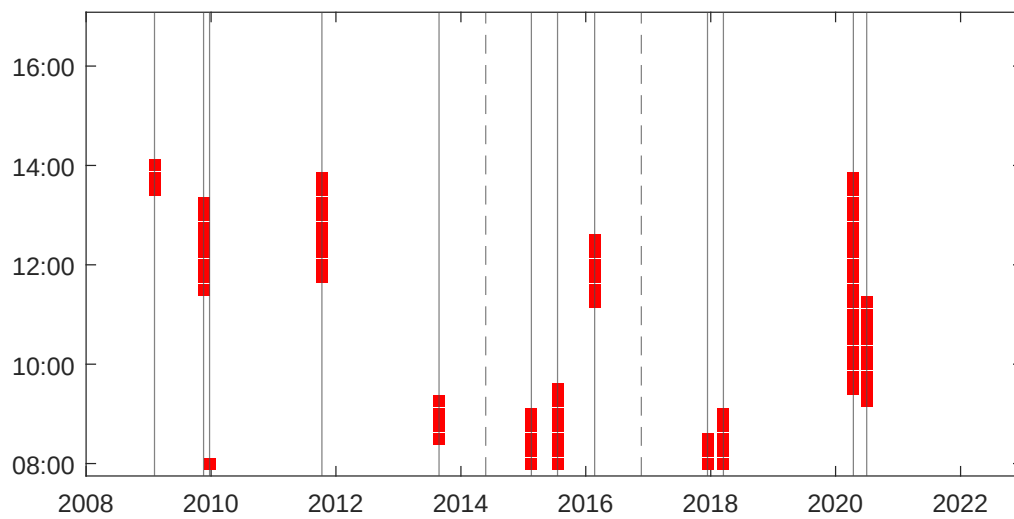


Figure A25: System-Wide Eurex Outages since 2008. For each trading day since 2008, red squares mark 15-minute intraday windows with no trading in the German 10-year bond future (FGBL), which line up perfectly with the twelve known outage days, marked by solid vertical lines. The two dashed lines mark days with partial outages, see [Figure A26](#). The vertical axis ranges from 7:45 a.m. to 5:00 p.m. local time.

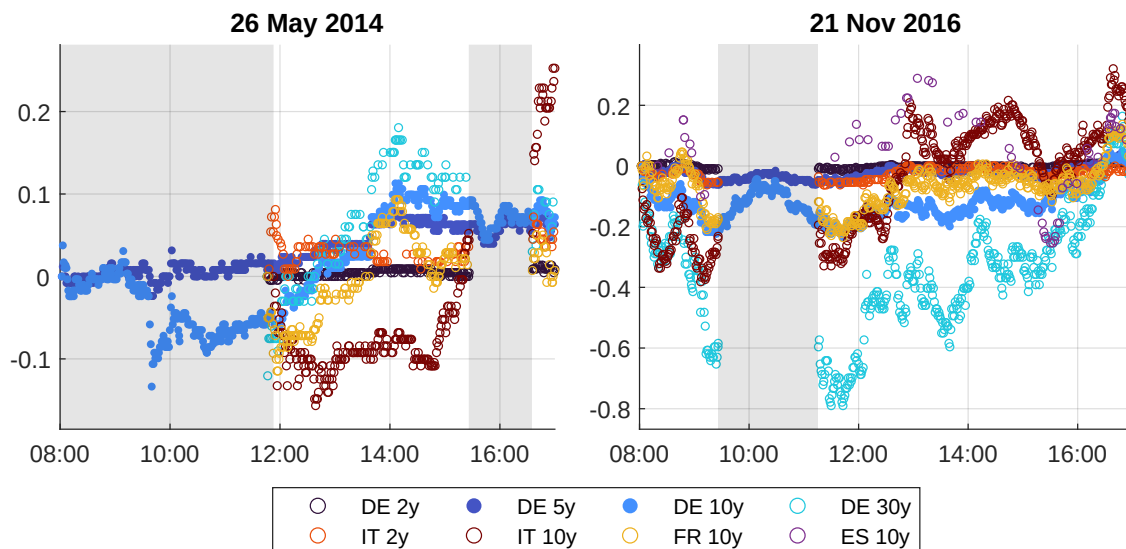
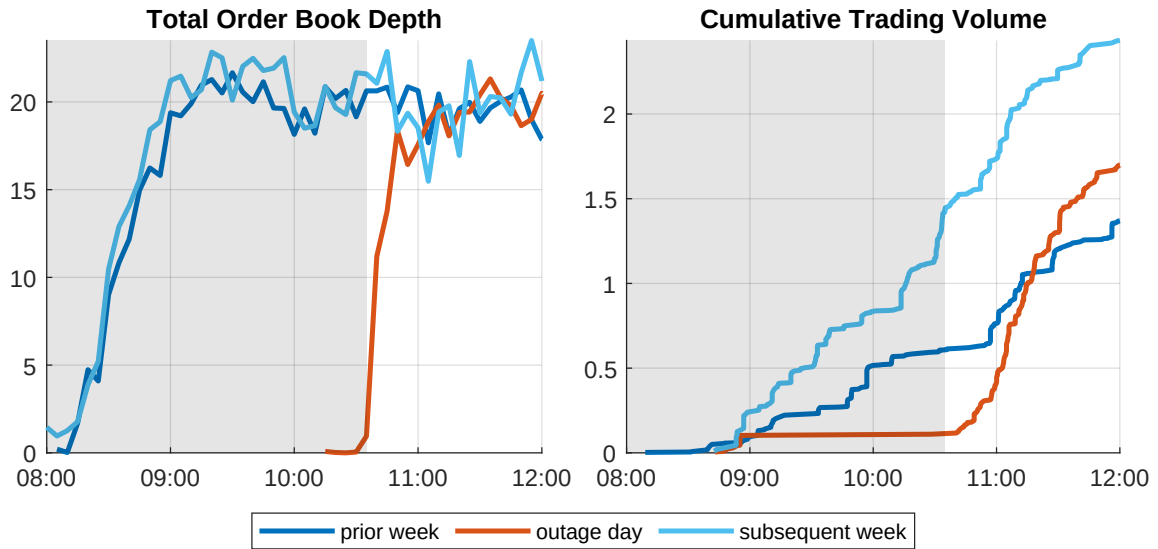


Figure A26: Partial Outages on Eurex. This figure shows intraday transaction prices of all euro area government bond futures (in percentage changes since 8:00 a.m. at minutely frequency). The grey areas mark the outage times that affected all futures except those for 5-year and 10-year German bonds. The Spanish 10-year future was introduced in October 2015, i.e. after the 2014 outage.

(a) Confirmed MTS Outage on 12 January 2010



(b) Suspected MTS Outage on 26 July 2019

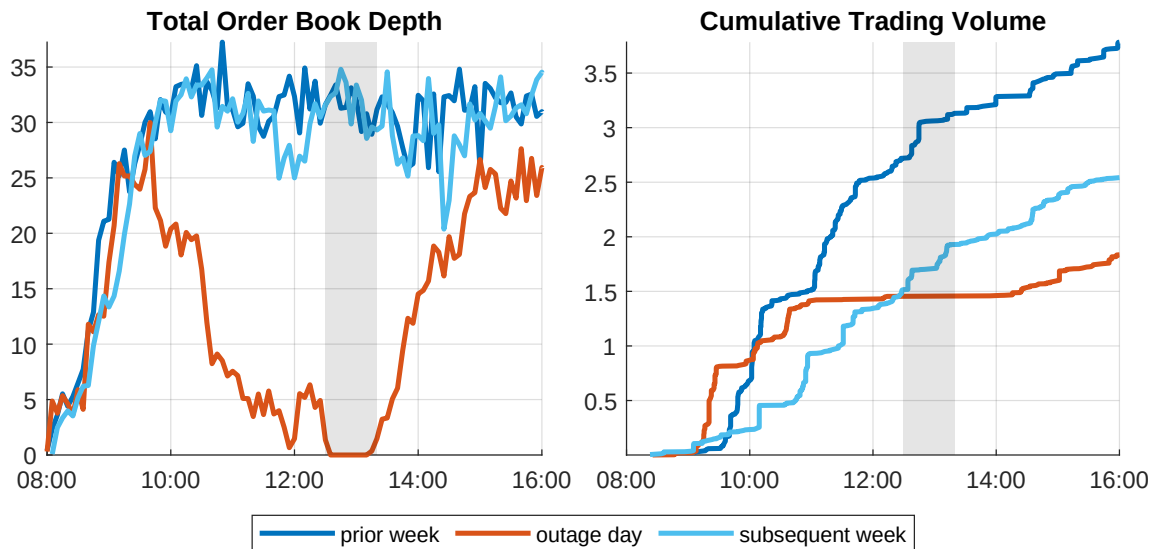


Figure A27: Outages on MTS. The left panels show the total quoted volume (in billion €), across all German, French, Italian and Spanish sovereign bonds and all market segments, in 5-minute intervals. The right panels show the total cumulative trading volume (in billion €). Red lines refer to the (suspected) outage day, dark and light blue lines refer to the same day in the previous and subsequent week. Grey areas mark the (suspected) outage times.

D.2 News Reports on Eurex Outages

Table A23 provides a timeline of the two Eurex outages in 2020 and related news.

Table A23: Timeline of Eurex outages in 2020 and same-day news.

14 April 2020	
9:25	outage start
10:32 @Eurex	Due to technical problems, the Eurex T7 system is not available at the moment. We are investigating and will keep you informed.
11:47 DowJones	Trading on Deutsche Boerse AG's Xetra is currently suspended due to a technical fault, a spokesman for the German stock exchange operator said on Tuesday. The spokesman said to Dow Jones Newswires that he couldn't yet comment on when the trading would resume and communication related to trading resumption would be released on the Xetra Newsboard.
13:19 @Eurex	Trading will resume according to the following schedule: 13:15 CEST Pre-Trading Instrument State BOOK, 13:45 CEST Trading Instrument State OPENING AUCTION, 13:50 CEST Trading Instrument State CONTINUOUS.
14:00	outage end
17:51 Reuters	The outage was caused by a malfunction in the internal communication of the trading system, Deutsche Boerse said in a statement, adding that trading was operating smoothly again on Tuesday afternoon. A Deutsche Boerse spokesman said the outage was not due to a hacker attack.
1 July 2020	
8:49	outage start
9:39 Reuters	Frankfurt-based electronic trading system Xetra was experiencing a 'technical issue', affecting all securities traded on the platform, a Deutsche Boerse spokesman said on Wednesday. 'I just can confirm that there is a technical issue on Xetra... we're currently investigating the failure,' Patrick Kalbhenn, a spokesman for the German stock exchange, told Reuters.
10:11 @Eurex	Due to technical problems the trading system T7 is not currently available.
11:54 @Eurex	Update on the disruption: continuous trading on Eurex resumed at 1130.
11:31	outage end
17:25 Bloomberg	The disruption in the T7 system in April and today's failure had the same origin. They were due to faulty third-party software that is part of the trading system. We now understand the exact cause and have eliminated the issue. The system is now running stably and we expect it to remain so. External causes can be ruled out.

An [ESMA report](#) contains the following paragraphs about the Eurex outages in 2020: *'The first two incidents reported related to an issue with the Deutsche Boerse T7 trading system. The first was reported on 14 April and trading was interrupted in the trading venue due to a software issue. This issue required the trading venue to stop and restart manually the system which was a heavy and time-consuming process. The second incident was reported on 1 July and due to a human error. Two failures of the trading venues' central network occurred which caused trading to be halted in Deutsche Boerse. In both circumstances the incident affected a significant number of trading venues given that the T7 trading system is widely used across the EU.'*

What follows are newspaper reports for all previous outages on Eurex, in chronological order:

[6 February 2004](#): Computer glitch caused Xetra failure

Deutsche Boerse AG said on Friday a computer failure led to a nearly hour-long outage on its Xetra electronic trading platform. The outage was the fourth on Xetra, which accounts for 90 percent of trade in German stocks, since the electronic trading platform was introduced six years ago. Pre-trading resumed at 1045 GMT after a 45-minute trading halt while full trading resumed at 1130 GMT, Deutsche Boerse said in a statement. The stock market operator said trade in some stocks had been delayed even longer with dealing halted from 0900 GMT, caused by a failure in a host database. Traders said the Xetra failure caused some concern. 'It was a little bit hectic when it first began but luckily there was not that much important company news during the trading halt,' said one trader.

19 November 2007: Rare glitch hits Frankfurt stock exchange

Trading in German stock exchange operator Deutsche Boerse's electronic order-matching system Xetra was interrupted early on Monday by technical problems, which traders and the bourse said were rare. The impact of the disruption of just over one hour was minor because early volume had been 'very low,' said one Frankfurt-based stock broker, who could not recall more than two or three short Xetra disruptions in the past two years. 'Normally they (Deutsche Boerse) say the system is running nearly 100 percent,' he said, adding from his own experience. 'If it's down two times in one year, it's a lot.' Xetra trading was halted around half an hour after the start at 0800 GMT and resumed at 0940 GMT. During the interruption, Deutsche Boerse said it was investigating 'technical problems' in the system, which handled 16 million transactions in October, up 74 percent year-on-year. A Deutsche Boerse spokesman said after the restart of trading that he did not yet have information about the exact nature of the problem.

4 February 2009: Derivatives exchange Eurex resumes trading

Derivatives exchange Eurex resumed trading on Wednesday after a one-hour shutdown that traders took with a shrug. Eurex said a technical glitch halted trading between 1218 GMT and 1315 GMT on what is the main market for Europe's most traded fixed income futures and stock index options and futures. Traders said such an event was rare. A Eurex spokesman said the last time something similar had occurred was in 2006. '(For us) there was no effect, no losses as a result of this Eurex downtime,' said Sebastian Qureshi, head of German hedge fund manager Varengold, which specialises in managed futures strategies and trades also on Eurex. 'One hour is too short, it's not really affecting our business,' he said. 'We just went out to get a coffee,' a Frankfurt-based Eurex market-maker added.

18 November 2009: Eurex trade suspended until further notice

Trade on the Eurex derivatives exchange has been suspended since 1100 GMT due to technical problems, a Eurex spokesman said on Wednesday, adding it was not foreseeable when trade would resume. 'We are working hard to solve the problem,' the spokesman said. Eurex is jointly operated by Deutsche Boerse and SIX Swiss Exchange.

23 December 2009: Eurex Exchange says Wednesday opening delayed.

11 October 2011: Eurex trade suspended until further notice

Deutsche Boerse AG interrupted trading on its electronic derivatives platform Eurex on Tuesday to investigate technical problems. Trading on Eurex had been interrupted 'to avoid any threat to the functioning of Exchange trading', the stock exchange operator said in a statement. Earlier, Deutsche Boerse said it had been experiencing technical problems. 'We are investigating and will keep you informed,' Deutsche Boerse said in an e-mailed statement.

26 August 2013: Eurex Restarts Trading After Market Halt

German exchange operator Deutsche Boerse AG's (DB1.XE, DBOEF) Eurex Exchange arm re-

opened trading early Monday after a brief halt to certain trading earlier in the session. At roughly 0642 GMT, Eurex posted a note on its website saying that it had halted trading on its New Trading Architecture platform—a new electronic trading platform that was introduced by the firm in December—in order to avoid a threat to the functioning of exchange trading. The halted trading announcement came roughly 12 minutes after an earlier posting said there were technical problems at the exchange. Trading in all Eurex products was restarted at roughly 0720 GMT, according to the company’s website.

26 May 2014 (partial outage): Btp and Oat futures, delayed opening due to technical problems

Italian, French bond futures trading delayed, Eurex recorded message citing technical problems at exchange

17 February 2015: Eurex Restarts Trading After Market Halt

There is no trading currently on any Eurex futures or option products due to technical issues, according to one London-based trader. According to one London-based trader, the earliest start of opening Auctions on Eurex will be 0915 CET or 0815 GMT, while netting will start as of 0920 CET or 0820 GMT. Adds these times ‘not set in stone though.’ Eurex spokesman not immediately available to comment.

20 July 2015: Eurex Restarts Trading After Market Halt

Europe’s largest derivatives market, Eurex, suffered technical issues on Monday that delayed trading of all futures and options contracts and took two hours to fully resolve. The outage caused little disruption to broader cash equity and bond markets, however, unlike April’s Bloomberg terminal outage, which delayed debt sales and exacerbated a spike in volatility. Traders said the outage had effectively choked off liquidity in the derivatives market, with only over-the-counter deals available. But receding fears over Greece and recent declines in volatility meant it had less impact than it might have. ‘This type of outage is usually significant, but because of the broader environment things were much calmer,’ said a London-based equity derivatives trader. ‘It was a minor event in the end, but it could have been a major one had it hit a few weeks back.’ Frankfurt-based Eurex said complete trading had resumed at 0810 GMT. Index futures trading usually begins around 0600 GMT. A bond trader based in Frankfurt said the mood was ‘quite relaxed’. Another London-based trader said there had not been a big impact.

22 February 2016: Eurex says continuous trading to resume at 1150 GMT

Deutsche Boerse’s Eurex says pre-trading to start at 1125 GMT, continuous trading from 1150 GMT.

21 November 2016 (partial outage):

«For this outage, the only real-time confirmation we could find is on [Twitter](#).»

16 March 2018: Eurex hit by technical issues, bond and stock futures trading delayed

Many key European bond and stocks futures, including German Bund futures and DAX futures, did not open for trading on Friday as the Eurex trading system was hit by technical issues. German Bund futures, which allow investors to hedge against German government bonds, Italian BTP futures and French OAT futures were all down. Many stock futures were also down, including were Eurostoxx futures and Dax futures ‘There are some technical problems for the T7 system which has caused some delays. It’s under investigation currently and we will have updates on our production newsboard. As of now I have no further details on when it will be resolved,’ said a Eurex representative.

As a result, trading in government bonds is extremely thin, most likely because investors are

unable to hedge their investments, DZ Bank strategist Daniel Lenz said.

D.3 News Reports on Cash Venue Outages

Bloomberg Outage on 17 April 2015

[Yahoo](#): In the City of London [...] and financial centres around the globe traders and analysts were left gnashing their teeth as their Bloomberg computer terminals crashed on Friday morning, with fixed income markets suffering an especially large impact. Bloomberg [...] confirmed that its terminal system was unavailable worldwide, with the outage first cited at around 08:20 in London. [...] by 09:10 London time the company said that some customers had reported the terminal was back online.

[The Globe and Mail](#): Bloomberg's trading terminals, which are used by most of the world's biggest financial firms, went down for two-and-a-half hours on Friday due to apparent technical problems.

[UK Debt Management Office](#): Due to ongoing technical issues with the third party platform supplier [...] this morning's UK Treasury Bill tenders will be postponed.⁴²

Brokertec Outage on 11 January 2019

[Reuters](#): BrokerTec, an interdealer broker of U.S. government debt, planned to reopen its U.S. market platform at 3:35 p.m. (2035 GMT) following an outage earlier Friday, a company spokeswoman said. 'All trading participants should be able to reconnect to the system,' the spokeswoman said in an email statement.

[Bloomberg](#): Treasuries Hit by One-Hour Outage on Biggest Electronic Platform. BrokerTec, the biggest electronic trading platform for Treasury securities, shut down for more than an hour on Friday because of a technical malfunction, an outage that several traders said caused a market-wide slowdown in one of the world's biggest assets.

[Fed New York Treasury Market Practices Group meeting minutes](#): Members [...] noted that the timing and nature of this particular outage had a relatively limited impact given that [...] the outage occurred during a period of low trading activity.

Frankfurt stock exchange (FWB) Outage on 27 May 2015

[Reuters](#): Deutsche Boerse said floor trading on the Frankfurt Stock Exchange on Wednesday would not resume before 11 Central European Time, or 0900 GMT. Trading on the Xetra electronic exchange remained intact.

[Die Welt](#): On Wednesday a technical error brought the floor trading at the Frankfurt Stock Exchange to a halt all morning. Only around 1300 CET the trading began running smoothly again. Frankfurt Stock Exchange's electronic securities trading system Xetra, on which more than 90% of trade is handled, was not affected by the breakdown. Overnight there were problems in the data processing, a spokesman for stock exchange operator Deutsche Boerse said. Such a long breakdown is the absolute exception, according to the spokesman.

⁴²The UK Debt Management Office runs its auctions on the Bloomberg Auction System (see [Guide to DMO Primary Dealers](#)). In this sense, the fact that the auction was postponed is no *prima facie* that UK officials necessarily feared an impaired price discovery process (as we argue for the cancelled auction of Dutch bonds during the Eurex outage on 14 April 2020).

MTS Outage on 12 January 2010

[Dow Jones](#): ‘MTS Group can confirm that the cash market facility experienced an outage at the beginning of the trading day,’ the company said in a statement. ‘The issue was resolved and normal market conditions resumed. The outage caused no unverified transactions.’ The problem resulted in users of MTS’ system being unable to log in. It took place roughly between 0830 GMT and 0930 GMT after which the system was up and running again, Patrick Humphris, a spokesman for LSE, said.

D.4 Narrative Evidence

Appendix [D.4.1](#) contains selected quotes from market participants regarding the effect of outages and the role of different trading venues in euro area government bond markets. Appendix [D.4.1](#) cites from an internal memo of one of the major dealers in this market. Overall, this narrative evidence is in line with our empirical result that proper market functioning for EGBs vitally depends on an active futures market.

D.4.1 ESMA Consultations

The European Securities and Markets Authority (ESMA) has published two consultation papers that contain valuable information about euro area bond market functioning. The below sections summarize the most informative responses by market participants.

Market Outages The below quotes refer to investor responses to the [ESMA Consultation Paper on Market Outages](#). The final report was published on 24 May 2023. The report’s focus is on stocks and the substitutability among stock exchanges, but some questions also concern the fixed-income market, e.g. question 12: ‘*Is there any particular issue relating to trading of non-equity instruments that should be taken into account in the case of an outage? Where possible please differentiate between bonds and derivatives.*’ The below quotes refer to investor responses to this question. We emphasize particularly informative bits in **bold font**.

[Electronic Debt Markets Association](#) (EDMA, 6 members are: BGC Fenics, Bloomberg, BrokerTec, MarketAxess, MTS and Tradeweb): ‘non-equity trading on EU trading venues is **less affected by outages, including fixed income markets, with trading more naturally moving to alternative platforms** [...] Given that end users have a plethora of alternative trading venues available to them trading the same financial securities, there is already appropriate (commercial) pressure on trading venues to reopen as quickly as it is safe to do so.’

[Euronext](#): ‘in the fixed income market, it is **quite fragmented and trading generally moves to other venues when there are outages** in any case so there is less of an impact.’

[Federation of European Securities Exchanges](#): ‘In particular for fixed income market, it is important to consider the different structure of the market, where there is **not the same reliance on the primary market as is the case for equity, and trading is distributed more widely across several trading venues and systematic internalisers, so there is less of an impact in case of an outage** of a single trading venue’

[The Investment Association](#): ‘From a fixed income, bonds perspective, we echo the observation outlined by ESMA [...] that the trading of bonds is less affected by an outage regardless of the type of trading venue the outage occurs on, as trading does more naturally gravitate towards an alternative platform. Partly this is due to the vast differences in the trading landscape of equity and bond instruments.’ ‘Furthermore, we recommend that ESMA consider the **impact of**

market outages on futures exchange markets [...] and the trading of bond futures as an outage would most likely have significant implications on the liquidity portfolio of many government securities’

Algorithmic Trading The below quotes refer to investor responses to the [ESMA Consultation Paper on Algorithmic Trading](#). While the report’s focus is on stocks, some questions are informative also for the fixed-income market, e.g. question 36 on market outages: *‘Do you believe any initiative should be put forward to ensure there is more continuity on trading in case of an outage on the main market, e.g. by requiring algo traders to use more than one reference data point?’* The below quotes refer to investor responses to this question. We emphasize particularly informative bits in **bold font**.

Deutsche Boerse: ‘Regarding, initiatives aiming at continuity of trading in case of an IT incident/outage, DBG does not believe that such an initiative should be put forward given a close to 100% system performance of main markets. We would caution against forcing algorithms to include different sources of information. **The underlying assumption seems to be that regulated markets, MTFs (multilateral trading facilities) and potentially SIs are set on the same level in terms of price formation and information, with easy switch from one to the other, putting aside respective market shares and the notion of reference market.** The explored initiative would hence introduce an artificial change to the current market structure which is at odd with MiFID. To the contrary, the flight to execution quality at the height of volatility in the COVID-19 crisis proved once more that there was a need by investors to trade on transparent regulated markets when looking at the migration of volumes from dark, SI, and OTC trading to regulated markets. Last but not least, **it should be up to the trading participants to decide if they see merit in connecting to more than one reference data point or not**, but they should not be forced upon by regulation.’

Federation of European Securities Exchanges: ‘FESE considers that **declines in trading following outages are linked to the importance of price formation.** Despite the ability to trade on alternative venues, the **low confidence of traders in the price formation on alternative venues may deter them from trading on those markets** during the outage period.’

German Banking Industry Committee: ‘**we strongly oppose the notion that outages might be compensated by obliging intermediaries, i.e. investment firms, to connect to more than one trading venue so that in the event of an outage, trading can continue seamlessly on another venue.** This would mean market participants would have to maintain double memberships in all relevant main markets. This proposal is completely unproportionate, it might also lead to liquidity reduction in those main markets and will increase costs for intermediaries and clients alike. Neither do we see how the notion to require algo traders to use more than one reference data point might solve the real problem, nor do alternative markets exist for all financial instruments.’

Virtu: ‘Virtu believes that it is in the best interests of the market and orderly trading for the appropriate amount of time to be taken to properly resolve an incident and to restart afterward, instead of forcing trading venues to adhere to an arbitrary two hours of restart time. [...] **The market would be better served by improved resilience across the system as a whole, with true alternatives to primary markets (or any individual venue),** rather than a specific and arbitrary focus on restart times.’

ACI Financial Markets Association: ‘If an outage suspends trading on the main market, it is

important to be able to migrate to another trading venue for the use of reference data points to ensure that market liquidity is not affected, since the **simultaneous suspension of Algorithmic trading by numerous market participants could result in high volatility and a drastic reduction in market liquidity.**

World Federation of Exchanges (WFE): ‘The question as to whether to require use of more than one reference data point will, however, be a function of how well the alternative data points reflect the market in question. **Fragmentation of markets can bring choice (and competition in terms of execution costs) but not all alternative venues generate (or are capable of generating) meaningful price discovery of their own, instead relying on ‘main’ venues to do so.**

Chicago Board Options Exchange (CBOE): ‘Technical outages by European trading venues are a reasonably regular occurrence and largely inevitable. When they do happen, they are highly disruptive, particularly when experienced by national stock exchanges (‘primary outages’), which facilitate trading in the stocks they list and are the sole operators of official opening and closing auctions for those stocks. **Primary outages in recent years have seen market-wide trading in instruments listed on those exchanges dry up to almost nothing.**

All Options International: ‘**The added liquidity and/or other fail-over benefits that the fragmented market structure and its supporters claim to have in practice is just false.** This is shown by the drop of liquidity of secondary venues the moment the primary venue has an outage. **We believe that the best way to maintain liquid markets is to concentrate all liquidity on one, single venue. Real price forming always happens on the primary market.** In order to generate a better, more liquid market in all scenarios the fragmentation of markets needs to be stopped and all off-book and SI trading needs to be prohibited.’

Citigroup incident In 2005, the UK’s Financial Services Authority (FSA) ordered Citigroup to pay almost £14 million in fines for a trading strategy the bank had executed in August 2004, see e.g. the [FSA’s final notice](#) for a detailed description of the trading strategy and a timeline of events and this [Guardian article](#) for further background information.

What makes this incident particularly informative for our purposes is an internal memo cited by the FSA and in news articles by the [Financial Times](#), [New York Times](#) and [Euroweek](#) (now GlobalCapital). This memo was titled ‘*Challenging the dominance of Eurex futures*’ and was written by a member of Citigroup’s European government bond trading desk in London.

According to the FSA, the memo summarized the goal of Citigroup’s trading strategy as ‘*imposing a cost on competitors for offering liquidity on MTS, reducing the attractiveness of using the Bund future to hedge, widening cash market bid-offer spreads, [...] and helping remove some of the smaller primary dealers from the cash market.*’ The news articles quote the memo as saying ‘*Overall, these trades may help to reduce the market’s reliance on the Bund future and turn the European Government bond market into one that more closely resembles the US government bond market.*’

These quotes are in line with our evidence that proper EGB market functioning vitally depends on an active futures market.